

Totally 2-closed finite groups with trivial Fitting subgroup

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A finite permutation group $G \leq \text{Sym}(\Omega)$ is called 2-closed if G is the largest subgroup of $\text{Sym}(\Omega)$ which leaves invariant each of the G -orbits for the induced action on $\Omega \times \Omega$. Introduced by Wielandt in 1969, the concept of 2-closure has developed as one of the most useful approaches for studying relations on a finite set invariant under a group of permutations of the set; in particular for studying automorphism groups of graphs and digraphs. The concept of total 2-closure switches attention from a particular group action, and is a property intrinsic to the group: a finite group G is said to be totally 2-closed if G is 2-closed in each of its faithful permutation representations. There are infinitely many finite soluble totally 2-closed groups, and these have been completely characterized, but up to now no insoluble examples were known. It turns out, somewhat surprisingly to us, that there are exactly 6 totally 2-closed finite nonabelian simple

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groups: the Janko groups J_1 , J_3 and J_4 , together with Ly , Th and the Monster M . Moreover, if a finite totally 2-closed group has no nontrivial abelian normal subgroup, then we show that it is a direct product of some (but not all) of these simple groups, and there are precisely 47 examples.

In the course of obtaining this classification, we develop a general framework for studying 2-closures of transitive permutation groups, which we hope will prove useful for investigating representations of finite groups as automorphism groups of graphs and digraphs, and in particular for attacking the long-standing polycirculant conjecture. In this direction, we apply our results, proving a dual to a 1939 theorem of Frucht from Algebraic Graph Theory. We also pose several open questions concerning closures of permutation groups.

Keywords: 2-closed permutation groups; graph representations of groups; simple groups; polycirculant conjecture; Fitting subgroup.

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1. Introduction

In 1969, Wielandt [39, Definition 5.3] introduced the concept of 2-closure for a permutation group G on a set Ω . The 2-closure $G^{(2)}$ of G is the set of all $g \in \text{Sym}(\Omega)$ such that g leaves invariant each G -orbit in the induced G -action on ordered pairs from Ω . The 2-closure $G^{(2)}$ is a subgroup of $\text{Sym}(\Omega)$ containing G [39, Theorem 5.4], and G is said to be 2-closed if $G^{(2)} = G$.

It is important to note that the 2-closure of a permutation group G on Ω is not intrinsic to the group G . It can depend on the nature of the action on Ω . For example, the symmetric group $\text{Sym}(3)$ acts faithfully and intransitively on $\{1, 2, 3, 4, 5\}$ with orbits $\{1, 2, 3\}$ and $\{4, 5\}$, and in this action its 2-closure is $\text{Sym}(3) \times C_2$; while $\text{Sym}(3)$ is 2-closed in its natural action on $\{1, 2, 3\}$. In 2016, Holt^a suggested a stronger concept independent of the permutation representation. This was adopted and explored by the first author with Abdollahi in [1] where it was called *total 2-closure*. To explain the concept we first enrich the notation for 2-closure by writing $G^{(2), \Omega}$ for the 2-closure of a permutation group $G \leq \text{Sym}(\Omega)$. We then define a group G to be *totally 2-closed* if $G^{(2), \Omega} = G$ whenever G is faithfully represented as a permutation group on Ω . It was proved in [1, Theorem 1] that a finite totally 2-closed group has cyclic center, and moreover [1, Theorem 2] that a finite nilpotent group is totally 2-closed if and only if it is either cyclic or a direct product of a generalized quaternion group with a cyclic group of odd order.

The first and fourth authors, with Abdollahi undertook a general study of total 2-closure in [2]. They proved [2, Theorem B] that a finite soluble group is totally 2-closed if and only if it is nilpotent, and hence is one of the groups identified in [1, Theorem 2]. Further they showed in [2, Theorem A] that the Fitting subgroup of a totally 2-closed finite group is itself totally 2-closed (and hence lies in the family just described). At the time of their writing of the paper [2], no examples of insoluble totally 2-closed groups were known. They studied the structure of an

^amathoverflow.net/questions/235114/2-closure-of-a-permutation-group.

insoluble example G of smallest order, showing that G modulo its cyclic center $Z(G)$ has a unique minimal normal subgroup which is nonabelian [2, Proposition 4.1].

The first purpose of this paper is to seek out finite insoluble totally 2-closed groups G . We focus on the case where the Fitting subgroup $F(G)$ is trivial, or equivalently, the case where G has no nontrivial abelian normal subgroup. We show first that such a group G is a direct product of nonabelian simple groups, each of which is itself totally 2-closed (see Theorem 2.5). In light of this result, we decided that our next challenge should be to find all totally 2-closed finite simple groups. The 2-closure condition seemed so strong that we initially believed that we would find no examples. Surprisingly (to us) such groups do indeed exist, but they are rare, and the only examples are six of the sporadic simple groups.

Theorem 1.1. *Let T be a nonabelian finite simple group. Then T is totally 2-closed if and only if T is one of the groups J_1 , J_3 , J_4 , Ly , Th or M .*

Theorem 1.1 gives us the building blocks of finite totally 2-closed groups with trivial Fitting subgroup. We were able to use this result as the basis of an inductive approach yielding the following complete classification of all such groups.

Theorem 1.2. *Let G be a nontrivial finite group with trivial Fitting subgroup. Then G is totally 2-closed if and only if each of the following holds:*

- (1) $G = T_1 \times \cdots \times T_r$, where the T_i are nonabelian finite simple groups and $r \leq 5$;
- (2) $T_i \not\cong T_j$ for each $i \neq j$; and
- (3) One of the following holds:
 - (i) $T_i \in \{J_1, J_3, J_4, Th, Ly\}$ for each $i \leq r$; or
 - (ii) $T_i \in \{J_1, J_3, J_4, Ly, M\}$ for each $i \leq r$.

Corollary 1.3. *There are precisely 47 finite totally 2-closed groups with trivial Fitting subgroup.*

Proving these results involved a wide variety of methods, and we needed to develop several tools which we hope will assist future analyses of finite totally 2-closed groups.

The second purpose of the paper is to develop general machinery to study 2-closures of transitive permutation groups. This is not only crucial for our proof of Theorem 1.2, but is also key in order to make further progress on the *polycirculant conjecture* (see Sec. 1.1 for more details).

1.1. Motivation from graph theory and the polycirculant conjecture

The concept of 2-closure is a very combinatorial notion, and there is indeed a close relationship between 2-closed permutation groups and digraphs: each subset $A \subseteq \Omega \times \Omega$ may be viewed as the arc-set of a digraph $\Gamma = (\Omega, A)$ with vertex-set Ω and with a directed edge (an arc) from α to β for each $(\alpha, \beta) \in A$. The automorphism group $\text{Aut}(\Gamma)$ of Γ is defined as the subgroup of all permutations of Ω which

leave the arc-set A invariant. In particular, each element of $(\text{Aut}(\Gamma))^{(2),\Omega}$ leaves A invariant and hence is an automorphism of Γ . Thus $(\text{Aut}(\Gamma))^{(2),\Omega} \subseteq \text{Aut}(\Gamma)$, and the reverse inclusion holds by the definition of 2-closure. Therefore $(\text{Aut}(\Gamma))^{(2),\Omega} = \text{Aut}(\Gamma)$. The same argument also shows that $(\text{Aut}(\Gamma))^{(2),\Omega} = \text{Aut}(\Gamma)$ when Γ is an undirected graph. Thus, to summarize, *for each (di)graph Γ with vertex set Ω , the automorphism group $\text{Aut}(\Gamma)$ is a 2-closed subgroup of $\text{Sym}(\Omega)$* . In fact, for each permutation group $G \leq \text{Sym}(\Omega)$, the 2-closure $G^{(2),\Omega}$ is equal to the intersection of the groups $\text{Aut}(\Gamma)$ over all digraphs $\Gamma = (\Omega, A)$ for which the arc-set A is a G -orbit in $\Omega \times \Omega$.

In particular, if $G \leq \text{Sym}(\Omega)$ is transitive on Ω and if A is a G -orbit in $\Omega \times \Omega$ not equal to the diagonal $\{(\alpha, \alpha) \mid \alpha \in \Omega\}$, then the digraph $\Gamma = (\Omega, A)$ is called a *G -orbital digraph*. The study of G -orbital digraphs goes back to the work of Higman [17] at approximately the same time as Wielandt's work on closures in the late 1960s.

Let $\Gamma = (\Omega, A)$ be a digraph as above, with vertex-set Ω and arc-set $A \subseteq \Omega \times \Omega$. An element of $\text{Sym}(\Omega)$ in which all cycles have the same length is called *semiregular*; while an automorphism α of Γ is called *semiregular* if the permutation induced by α on the vertices of Γ is semiregular. The graph Γ is said to be *polycirculant* if $\text{Aut}(\Gamma)$ contains a nontrivial semiregular element.

In 1981 [29], Marušič made the following well-known conjecture: *Every finite connected vertex-transitive digraph is a polycirculant*. This is known in the literature as the *polycirculant conjecture*, and was also made (independently) by Jordan in 1988 [21]. The conjecture was generalized to the following by Klin at the British Combinatorial Conference in 1995: *Every finite transitive 2-closed permutation group of degree greater than 1 contains a semiregular element*. As explained above, the automorphism group of every finite digraph Γ with vertex set Ω is a 2-closed subgroup of $\text{Sym}(\Omega)$. Thus, Klin's generalization is stronger than the polycirculant conjecture. We point the reader to the beautiful survey [24] by Kutnar and Marušič for further observations on the underlying philosophy behind the conjecture, and its importance in graph theory. See also [8, Sec. 1.3.4] and the more recent survey [3].

There have been two main approaches to proving the polycirculant conjecture (or rather, Klin's generalized conjecture). The first has been to study those transitive permutation groups which do not contain a semiregular element (such groups are called *elusive*), and show that no such group can be 2-closed. This approach has led to some outstanding breakthroughs, one of the most significant of which was Giudici's theorem [15]. There, it is proved that apart from a family of groups of product action type associated to the Mathieu group M_{11} , all quasiprimitive permutation groups contain semiregular elements (a permutation group is called *quasiprimitive* if every nontrivial normal subgroup is transitive). Further, of those quasiprimitive groups which fail to have semiregular elements, none are 2-closed. This, combined with [10, Theorem 5.5], yields the proof of the polycirculant conjecture in the case where the automorphism group of the digraph in question is almost simple.

The second main approach is the dual to the first approach: to study the 2-closed transitive permutation groups, and prove that all such groups contain semiregular elements. The first step in such an approach is to develop machinery to determine the 2-closure of a given transitive permutation group. When the given transitive group is primitive, a powerful theorem of Liebeck *et al.* [26] shows that, apart from a finite list of exceptions and two known infinite families, the 2-closure of an almost simple simply primitive group $G \leq \text{Sym}(\Omega)$ is also almost simple, with the same socle as G . Further progress on the primitive case was made by Praeger and Saxl in [31], where it is shown that if G is an arbitrary primitive group, then $\text{soc}(G) = G^{(2),\Omega}$, apart from some known families of exceptions.

These results greatly facilitate the computation of the 2-closure of a primitive permutation group. On the other hand, there has been little to no progress in building general techniques to determining the 2-closure of an imprimitive group. Since Giudici's theorem proves the polycirculant conjecture in the primitive case, building such techniques is key for further progress.

With this in mind, part of the motivation for this paper was to study 2-closures of imprimitive permutation groups which induce a faithful 2-closed action on a set of maximal blocks. Our main results are Theorems 3.8 and 3.10 (for notational convenience, we postpone their statements until Sec. 3).

1.2. An application: Frucht representations

Before describing our approach to proving Theorems 1.1 and 1.2, we would like to draw attention to a graph theoretic consequence of our results, which may be of independent interest. In 1939, Frucht [14] confirmed a conjecture of König that every finite group G can be written as the automorphism group of a finite simple undirected graph Γ . We will say that a faithful permutation representation $G \hookrightarrow G^\Omega \leq \text{Sym}(\Omega)$ of G on a finite set Ω is a *Frucht representation* of G if there exists a set E of unordered pairs from Ω such that G^Ω is the automorphism group of the finite simple undirected graph $\Gamma = \Gamma(\Omega, E)$. Since $\text{Aut}(\Gamma)$ is 2-closed, Theorem 1.2 implies the following, which can be viewed as a dual to Frucht's theorem in the case where $F(G) = 1$.

Corollary 1.4. *Let G be a nontrivial finite group with trivial Fitting subgroup, and assume that every faithful permutation representation of G is a Frucht representation. Then $G = T_1 \times \cdots \times T_r$, where the T_i are pairwise non-isomorphic finite simple groups; $r \leq 5$; and either*

- (i) *for each $i \leq r$, $T_i \in \{J_1, J_3, J_4, \text{Th}, \text{Ly}\}$; or*
- (ii) *for each $i \leq r$, $T_i \in \{J_1, J_3, J_4, \text{Ly}, \mathbb{M}\}$.*

It is an open question as to whether the converse of Corollary 1.4 holds. This, and further open questions concerning closures of permutation groups are posed below.

1.3. Some open questions

We mention a few open questions about closures of permutation groups. Firstly, since we have considered only groups with trivial Fitting subgroup in this paper, we ask.

Question 1. Are there any finite insoluble totally 2-closed groups with nontrivial Fitting subgroup? If so, can we classify them?

One of the ultimate goals of this programme of work is to classify all finite totally 2-closed groups. Given the results in this paper, this goal would be realized with a complete answer to Question 1. See Sec. 8 for some remarks on the obstacles involved in answering this question.

The analogous concept of total k -closure may be defined for arbitrary positive integers k , namely, G is *totally k -closed* if, for every faithful representation $G \leq \text{Sym}(\Omega)$, G is equal to the set of all elements $g \in \text{Sym}(\Omega)$ such that g leaves invariant each G -orbit in the induced G -action on ordered k -tuples from Ω . This concept has been explored by Churikov and the third author in [11]. For each $k \geq 2$, it follows from [39, Theorem 5.8] that, if G is totally $(k-1)$ -closed then G is also totally k -closed. On the other hand, by [11, Theorem 1.2], there are infinitely many finite totally k -closed groups which are not totally $(k-1)$ -closed, and this is true even for abelian p -groups for each prime p . The paper [11] gives a discussion of the topic and poses three open problems about total k -closure (where $k \geq 3$) for finite nilpotent groups, soluble groups, and sporadic simple groups — all problems which have been resolved for $k = 2$, in [1, 2] and this paper. It follows from [39, Theorem 5.11] that an alternating group A_n (with $n \geq 5$) is not totally k -closed for any $k \leq n-2$, so we further ask.

Question 2. Are there any finite simple groups of Lie type which are totally 3-closed? If so, find them all.

The example of the alternating groups above shows that there is no fixed integer k such that all nonabelian finite simple groups are totally k -closed. It is worthwhile to note, however, that for any faithful representation of a finite nonabelian simple group G , there is a G -orbit on which G acts non-trivially and hence faithfully; applying [39, Theorem 5.12] to the G -action on this orbit we deduce that G is totally n -closed, where $n = |G|$. Since the alternating groups seem like exceptional cases in practice, we therefore ask.

Question 3. Is there a fixed integer k such that all nonabelian finite simple groups which are not alternating groups are totally k -closed?

We remark that we submitted Questions 1–3 to the latest (20th) edition of the Kourovka notebook (a collection of unsolved problems in Group Theory) which has just been released [23]. They appear as Questions 20.1–20.3 — the first three of 126 new open problems.

Finally, and as mentioned above, it is not clear whether or not the converse to Corollary 1.4 holds. So we ask.

Question 4. Let $G = T_1 \times \cdots \times T_r$, where the T_i are pairwise non-isomorphic finite simple groups; $r \leq 5$; and either

- (i) for each $i \leq r$, $T_i \in \{J_1, J_3, J_4, \text{Th}, \text{Ly}\}$; or
- (ii) for each $i \leq r$, $T_i \in \{J_1, J_3, J_4, \text{Ly}, \mathbb{M}\}$.

Then is it true that every faithful permutation representation of G is a Frucht representation? If not, then which (if any) of the listed 47 groups does have this property?

1.4. Structure of the paper

In the remained of this section, we outline our approach to the proofs of Theorems 1.1 and 1.2, and detail the structure of the paper. The fundamental starting point is Wielandt's Dissection Theorem [39, Dissection Theorem 6.5] which for convenience we state as Theorem 2.1. It is an invaluable tool, and in particular it underpins our proof of Theorem 2.5, which we present in Sec. 2.

As mentioned above, Theorem 2.5 reduces our classification of finite totally 2-closed groups G with trivial Fitting subgroup to the case where G is a direct product of nonabelian simple groups, which are 'independent' of each other in the sense that each of the simple direct factors is not isomorphic to a section of the direct product of the others. (We comment on this condition in relation to the sporadic simple groups in Remark 2.6). Our next step is to reduce further: in Proposition 2.10, we show that such a group G is totally 2-closed if and only if no simple direct factor of G admits a nontrivial factorization, and each (not necessarily faithful) *transitive* permutation representation of G is 2-closed.

This suggests two natural avenues to pursue. The first is to use the classification of the maximal factorizations of the nonabelian simple groups in [27] to eliminate many families of simple groups from being direct factors of G . We do this in Sec. 2.3, where we show, in particular, that the simple direct factors of G are either sporadic groups or exceptional groups of Lie type.

Our second natural avenue to pursue concerns transitive actions of such groups G . Strong restrictions on the 2-closures of primitive almost simple groups were obtained in [26]. With these restrictions in mind for the case where G is simple, we develop powerful reduction theorems in Sec. 3 for general transitive imprimitive permutation groups G which induce a faithful 2-closed action on a set of nontrivial blocks of imprimitivity: we show that either G itself is 2-closed, or the 2-closure of G has a very particular structure. This provides further criteria to determine whether or not such permutation groups are 2-closed (see, in particular, Theorem 3.10).

The tools developed in Secs. 2 and 3 are sufficient to begin our analysis of the 2-closures of permutation representations of the finite simple groups. In Secs. 4 and 6, we deal with the sporadic groups other than J_4 , and the exceptional groups of Lie

type, respectively. The case of $G = J_4$ is delicate, and requires a nuanced approach, see Sec. 5. These three sections complete the proof of Theorem 1.1.

Section 7 is reserved for the study of 2-closures of direct products of finite simple groups, and the proof of Theorem 1.2.

Finally, in Sec. 8, we make some concluding remarks on the question of classifying the finite insoluble totally 2-closed groups with nontrivial Fitting subgroup. We also prove a useful reduction theorem for this problem.

Notation: Most of our notation is standard. For a finite group G , we write $Z(G)$, $F(G)$, $\Phi(G)$ and $[G, G]$ for the center, Fitting subgroup, Frattini subgroup, and derived subgroup of G , respectively.

For a set Ω , we write $\text{Sym}(\Omega)$ and $\text{Alt}(\Omega)$ for the symmetric and alternating group on Ω , respectively. If a finite group G acts on Ω , we write G^Ω for the image of the induced action of G on Ω .

For group names, we use conventions from the Atlas [12, especially p. xx], except that we write $\text{Alt}(n)$ and $\text{Sym}(n)$ for the alternating and symmetric groups of degree n .

2. Applying Wielandt's Dissection Theorem

As we mentioned in Sec. 1, Wielandt's Dissection Theorem [39, Dissection Theorem 6.5] is a fundamental tool for studying 2-closure. For convenience we state it here. Note that, for $G \leq \text{Sym}(\Omega)$ and $\delta \in \Omega$, we denote by G_δ the stabilizer of δ in G , and by δ^G the G -orbit in Ω containing δ . Also if $\Gamma \subseteq \Omega$ and Γ is G -invariant, then G^Γ denotes the permutation group on Γ induced by G .

Theorem 2.1 (Wielandt's Dissection Theorem). *Let $G \leq \text{Sym}(\Omega)$, and suppose that Ω is the disjoint union $\Omega = \Gamma \cup \Delta$ such that both Γ and Δ are G -invariant. Then the following are equivalent:*

- (a) $G^\Gamma \times G^\Delta \leq G^{(2), \Omega}$;
- (b) $G = G_\gamma G_\delta$ for all $\gamma \in \Gamma, \delta \in \Delta$;
- (c) G_δ is transitive on γ^G for all $\gamma \in \Gamma, \delta \in \Delta$.

In this section, we discuss some ways in which this result can be applied. To begin, we state the following helpful application given in [2]. Here, for a subgroup H of a group G , we denote by $\text{Core}_G(H)$ the *core* of H in G , that is, $\text{Core}_G(H) = \bigcap_{g \in G} H^g$, the largest normal subgroup of G contained in H . By a *nontrivial factorization* of G we mean a factorization $G = HK$ with both H and K proper subgroups of G .

Lemma 2.2 ([2, Lemma 3.1]). *Let G be a finite totally 2-closed group such that G admits a nontrivial factorization $G = HK$ with*

$$\text{Core}_G(H) \cap \text{Core}_G(K) = 1.$$

Then $G = \text{Core}_G(H) \times \text{Core}_G(K)$ and both $\text{Core}_G(H)$ and $\text{Core}_G(K)$ are totally 2-closed. In particular, if $H \cap K = 1$, then $G = H \times K$ and both H and K are totally 2-closed.

Lemma 2.2 has the following important consequence for simple groups.

Corollary 2.3. *Let T be a finite nonabelian simple group. If T is totally 2-closed, then T admits no nontrivial factorization.*

Proof. Suppose that T is totally 2-closed. If $T = HK$ is a nontrivial factorization, then the cores $\text{Core}_T(H) = \text{Core}_T(K) = 1$ since T is simple, and this contradicts Lemma 2.2. Hence T has no nontrivial factorizations. $\square$

The following result from Wielandt [39] is also useful when making a detailed analysis of 2-closures.

Theorem 2.4 ([39, Theorem 5.6]). *Let $G \leq \text{Sym}(\Omega)$ and $x \in \text{Sym}(\Omega)$. Then $x \in G^{(2),\Omega}$ if and only if, for all $\alpha, \beta \in \Omega$, there exists $g \in G$ such that $\alpha^x = \alpha^g$ and $\beta^x = \beta^g$.*

2.1. Reduction to direct products of nonabelian finite simple groups

In this section, we reduce our proof of Theorem 1.2 to particular direct products of finite simple groups. Before we state our main result, we remind the reader of some standard terminology: a nontrivial group X is said to be a *section* of a group G if there are subgroups $L \trianglelefteq H \leq G$ such that $H/L \cong X$. Our reduction theorem can now be stated as follows.

Theorem 2.5. *Let G be a nontrivial finite totally 2-closed group such that $F(G) = 1$. Then $G = T_1 \times \cdots \times T_r$ for some $r \geq 1$, where for each $i \leq r$, T_i is a finite nonabelian simple group which is totally 2-closed, and T_i is not a section of $\prod_{j \neq i} T_j$.*

Proof. Since G is finite, nontrivial, and $F(G) = 1$, a minimal normal subgroup M of G is a direct product of nonabelian finite simple groups. If $M = G$ then, as M is minimal normal, the group G would itself be a nonabelian simple group which is totally 2-closed, and the conclusion holds with $r = 1$. Thus, we may assume that $M \neq G$, and $M = S_1 \times \cdots \times S_s$ with each S_i a finite nonabelian simple group. We note that the minimal normal subgroups of M are precisely the subgroups

$$\widehat{S}_i = \{(x_1, \dots, x_s) \mid x_j = 1 \text{ for } j \neq i, \text{ and } x_i \in S_i\} \cong S_i,$$

for $i = 1, \dots, s$ (see, for example [32, Theorem 4.16(iv)]). By the ‘Odd Order Theorem’, since each of the $\widehat{S}_i$ is a finite nonabelian simple group, each $\widehat{S}_i$ has a nontrivial Sylow 2-subgroup, P_i say, and hence $P = P_1 \times \cdots \times P_s$ is a (nontrivial) Sylow 2-subgroup of M . By the Frattini argument (see, for example [34, 5.2.14]),

$G = N_G(P)M$. Now, $N_M(P) = N_{\widehat{S}_1}(P_1) \times \cdots \times N_{\widehat{S}_s}(P_s)$, and $N_M(P)$ projects to a proper subgroup $N_{\widehat{S}_i}(P_i)$ of $\widehat{S}_i$, for each i . In particular $M \not\leq N_G(P)$. Hence the core $X = \text{Core}_G(N_G(P))$ does not contain M , and so $X \cap M = 1$ since M is a minimal normal subgroup of G . Also since M is normal in G , $\text{Core}_G(M) = \bigcap_{g \in G} M^g = M$. Then, since G is totally 2-closed, it follows from Lemma 2.2 that $G = X \times M$ and both X and M are totally 2-closed. Since $M \neq G$ it follows that $X \neq 1$ and since M is a minimal normal subgroup of G we conclude that $M = \widehat{S}_1$ is a finite nonabelian simple group.

Thus each minimal normal subgroup of G is a totally 2-closed, finite nonabelian simple group and is a direct factor of G . Applying this argument recursively we conclude that $G = T_1 \times T_2 \times \cdots \times T_r$ for some $r \geq 2$, where each T_i is a totally 2-closed, finite nonabelian simple group. For $1 \leq j \leq r$ let π_j denote the projection $\pi_j : G \rightarrow T_j$ given by $(t_1, \dots, t_r) \rightarrow t_j$, and let $\widehat{T}_j$ denote the minimal normal subgroup of G defined similarly to the $\widehat{S}_i$ above.

Let i, j be distinct integers between 1 and r . We show next that T_i is not a section of T_j . Suppose that this is false, so there exist subgroups $L \trianglelefteq H \leq T_j$ such that $H/L \cong T_i$, or equivalently, there exists an epimorphism $\varphi : H \rightarrow T_i$ with kernel L . Without loss of generality, assume that $i = 1$ and $j = 2$, and let $J = \{(\varphi(h), h, 1, \dots, 1) \mid h \in H\}$ and $K = 1 \times T_2 \times \cdots \times T_r$. Clearly both J and K are proper subgroups of G . Moreover, since φ is an epimorphism it follows that $G = JK$ is a nontrivial factorization, and since K is normal in G we have $\text{Core}_G(K) = K$. We claim that $\text{Core}_G(J) = 1$. Now, $\text{Core}_G(J) \leq J < \widehat{T}_1 \times \widehat{T}_2 \times 1 \cdots \times 1$, and if $\text{Core}_G(J) \neq 1$ then $\text{Core}_G(J)$ would contain at least one of the minimal normal subgroups of G , namely one of $\widehat{T}_1, \dots, \widehat{T}_r$, so $\text{Core}_G(J)$ would contain $\widehat{T}_1$ or $\widehat{T}_2$. Let $x = (\varphi(h), h, 1, \dots, 1) \in \text{Core}_G(J)$. If $x_1 = \varphi(h) \neq 1$ then $h \neq 1$, so $\text{Core}_G(J) \cap \widehat{T}_1 = 1$. Also if $x \in \widehat{T}_2$, then $x_1 = \varphi(h) = 1$ so $h \in \text{Ker}(\varphi) = L$. Since L is a proper subgroup of T_2 it follows that $\text{Core}_G(J)$ does not contain $\widehat{T}_2$ either. Hence $\text{Core}_G(J) = 1$ as claimed. It now follows from Lemma 2.2 that $G = 1 \times K$, which is a contradiction. Thus none of the T_i is a section of any of the other T_j (and in particular the T_i are pairwise non-isomorphic).

Finally, suppose that, for some $i \leq r$, T_i is a section of $K := \prod_{j \neq i} T_j$. Then there are subgroups $L \trianglelefteq H \leq K$ such that $H/L \cong T_i$. Let H, L be subgroups of K with this property such that $|H|$ is as small as possible. Identify K with the normal subgroup $\prod_{j \neq i} \widehat{T}_j$ of G . Let $1 \leq j \leq r$ with $j \neq i$, and consider $H_j := H \cap (\prod_{k \neq j} \widehat{T}_k)$, the kernel of the restriction $\pi_j|_H$ to H of the projection π_j . We will identify $\pi_j(H)$ with the quotient H/H_j so that, for each subgroup $J \leq H$, the image $\pi_j(J)$ is identified with JH_j/H_j . In particular $\pi_j(L) = LH_j/H_j$. Then

$$Q := \pi_j(H)/\pi_j(L) = (H/H_j)/(LH_j/H_j) \cong H/(LH_j) \cong (H/L)/(LH_j/L),$$

a quotient of the simple group $H/L \cong T_i$. Thus $Q \cong T_i$ or 1. Now, $\pi_j(L) \trianglelefteq \pi_j(H) \leq T_j$, and by the previous paragraph T_i is not a section of T_j , and hence the quotient $\pi_j(H)/\pi_j(L) \not\cong T_i$. Thus, Q must be trivial, and hence $H = LH_j$. This, however, implies that $T_i \cong H/L = (LH_j)/L \cong H_j/(L \cap H_j)$, and so, by the minimality of

$|H|$, we conclude that $H = H_j$, that is to say, $\pi_j(H) = 1$. Since this holds for each $j \neq i$, and since $H \leq \prod_{j \neq i} \hat{T}_j = K$, it follows that $H = 1$, which is a contradiction. Thus T_i is not a section of K , and the proof is complete. $\square$

Remark 2.6. We make some comments about the condition in Theorem 2.5 that each T_i is not isomorphic to a section of the product $\prod_{j \neq i} T_j$. Since the T_i are simple groups, this condition is equivalent to requiring that, for all distinct i, j , the group T_i is not isomorphic to a section of T_j . Deciding precisely when this condition holds was not easy, but in the case where both T_i and T_j are sporadic simple groups (which is the case with the examples in Theorem 1.2), an almost complete answer is given in [12, p. 238], which provides a beautiful diagram showing exactly which sporadic simple groups are involved in this way in other sporadic simple groups. The only unresolved case in this diagram is whether or not the small Janko group J_1 occurs as a section of the Monster $\mathbb{M}$, but it is now known that J_1 is not a section of the Monster, see [40, Sec. 5.8] for a discussion. Considering this diagram (and the situation for $J_1, \mathbb{M}$) we see that, for T_i, T_j any distinct pair of sporadic groups occurring in the statement of Theorem 1.2, if T_i is a section of T_j then $T_i = Th$ and $T_j = \mathbb{M}$ (which explains the need for two parts in Theorem 1.2(3)).

2.2. Totally 2-closed direct products of simple groups

By Theorem 2.5, we need to analyze direct products of totally 2-closed simple groups. With this in mind we introduce the following hypothesis and notation.

Hypothesis 2.7. Let $G = T_1 \times T_2 \times \cdots \times T_r$ for some $r \geq 1$, where each T_i is a finite nonabelian simple group, and suppose that, for $1 \leq i \leq r$, T_i is not a section of $\prod_{j \neq i} T_j$. For $1 \leq i \leq r$, let $\pi_i : G \rightarrow T_i$ denote the projection map $\pi_i : (t_1, \dots, t_r) \rightarrow t_i$, and define

$$\hat{T}_i = \{g \in G \mid \pi_j(g) = 1 \text{ for all } j \neq i\} \cong T_i.$$

As mentioned earlier, (see, for example [32, Theorem 4.16(iv)]), $\{\hat{T}_1, \dots, \hat{T}_r\}$ is the complete set of minimal normal subgroups of G .

This notation helps avoid possible confusion by distinguishing between the abstract group T_i and the simple normal subgroup $\hat{T}_i$ of G . Note that Hypothesis 2.7 holds, by Theorem 2.5, if G is totally 2-closed with $F(G) = 1$. We extend the argument in the last paragraph of the proof of Theorem 2.5 to obtain the following characterization of certain subgroups of G .

Lemma 2.8. *Let G, r be as in Hypothesis 2.7, and let $i \leq r$. If $H \leq G$ has T_i as a section, then H contains $\hat{T}_i$.*

Proof. Suppose that $H \leq G$ and H has a section T_i , that is to say, there exist $L \trianglelefteq K \leq H$ such that $K/L \cong T_i$. We need to show that $\hat{T}_i \leq H$. Without loss of generality, we may assume that no proper subgroup of K has T_i as a section. Let $1 \leq$

$j \leq r$ with $j \neq i$, and let $J = K \cap (\prod_{k \neq j} \widehat{T}_k)$, the kernel of the restriction $\pi_j|_K$ of π_j to K . Then $\pi_j(L) \leq \pi_j(K) \leq T_j$ and $\pi_j(K)/\pi_j(L) \cong (K/J)/(LJ/J) \cong K/(LJ) \cong (K/L)/(LJ/L)$, a quotient of the simple group $K/L \cong T_i$. By Hypothesis 2.7, T_i is not a section of $\prod_{k \neq i} T_k$, and in particular, T_i is not a section of T_j . Hence $\pi_j(K)/\pi_j(L) \not\cong T_i$. This implies that $\pi_j(K) = \pi_j(L)$, or equivalently, $K = LJ$. Thus $T_i \cong K/L = (LJ)/L \cong J/(L \cap J)$, and since T_i is not a section of any proper subgroup of K , we conclude that $K = J$, that is, $\pi_j(K) = 1$. Since this holds for each $j \neq i$, it follows that $K \leq \widehat{T}_i$, and since T_i is a section of K this means that $K = \widehat{T}_i$. Hence $\widehat{T}_i \leq H$. $\square$

Lemma 2.8 has an important consequence for factorizations of groups satisfying Hypothesis 2.7. The proof uses the notion of a strip. For $G = T_1 \times T_2 \times \cdots \times T_r$ as in Hypothesis 2.7, a subgroup D is called a *strip* if $D \neq 1$, and for each $i \leq r$, either $\pi_i(D) = 1$ or $D \cap \text{Ker}(\pi_i) = 1$. Note that $D \cap \text{Ker}(\pi_i) = 1$ if and only if the restriction $\pi_i|_D$ of π_i to D is an isomorphism from D to $\pi_i(D)$. The *support* of a strip is the set $\{i \mid \pi_i(D) \neq 1\}$, and two strips are said to be *disjoint* if their supports are disjoint. A strip D is *full* if $\pi_i(D) = T_i$ for all i in the support of D . A subgroup H of a group G with a given direct decomposition $G = T_1 \times T_2 \times \cdots \times T_r$ is called a *subdirect subgroup* if $\pi_i(H) = T_i$ for each $i = 1, \dots, r$.

Lemma 2.9. *Let G be as in Hypothesis 2.7, and such that each T_i admits no nontrivial factorization. If $G = HK$ is a nontrivial factorization, then $\text{Core}_G(H) \neq 1$ and $\text{Core}_G(K) \neq 1$.*

Proof. Suppose that $G = HK$ with proper subgroups H, K , and suppose that one of H, K is core-free, say $\text{Core}_G(H) = 1$. By the assumption on the T_i , we must have $r \geq 2$. Let $i \leq r$. If $\pi_i(H) = T_i$, then T_i is a quotient of H , and hence by Lemma 2.8, H contains $\widehat{T}_i$, contradicting the fact that $\text{Core}_G(H) = 1$. Thus $\pi_i(H) < T_i$. Now, the factorization $G = HK$ implies that $T_i = \pi_i(H)\pi_i(K)$, and since $\pi_i(H) < T_i$, and by assumption the group T_i has no nontrivial factorization, it follows that $\pi_i(K) = T_i$. Thus K is a subdirect subgroup of $G = T_1 \times T_2 \times \cdots \times T_r$. By ‘Scott’s Lemma’ [32, Theorem 4.16(iii)], K is a direct product of pairwise disjoint full strips of G . Since K is a proper subgroup of G , at least one of these strips, say D , has support of size greater than 1. Let i, j be distinct elements of the support of D . By the definition of a strip, $D \cap \text{Ker}(\pi_i) = 1$ and $D \cap \text{Ker}(\pi_j) = 1$, and as we noted above, this implies that $D \cong \pi_i(D) = T_i$ and $D \cong \pi_j(D) = T_j$. This implies that T_i is a section of T_j , and hence also of $\prod_{k \neq i} T_k$, contradicting Hypothesis 2.7. $\square$

The next proposition assists us to decide which groups satisfying Hypothesis 2.7 are in fact totally 2-closed by reducing consideration to transitive (not necessarily faithful) permutation representations.

Proposition 2.10. *Let G be as in Hypothesis 2.7. Then G is totally 2-closed if and only if, the following two conditions hold:*

- (a) *For $i = 1, \dots, r$, the group T_i admits no nontrivial factorization; and*
- (b) *For each permutation representation $\varphi : G \rightarrow \text{Sym}(\Omega)$ with $|\Omega| > 1$ and $\varphi(G)$ transitive, we have $\varphi(G)^{(2),\Omega} = \varphi(G)$.*

Proof. Suppose first that G is totally 2-closed. Then condition (a) holds by Theorem 2.5. Let $\varphi : G \rightarrow \text{Sym}(\Omega)$ with $|\Omega| > 1$ and $\varphi(G)$ transitive. By [32, Theorem 4.16(iv)], there is a proper subset $I \subseteq \{1, \dots, r\}$ such that $\text{Ker}(\varphi) = \prod_{i \in I} \hat{T}_i$, and so $\varphi(G) \cong G_0$ where $G_0 := \prod_{j \in J} \hat{T}_j$ with $J = \{1, \dots, r\} \setminus I$. Hence $G = \text{Ker}(\varphi) \times G_0$ and by Lemma 2.2, G_0 is totally 2-closed, whence $\varphi(G)^{(2),\Omega} = \varphi(G)$ holds. Thus condition (b) holds also.

Conversely suppose that conditions (a) and (b) hold for G . In particular each nontrivial transitive permutation representation of G is 2-closed. In order to prove that G is totally 2-closed we must therefore show that each faithful intransitive permutation representation of G is also 2-closed. Suppose that this is not true. Then we may suppose that $G \leq \text{Sym}(\Omega)$ with orbits $\Omega_1, \dots, \Omega_m$ and $m \geq 2$, such that $X := G^{(2),\Omega}$ contains G as a proper subgroup. Suppose moreover that m is minimal, over all faithful non-2-closed intransitive representations of all groups G satisfying Hypothesis 2.7 and satisfying conditions (a) and (b).

For each $i \leq m$, let $G_i = G^{\Omega_i}$, the transitive permutation group on Ω_i induced by G . It follows from the minimality of m that each of the $|\Omega_i| \geq 2$ and hence each of the G_i is a nontrivial transitive permutation representation of G . Thus by condition (b), each G_i is 2-closed, that is $G_i^{(2),\Omega_i} = G_i$. Let $\Delta = \bigcup_{i=2}^m \Omega_i$, and let $Y = G^\Delta$, the permutation group on Δ induced by G . Then, arguing as in the first paragraph, $G_1 \cong \prod_{i \in I} T_i$ and $Y \cong \prod_{j \in J} T_j$ for some non-empty subsets $I, J \subseteq \{1, \dots, r\}$, and since G is faithful on Ω , $I \cup J = \{1, \dots, r\}$. Further, Hypothesis 2.7 holds for Y , as does condition (a) since it holds for G . Also, condition (b) on nontrivial transitive permutation representations for Y holds since each such representation for Y is also a nontrivial transitive permutation representation of G . Since Y has only $m - 1$ orbits in Δ , it follows that $Y^{(2),\Delta} = Y$ (either from the minimality of m if $m \geq 3$, or from the condition on nontrivial transitive permutation representations of G if $m = 2$). Now, $\Omega = \Omega_1 \cup \Delta$, and by Theorem 2.4, $X = G^{(2),\Omega} \leq G_1^{(2),\Omega_1} \times Y^{(2),\Delta} = G_1 \times Y \cong (\prod_{i \in I} T_i) \times (\prod_{j \in J} T_j)$. If $I \cap J = \emptyset$, then since $I \cup J = \{1, \dots, r\}$ it would follow that $X \cong G$, which is a contradiction. Hence $I \cap J \neq \emptyset$, X is a subdirect subgroup of $G_1 \times Y$, and

$$G_1 \times Y \cong \prod_{i=1}^r T_i^{\delta_i}, \quad \text{where } \delta_i \in \{1, 2\} \quad \text{and} \quad \delta_i = 2 \quad \text{if and only if } i \in I \cap J.$$

Hence $X \cong \prod_{k=1}^r T_k^{\alpha_k}$, where for each k , $1 \leq \alpha_k \leq \delta_k$, and since $G < X$ there exists i such that $\alpha_i = \delta_i = 2$. Such an i must lie in $I \cap J$, and without loss of generality,

we may assume that $i = 1$ so $1 \in I \cap J$. Hence $\widehat{T}_1$ acts non-trivially on both Ω_1 and Δ , and X contains $\widehat{T}_1^{\Omega_1} \times \widehat{T}_1^\Delta \cong T_1^2$. Again, without loss of generality, $\widehat{T}_1$ acts non-trivially on Ω_2 , and $\widehat{T}_1^{\Omega_2} \cong \widehat{T}_1^\Delta$.

Let $K = \prod_{j \neq 1} \widehat{T}_j$, so that $G = \widehat{T}_1 \times K$. If K is transitive on Ω_1 then $G = KG_\omega$, where $\omega \in \Omega_1$, and $G_\omega/(K \cap G_\omega) \cong G_\omega K/K = G/K \cong T_1$. Then, by Lemma 2.8, $\widehat{T}_1 \leq G_\omega$, which implies that $\widehat{T}_1$ fixes Ω_1 pointwise, a contradiction. Hence K is intransitive on Ω_1 , and the same argument shows that K is also intransitive on Ω_2 . Let Σ_i denote the set of K -orbits in Ω_i , for $i = 1, 2$. Since $G = \widehat{T}_1 \times K$, the group $\widehat{T}_1$ is transitive on both Σ_1 and Σ_2 , and since X contains $\widehat{T}_1^{\Omega_1} \times \widehat{T}_1^\Delta \cong T_1^2$, it follows that X is transitive on $\Sigma_1 \times \Sigma_2$ in product action, that is, $x : (\sigma_1, \sigma_2) \rightarrow (\sigma_1^x, \sigma_2^x)$ for $\sigma_1 \in \Sigma_1, \sigma_2 \in \Sigma_2, x \in X$.

We claim that the subgroup $\widehat{T}_1$ is also transitive on $\Sigma_1 \times \Sigma_2$. Choose $(\sigma_1, \sigma_2) \in \Sigma_1 \times \Sigma_2$, and let $\beta_1 \in \sigma_1, \beta_2 \in \sigma_2$. Since $X = G^{(2), \Omega}$, the G -orbit $(\beta_1, \beta_2)^G$ is equal to the X -orbit $(\beta_1, \beta_2)^X$. First, we note that, since X is transitive on $\Sigma_1 \times \Sigma_2$, the stabilizer X_{σ_1} is transitive on Σ_2 . Now, σ_1 is a K -orbit in Ω_1 and so $X_{\sigma_1} = KX_{\beta_1}$. It follows that the X_{β_1} -orbit containing β_2 must contain points from each element of Σ_2 . On the other hand, $(\beta_1, \beta_2)^G = (\beta_1, \beta_2)^X$ implies that the X_{β_1} -orbit containing β_2 is the same as the G_{β_1} -orbit containing β_2 , and therefore this G_{β_1} -orbit contains points from each element of Σ_2 , and hence G_{β_1} is transitive on Σ_2 . Since $G_{\beta_1} < G_{\sigma_1}$, also G_{σ_1} is transitive on Σ_2 and hence G is transitive on $\Sigma_1 \times \Sigma_2$. Since $G = \widehat{T}_1 \times K$ and K acts trivially on $\Sigma_1 \times \Sigma_2$, we conclude that $\widehat{T}_1$ is transitive on $\Sigma_1 \times \Sigma_2$, proving the claim.

Finally, for $i = 1, 2$, let $S_i = (\widehat{T}_1)_{\sigma_i}$. Then since $\widehat{T}_1$ is transitive on $\Sigma_1 \times \Sigma_2$, S_1 is transitive on Σ_2 and hence $\widehat{T}_1 = S_1 S_2$, a nontrivial factorization, which is a contradiction to condition (a). $\square$

For proving that a simple group is totally 2-closed, the following immediate corollary of Proposition 2.10 gives a reduction to consideration of transitive actions.

Corollary 2.11. *Let T be a finite nonabelian simple group. Then T is totally 2-closed if and only if T admits no nontrivial factorizations and, for each permutation representation $\varphi : G \rightarrow \text{Sym}(\Omega)$ with $|\Omega| > 1$ and $\varphi(G)$ transitive, we have $\varphi(G)^{(2), \Omega} = \varphi(G)$.*

2.3. Simple groups, factorizations, and rank

In Proposition 2.12, we consider the totally 2-closed simple groups G in the light of Corollary 2.11, applying the classification of their maximal factorizations [27], and considering their low rank permutation representations. For example, if a simple group G is 2-transitive on Ω then $G^{(2), \Omega} = \text{Sym}(\Omega)$, by [39, Theorem 5.11], and hence G is not totally 2-closed.

We denote the *socle* of G by $\text{soc}(G)$, that is, the product of the minimal normal subgroups of G . We denote the set of right cosets of a subgroup H in G by $[G :$

$H] = \{Hg \mid g \in G\}$, and note that G induces by right multiplication a transitive action on $[G : H]$ with kernel $\text{Core}_G(K)$, and such that H is the stabilizer of the ‘point’ H .

Proposition 2.12. *Let G be a finite nonabelian simple group that is totally 2-closed. Then one of the following is true.*

- (a) G lies in one of the following infinite families: ${}^3D_4(q)$, $E_6(q)$, ${}^2E_6(q)$, $E_7(q)$, $E_8(q)$, $F_4(q)$ (with q odd), ${}^2F_4(q)'$ (with $q = 2^{2c+1}$); or
- (b) G is one of $J_1, J_3, J_4, \text{Ly}, \text{ON}, \text{Co}_2, \text{Fi}_{23}, \text{HN}, \text{Th}, \mathbb{B}, \mathbb{M}$.

Proof. By Corollary 2.11, G admits no nontrivial factorization. We treat the simple groups by families. First, if $G = A_n$, then G is 2-transitive of degree n and so, as we observed above, G is not totally 2-closed, a contradiction. Thus $G \not\cong A_n$.

If G is a totally 2-closed sporadic simple group, then this implies, by [27, Theorem C], that either G is as in part (b), or $G \in \{\text{Co}_3, \text{McL}, \text{Fi}'_{24}\}$. Now $G \neq \text{Co}_3$ since Co_3 has a 2-transitive representation of degree 276 (see, for example [13, Sec. 7.7]). If G is McL or Fi'_{24} , then $A = \text{Aut}(G) = G.2$, and the coset action of A on the set $\Omega = [A : H]$ of right cosets of $H = \text{PSU}(4, 3) : 2_3$ or $\text{Fi}_{23} \times 2$, respectively, is such that both A and G have rank 3 (see [12, pp. 100, 207]). It follows that $A \leq G^{(2), \Omega}$ by Theorem 2.4, so that $G \neq G^{(2), \Omega}$, which is a contradiction.

Thus, we may assume that G is a totally 2-closed simple group of Lie type. Suppose first that G is a classical simple group. Then $G \neq \text{PSL}_n(q)$ or $\text{PSU}_3(q)$ since these groups have a 2-transitive representation, (see, for example [13, Sec. 7.7]). Among the other simple classical groups, the only ones which have no nontrivial factorizations are, by [27, Theorem A],

- $G = \text{PSU}_n(q)$ with n odd, $n \geq 5$, and $(n, q) \neq (9, 2)$; and
- $G = \text{P}\Omega_{2m}^-(q)$ with m even, $m \geq 4$, $q \neq 2, 4$.

For G one of these groups, it follows from [22, Theorem 1.1 and Table IV on p. 45], that G has a rank 3 action on the set Ω of totally singular (or isotropic) 1-spaces of the natural module, of degree

$$\frac{(q^n + 1)(q^{n-1} - 1)}{q^2 - 1} \quad \text{or} \quad \frac{(q^m + 1)(q^{m-1} - 1)}{q - 1}, \quad \text{respectively.}$$

However, the group $A = \text{P}\Gamma\text{U}_n(q)$ or $\text{P}\Gamma\text{O}_{2m}^-(q)$, respectively, contains G as a proper subgroup (see [27, Table 2.1]), and also acts on Ω with rank 3. Hence $A \leq G^{(2), \Omega}$ by Theorem 2.4, and so $G \neq G^{(2), \Omega}$, which is a contradiction.

Finally, suppose that G is an exceptional simple group of Lie type. Then $G \neq \text{Sz}(q)$ or $\text{Ree}(q)$ since each of these groups has a 2-transitive representation, (see, for example [13, Sec. 7.7]). Among the other exceptional simple groups of Lie type, the only ones which do not appear in part (a), and which have no nontrivial factorizations are, by [27, Theorem B], the groups $G = G_2(q)$ (with $q > 2$, $q \neq 4, 3^c$).

However, it follows from [26, Theorem 1], that such a group G has a primitive permutation representation on a set Ω of size $q^3(q^3 - 1)/2$ such that $G^{(2),\Omega}$ has socle $\Omega_7(q)$, so $G \neq G^{(2),\Omega}$, which is a contradiction. $\square$

To summarize this section, we have proved that to determine the finite totally 2-closed groups G with $F(G) = 1$, we need only consider the groups G which are a direct product of a subset of the nonabelian finite simple groups listed in Proposition 2.12. Furthermore, the direct factors must satisfy the conditions listed in Theorem 2.5. Finally, by Corollary 2.11, such a group is totally 2-closed if and only if it is 2-closed in each of its (not necessarily faithful) transitive permutation representations. We investigate the transitive representations of these, and other more general, groups in Sec. 3.

3. Tools for Analyzing Imprimitive Actions

In this section, we introduce some group theoretic and combinatorial tools for studying 2-closures of transitive group actions.

Let Ω be a finite set, and let G be a subgroup of $\text{Sym}(\Omega)$. A partition Σ of Ω is G -invariant if each element of G permutes the parts of Σ among themselves. A partition is said to be *trivial* either if it consists of a single part, namely, Ω , or if each of its parts consists of a single point. All other partitions are nontrivial. Each transitive permutation group G leaves invariant these two trivial partitions, and G is called *primitive* if the only G -invariant partitions are the two trivial partitions; otherwise G is said to be *imprimitive*.

Higman's landmark result [17, (1.12)] in this area (which we will use) is that a finite transitive permutation group G is primitive if and only if all the nondiagonal G -orbital digraphs are connected.

Lemma 3.1. *Let $G \leq \text{Sym}(\Omega)$ be transitive on Ω , and let $X = G^{(2),\Omega}$. Suppose that Σ is a nontrivial G -invariant partition of Ω , and let $\overline{G} = G^\Sigma$ denote the subgroup of $\text{Sym}(\Sigma)$ induced by G .*

- (a) *Then Σ is also X -invariant, and the induced group $\overline{X} = X^\Sigma$ satisfies $\overline{X} \leq \overline{G}^{(2),\Sigma}$.*
- (b) *Also, for $B \in \Sigma$, $X_B^B \leq (G_B^B)^{(2),B}$.*

Proof. (a) For $\alpha \in \Omega$, let B_α denote the (unique) part of Σ that contains α , and let $\varphi : \Omega \times \Omega \rightarrow \Sigma \times \Sigma$ be the map $(\alpha, \beta) \rightarrow (B_\alpha, B_\beta)$. By [39, Theorem 4.11], G and X have the same invariant partitions in Ω , so Σ is X -invariant. This implies that φ is X -invariant in the sense that, for all $\alpha, \beta \in \Omega$ and $x \in X$, $\varphi(\alpha^x, \beta^x) = (B_\alpha^x, B_\beta^x)$. We show that $\overline{X}$ leaves invariant each $\overline{G}$ -invariant subset of $\Sigma \times \Sigma$: let $\Delta \subseteq \Sigma \times \Sigma$ be $\overline{G}$ -invariant, and let $\Delta' \subseteq \Omega \times \Omega$ be the set of all ordered pairs (α, β) such that $(B_\alpha, B_\beta) \in \Delta$. Then Δ' is the full pre-image of Δ under φ , that is, $\varphi(\Delta') = \Delta$, and it is straightforward to check that Δ' is a G -invariant subset of $\Omega \times \Omega$. Hence Δ'

is also preserved by $X = G^{(2),\Omega}$, and it follows that $\Delta = \varphi(\Delta')$ is preserved by $\overline{X}$. Since this holds for each $\overline{G}$ -invariant Δ , it follows that $\overline{X} \leq \overline{G}^{(2),\Sigma}$.

(b) Let Δ be a subset of $B \times B$ preserved by G_B^B , and let Δ' be the set of all pairs $(\alpha, \beta) \in \Omega \times \Omega$ such that $(\alpha^g, \beta^g) \in \Delta$ for some $g \in G$. Then Δ' is a G -invariant subset of $\Omega \times \Omega$. Moreover, from the definition of Δ' the set $\Delta \subseteq \Delta' \cap (B \times B)$, and to see that equality holds let $(\alpha, \beta) \in \Delta' \cap (B \times B)$, so $\alpha, \beta \in B$ and $(\alpha^g, \beta^g) \in \Delta$ for some $g \in G$. Then, since $\Delta \subseteq B \times B$, also $\alpha^g \in B$ so $\alpha^g \in B \cap B^g$ and hence $B^g = B$, since Σ is a G -invariant partition. Thus $g \in G_B$, so also $g^{-1} \in G_B$. Since Δ is G_B -invariant it follows that $(\alpha, \beta) = (\alpha^g, \beta^g)^{g^{-1}} \in \Delta^{g^{-1}} = \Delta$. So, we have proved that $\Delta' \cap (B \times B) = \Delta$.

Since $X = G^{(2),\Omega}$, Δ' is also X -invariant, and hence $\Delta' \cap (B \times B) = \Delta$ is X_B -invariant. Thus $X_B^B \leq (G_B^B)^{(2),B}$, as required. $\square$

Remark 3.2. We note that the inequalities in Lemma 3.1 may be strict. For example, let $G = \Gamma L_1(16)$ of degree 15 on $\Omega = \mathbb{F}_{16}^\times$. By [33, Corollary 4.1], the 2-closure $X = G^{(2),\Omega}$ is equal to G . Moreover, G preserves two nontrivial partitions of Ω , one of them Σ with 5 blocks of size 3, and the other Σ' with 3 blocks of size 5. (The partitions Σ, Σ' are the orbit sets of cyclic normal subgroups of G of orders 3, 5, respectively.) Now, $G^\Sigma \cong F_{20}$ is a sharply 2-transitive Frobenius group of order 20. Hence $\overline{G}^{(2),\Sigma} = \text{Sym}(5)$, while $\overline{X} = \overline{G} = F_{20}$. Also, for $B \in \Sigma'$, $G_B^B \cong F_{20}$ is sharply 2-transitive, and so $(G_B^B)^{(2),B} = \text{Sym}(5)$, while $X_B^B = G_B^B = F_{20}$.

For a more general example of inequality in part (a), suppose that G is a finite transitive permutation group on a set Σ such that G is not 2-closed, that is, $G^{(2),\Sigma} \neq G$. Let $B \in \Sigma$ and let $H = G_B$, so that we may identify Σ with $[G : H] = \{Hg \mid g \in G\}$ and G acts by right multiplication. Consider the regular action of G on $\Omega = G$ by right multiplication. Then $X = G^{(2),\Omega} = G$ by [39, Theorem 5.12], and $\Sigma = [G : H]$ is a G -invariant partition of Ω , so $\overline{G} = G^\Sigma \cong G$. Thus $\overline{X} = \overline{G} < \overline{G}^{(2),\Sigma}$.

By Theorem 2.5 and Proposition 2.10, the classification of finite totally 2-closed groups with trivial Fitting subgroup is reduced to the study of nontrivial transitive permutation representations of groups G satisfying Hypothesis 2.7 such that the simple direct factors T_i of G admit no nontrivial factorizations. Although it would be desirable to reduce further to primitive actions, this seems difficult (and perhaps not possible). Loosely speaking, our goal in this section is to prove that if such groups G are 2-closed in all of their primitive permutation representations, then it is ‘likely’ that they are totally 2-closed.

More precisely, we will explicitly construct the 2-closure of a finite group G acting imprimitively on a set Ω , with the property that its induced action on a nontrivial G -invariant partition Σ of Ω is faithful and 2-closed. To state our results, we require some notation.

Notation 3.3. Assume that G is a finite group, and that:

- (1) G acts faithfully, transitively, and imprimitively on a finite set Ω .

- (2) $\Sigma = \{\Delta_1, \Delta_2, \dots, \Delta_s\}$ is a nontrivial G -invariant partition of Ω , and $L = G^\Sigma$, the (transitive) subgroup of $\text{Sym}(\Sigma)$ induced by G , where $1 < s < |\Omega|$.
- (3) $\Delta := \Delta_1 \in \Sigma$ and $\omega \in \Delta$.
- (4) $M = G_\Delta$ (setwise stabilizer, so that $|G : M| = s$), $H = G_\omega$ (so $H < M$ with $|M : H| = |\Delta|$), and $R = M^\Delta$, the (transitive) subgroup of $\text{Sym}(\Delta)$ induced by M (so $R_\omega = H^\Delta$).
- (5) By the Imprimitve Wreath Embedding Theorem [32, Theorem 5.5], we may assume that $G \leq W = R \wr L$ acting on $\Omega = \Delta \times \{1, 2, \dots, s\}$; here we identify Σ with $\{1, 2, \dots, s\}$, and Δ_i with $\Delta \times \{i\}$ for each $i \leq s$.
- (6) $W = B_R \rtimes L$ is a semidirect product, where $B_R = \{(x_1, \dots, x_s) : x_i \in R\} \cong R^s$ is called the *base group* of W , and the subgroup L is called the *top group* of W . Typical elements $(x_1, x_2, \dots, x_s) \in B_R$ and $\sigma \in L$ act on a point $(\delta, i) \in \Omega$ by

$$(\delta, i)^{(x_1, x_2, \dots, x_s)} = (\delta^{x_i}, i) \quad \text{and} \quad (\delta, i)^\sigma = (\delta, i^\sigma).$$

- (7) We write $B_R = R_1 \times \dots \times R_s$, where

$$R_i = \{(x_1, \dots, x_s) : x_j = 1 \text{ if } j \neq i\} \cong R,$$

so R_i induces the group R on Δ_i and fixes Δ_j pointwise for all $j \neq i$. We extend this notation so that, for any subgroup $A \leq \text{Sym}(\Delta)$, we write $B_A = A_1 \times \dots \times A_s \cong A^s$ for the naturally embedded subgroup of $\text{Sym}(\Delta) \wr \text{Sym}(s)$. We refer to such subgroups B_A as ‘base groups’.

- (8) For $A \leq \text{Sym}(\Delta)$, and for each $i \leq s$, the (setwise) stabilizer of Δ_i in $A \wr L = B_A \rtimes L$ is the subgroup $W_i := B_A \rtimes L_i$ (where L_i is the stabilizer of i in L) and the map

$$\rho_i : W_i \rightarrow A \quad \text{given by } \rho_i : (x_1, x_2, \dots, x_s)\sigma \rightarrow x_i$$

is a group epimorphism; similarly, for distinct i, j the map

$$\rho_{i,j} : W_i \cap W_j \rightarrow A \times A \quad \text{given by } \rho_{i,j} : (x_1, x_2, \dots, x_s)\sigma \rightarrow (x_i, x_j)$$

is a group epimorphism, and its image $\rho_{i,j}(W_i \cap W_j) = \rho_{i,j}(A_i \times A_j)$ is the group induced by $W_i \cap W_j$ (and by $A_i \times A_j$) on $\Delta_i \cup \Delta_j$.

The group $A_i \times A_j$ fixes Δ_ℓ pointwise, for all $\ell \notin \{i, j\}$, and so acts faithfully on $\Delta_i \cup \Delta_j$; it also has a faithful *product action* on $\Delta_i \times \Delta_j$ given by $(\delta_i, \delta_j)^{(x_i, x_j)} = (\delta_i^{x_i}, \delta_j^{x_j})$. In our proofs, we will meet the 1-closure of a subgroup $S \leq A_i \times A_j$ in this product action: the 1-closure of S is denoted $S^{(1), \Delta_i \times \Delta_j}$, and is the largest subgroup of $\text{Sym}(\Delta_i \times \Delta_j)$ with the same orbits as S . To work with the group G we consider restrictions of the maps in Notation 3.3.8 to the corresponding subgroups of G . We also impose an additional condition on M .

- (9) For $i = 1, \dots, s$, let $M_i = W_i \cap G$, the setwise stabilizer in G of Δ_i . Note that the subgroups M_i , $1 \leq i \leq s$ form the complete G -conjugacy class of $M = M_1$. It follows from parts 4 and 5 that $\rho_i(M_i) = M_i^{\Delta_i} = R_i \cong R$, for each i , while $\rho_{i,j}(M_i \cap M_j) \leq R_i \times R_j$ is the subgroup induced by G on $\Delta_i \cup \Delta_j$, for distinct

i, j . We will meet the 1-closure $(\rho_{i,j}(M_i \cap M_j))^{(1), \Delta_i \times \Delta_j}$ of the product action of this subgroup (as described immediately above).

(10) M is assumed to be core-free in G , (equivalently $G \cong G^\Sigma = L$).

Condition 10 is vacuously true if G is simple. But it will also suffice (for us) to study 2-closures of groups acting faithfully on a nontrivial invariant partition, as we will see in Sec. 7. Note also that if Δ is chosen to be a maximal block of imprimitivity, then M is a maximal subgroup of G (we will often be interested in this case in our analysis below). Sometimes we will identify Ω with $[G : H]$; Σ with $[G : M]$; and $\Delta = \Delta_1$ with $[M : H]$.

In the following series of results, we will study the cases where G^Σ is 2-closed, but G^Ω is not 2-closed. We will see that this seems like quite an unusual phenomenon. See in particular our main results in this section: Theorems 3.8 and 3.10 (for notational convenience, we will postpone the statements of these theorems until slightly further on in the section).

We begin with a straightforward lemma from representation theory.

Lemma 3.4. *Let G, Σ, M, s be as in Notation 3.3, let p be a prime, and let*

$$V = \{(y_1, \dots, y_s) : y_i \in \mathbb{F}_p\},$$

be the permutation module for G on Σ over the field $\mathbb{F}_p$. Suppose that M is not contained in a proper normal subgroup of G . Then the only 1-dimensional $\mathbb{F}_p[G]$ -submodule of V is the trivial submodule $\{(y, \dots, y) : y \in \mathbb{F}_p\}$.

Proof. Let U be a 1-dimensional $\mathbb{F}_p[G]$ -submodule of V . Then there is an injective homomorphism from U to V , that is, the $\mathbb{F}_p$ -space $\text{Hom}_G(U, V)$ of $\mathbb{F}_p[G]$ -homomorphisms from U to V is nonzero. Since V is the induced module $(1_M) \uparrow^G$, where 1_M denotes the trivial $\mathbb{F}_p[M]$ -module, it follows from Frobenius reciprocity [4, Proposition 3.3.1] that $\text{Hom}_M(U \downarrow_M, 1_M)$ is also nonzero. Thus U has a nonzero quotient $\mathbb{F}_p[M]$ -module acted upon trivially by M , and since $\dim(U) = 1$, it follows that M acts trivially on U . Thus, M is contained in the kernel of the action of G on U . Since M is not contained in a proper normal subgroup of G , we deduce that G acts trivially on U . It is then an elementary exercise to prove that $\{(y, \dots, y) : y \in \mathbb{F}_p\}$ is the only nonzero trivial $\mathbb{F}_p[G]$ -submodule of V . This completes the proof. $\square$

The next lemma is a crucial tool in our treatment of imprimitive groups G as in Notation 3.3. It makes use of Lemma 3.1. Recall in particular the definition of base groups in Notation 3.3.7.

Lemma 3.5. *Let $G, \Omega, \Sigma, \Delta, L, R, M, M_i, s, \rho_{i,j}$ be as in Notation 3.3, let $X := G^{(2), \Omega}$, and $Y := X_\Delta^\Delta$. Suppose that $G \neq X$ but $L = L^{(2), \Sigma}$. Then the following hold:*

(1) $X \leq Y \wr L = B_Y \rtimes L$, and $R \leq Y \leq R^{(2), \Delta}$;

- (2) $X = N \rtimes G$, where $N = B_Y \cap X > 1$, and N consists precisely of all the elements $(y_1, \dots, y_s) \in B_Y$ such that

$$(y_i, y_j) \in (\rho_{i,j}(M_i \cap M_j))^{(1), \Delta_i \times \Delta_j} \quad \text{for all } 1 \leq i < j \leq s. \quad (3.1)$$

- (3) If $p := |R|$ is prime then $Y = R^{(2), \Delta} = R = C_p$ and, for $2 \leq j \leq s$,

$$(\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j} = (Y \times Y) \cap (\rho_{1,j}(M \cap M_j))^{(1), \Delta_1 \times \Delta_j} \quad (3.2)$$

is equal to either $Y \times Y$ or a diagonal subgroup of $Y \times Y$. In particular, there exists $(y_1, \dots, y_s) \in N$ with both y_1 and y_j nontrivial.

Proof. (a) By Lemma 3.1(a), $L \leq X^\Sigma \leq L^{(2), \Sigma} = L$, and hence $X^\Sigma = L$. By Lemma 3.1(b), for each $\Delta_i \in \Sigma$, $R = G_{\Delta_i}^{\Delta_i} \leq X_{\Delta_i}^{\Delta_i} \leq (G_{\Delta_i}^{\Delta_i})^{(2), \Delta_i} = R_i^{(2), \Delta_i}$, and the last subgroup is Y : here we identify the set $\Delta_i = \Delta \times \{i\}$ with Δ as in Notation 3.3.5. Thus, it follows from Lemma 3.1 that $X \leq Y \wr L = B_Y \rtimes L$ and $R \leq Y \leq R^{(2), \Delta}$.

(b) Let $N := B_Y \cap X$. Then $N \trianglelefteq X$, and it follows from Notation 3.3.10 that $G \cong G^\Sigma$, equivalently, $N \cap G = 1$. Thus, $|X/N| = |X^\Sigma| = |L| = |G^\Sigma| = |G|$, and therefore $X = N \rtimes G$. Since $X > G$ we have $N > 1$.

Let $y \in B_Y$, so $y = (y_1, \dots, y_s)$, where each $y_i \in Y$. Then $y \in N$ if and only if $y \in X = G^{(2), \Omega}$ and, by Theorem 2.4, this holds if and only if, for all $\alpha, \beta \in \Omega$, there exists $g \in G$ such that $(\alpha, \beta)^y = (\alpha, \beta)^g$. Suppose first that α, β lie in the same part Δ_i of Σ . Then $(\alpha, \beta)^y = (\alpha, \beta)^{y_i}$, and since $y_i \in Y = R^{(2), \Delta}$ it follows that there exists $r \in R$ with $(\alpha, \beta)^{y_i} = (\alpha, \beta)^r$, and as $R = G_{\Delta_i}^{\Delta_i}$ there exists $g \in G_{\Delta_i}$ such that $g^{\Delta_i} = r$. Hence $(\alpha, \beta)^y = (\alpha, \beta)^g$. Thus, we may assume that $\alpha \in \Delta_i, \beta \in \Delta_j$ with $1 \leq i < j \leq s$. Then $(\alpha, \beta)^y = (\alpha^{y_i}, \beta^{y_j}) \in \Delta_i \times \Delta_j$, and a suitable element g would need to lie in $M_i \cap M_j$ (the setwise stabilizer in G of Δ_i and Δ_j), and the image $(\alpha, \beta)^g$ would therefore be equal to $(\alpha, \beta)^a$ for some $a \in \rho_{i,j}(M_i \cap M_j)$. By the definition of the 1-closure of $\rho_{i,j}(M_i \cap M_j)$ in its action on $\Delta_i \times \Delta_j$ (see Notation 3.3.9), such an $a = a(\alpha, \beta)$ can be found, for all $(\alpha, \beta) \in \Delta_i \times \Delta_j$, if and only if $(y_i, y_j) \in (\rho_{i,j}(M_i \cap M_j))^{(1), \Delta_i \times \Delta_j}$.

(c) Suppose now that $|R| = p$ is a prime, so $|\Delta| = p$ and $R = C_p$ acting regularly. Then, by [39, Theorem 5.12], $R^{(2), \Delta} = R$, and hence by part (a), also $Y = C_p$. Fix an integer j with $2 \leq j \leq s$, and recall that $M = M_1$. By part (b), $N \neq 1$, and hence, since N is normal in X , there exists $y = (y_1, \dots, y_s) \in N$ such that $y^{\Delta_1 \cup \Delta_j} = (y_1, y_j) \neq 1$ with both y_1 and y_j nontrivial. In particular, $1 \neq (\rho_{1,j}(y))^{\Delta_1 \times \Delta_j} \in (\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j}$. On the other hand, $\rho_{1,j}(M \cap M_j) \leq R_1 \times R_j$ by Notation 3.3.9, and in this case $R_1 \times R_j = C_p \times C_p$. Thus $(\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j}$ is either $C_p \times C_p$ or a diagonal subgroup C_p . In the first case $\rho_{1,j}(M \cap M_j)$ is transitive on $\Delta_1 \times \Delta_j$, and hence its 1-closure is $(\rho_{1,j}(M \cap M_j))^{(1), \Delta_1 \times \Delta_j} = \text{Sym}(p^2)$, and (3.2) holds with both sides equal to $C_p \times C_p$. Suppose then that $(\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j}$ is a diagonal subgroup of $C_p \times C_p$, hence of order p and intransitive on $\Delta_1 \times \Delta_j$. It follows that $(Y \times Y) \cap (\rho_{1,j}(M \cap M_j))^{(1), \Delta_1 \times \Delta_j}$ is also intransitive, and hence is a

proper subgroup of $Y \times Y = C_p \times C_p$. Since this group contains $(\rho_{1,j}(y))^{\Delta_1 \times \Delta_j} \neq 1$, it follows that (3.2) holds with both sides equal to C_p . $\square$

First, we derive a useful consequence of Lemma 3.5.

Lemma 3.6. *Let $G, \Omega, \Sigma, M, R, L, \Delta$, etc. be as in Notation 3.3, and let $X := G^{(2), \Omega}$ and $Y = X_{\Delta}^{\Delta}$. Suppose also that $G \neq X$, that $L = L^{(2), \Sigma}$, that $p := |R|$ is prime, and that M is not contained in a proper normal subgroup of G . Let J consist of representative elements $j \in \{2, 3, \dots, s\}$, where $s = |\Sigma|$, such that*

$$(\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j} \cong C_p,$$

and such that, for distinct $j, j' \in J$, the parts $\Delta_j, \Delta_{j'} \in \Sigma$ lie in distinct M -orbits in $\Sigma \setminus \{\Delta\}$. If $\sum_{j \in J} |M : M \cap M_j| \geq s/2$, then the subgroup N from Lemma 3.5 is the diagonal subgroup $\{(x, \dots, x) : x \in R\}$ of $B_R = R^s \cong C_p^s$.

Proof. Write the cyclic group $R \cong C_p$ additively, and let N be the normal subgroup of X from Lemma 3.5. By Lemma 3.5 parts (b) and (c), $X = N \rtimes G$, N is an elementary abelian p -subgroup of $B_Y = B_R = R^s = C_p^s$, and by Notation 3.3.10, $G \cong G^{\Sigma} = L$. Thus B_R may be viewed as the permutation module V for the G -action on Σ over the field $\mathbb{F}_p$, and N as a nonzero G -submodule. Note that V is the induced module $1 \uparrow_M^G$, and by assumption M is not contained in a proper normal subgroup of G . Hence, by Lemma 3.4, it suffices to prove that $|N| = p$.

Assume to the contrary that $|N| \geq p^2$. Then there exists a nontrivial element $y = (y_1, \dots, y_s) \in N$ with the property that $y_e = 0$ for some $e \leq s$. Since G^{Σ} is transitive, we may assume, without loss of generality, that $y_1 = 1$. Let $\mathcal{O}$ be the union of those M -orbits in Σ containing some Δ_j with $j \in J$. Since $M \cap M_j$ is the setwise stabilizer in M of Δ_j , the set $\mathcal{O}$ has length $\sum_{j \in J} |M : M \cap M_j| \geq s/2$. By Lemma 3.5(c) and the definition of J , $\rho_{1,i}(N)$ is a full diagonal subgroup of $R \times R \cong C_p^2$, for all $i \in \mathcal{O}$. Thus, there exist automorphisms $\theta_i \in \text{Aut}(R)$ for $i \in \mathcal{O}$ such that, if $z = (z_1, \dots, z_s) \in N$, then $z_i = z_1^{\theta_i}$ for all $i \in \mathcal{O}$.

Since $y_1 \neq 0$, the number of nonzero entries in y is therefore at least $1 + |\mathcal{O}| > s/2$, that is, strictly more than half the entries of y are nonzero. On the other hand, since y also contains zero entries, and G^{Σ} is transitive, it follows that, for some $g \in G$, the conjugate y^g has first entry $(y^g)_1 = 0$. Then, since $y^g \in N$, it follows from the previous paragraph that $(y^g)_i = 0$ for all $i \in \mathcal{O}$, that is, strictly more than half the entries of y^g are zero. However, since y^g is conjugate to y , this implies that strictly more than half the entries of y are zero. This is a contradiction. Hence, $|N| = p$ and the result follows. $\square$

We next note a particular example for which the condition of Lemma 3.6 on J holds. This result will be crucial in our proof that the Thompson group Th is totally 2-closed (see Propositions 4.16 and 4.17).

Proposition 3.7. *Suppose that $G = \text{Th}$, $M = {}^3D_4(2).3$ and $H = [M, M]$, and note that all the conditions of Notation 3.3 hold with $\Omega = [G : H]$ and $\Sigma = [G : M]$. Let*

J be a subset consisting of representative elements $j \in \{2, 3, \dots, s\}$, where $s = |\Sigma|$, such that

$$(\rho_{1,j}(M \cap M_j))^{\Delta_1 \times \Delta_j} \cong C_3,$$

and such that, for distinct $j, j' \in J$, the parts $\Delta_j, \Delta_{j'} \in \Sigma$ lie in distinct M -orbits in $\Sigma \setminus \{\Delta\}$. Then $\sum_{j \in J} |M : M \cap M_j| > s/2$.

Proof. The group G^Σ has rank 11. We may assume that, for $i = 2, \dots, 11$, the subgroups $M \cap M_i$ are the stabilizers in M of parts $\Delta_i \in \Sigma$ from pairwise distinct M -orbits in $\Sigma \setminus \{\Delta\}$. These point stabilizers $(M \cap M_i)^\Sigma$ are given in [30], and are (in some order) $M \cap M_2 := 2^2.[2^9].(\text{Sym}(3) \times 3)$, $M \cap M_3 := 2^{1+8}.9.3$, $M \cap M_4 := 7^2.(3 \times \text{SL}_2(3))$, $M \cap M_5 := [3^5].2$, $M \cap M_6 := 2 \times \text{Alt}(4) \times \text{Alt}(4)$, $M \cap M_7 := (7.6) \times 3$, $M \cap M_8 := 13.6$, $M \cap M_9 := 3 \times \text{SL}_2(3)$, $M \cap M_{10} := 3^2.6$, and $M \cap M_{11} := \text{Sym}(3)$. The groups $M \cap M_i$, for $i \in \{2, 8, 11\}$ clearly have no elementary abelian quotient of order 9. Furthermore, it is proved in [30] that the 3^2 in $M \cap M_{10} := 3^2.6$ is self-centralizing, and the ‘6 on top’ acts as the unique subgroup of $GL_2(3)$ of order 6 (up to conjugacy). In particular $M \cap M_{10}$ has no elementary abelian quotient of order 9. Since $\sum_{i \in \{2, 8, 10, 11\}} |M : M \cap M_i| > |\Sigma|/2$, the result is proved. $\square$

The next result is our first main theorem in this section. In general, and as mentioned in Sec. 1.1, it is a powerful tool for studying 2-closures of those imprimitive permutation groups which induce a faithful action on a set of maximal blocks. For our study of totally 2-closed groups, it will allow to reduce many of our considerations to primitive actions. We summarize some concepts used in the statement and proof: For a finite group E and a positive integer f , a subgroup $D \leq E^f$ is called a *diagonal subgroup* if D is of the form $D = \{(e, e^{\theta_2}, \dots, e^{\theta_f}) : e \in F\}$ for some subgroup F of E and automorphisms θ_i of F ; D is called a *full diagonal subgroup* if $F = E$. If $f = 2$ and D is a subdirect subgroup of $E_1 \times E_2$, for finite groups E_1, E_2 , then by [32, Theorem 4.8] (sometimes called ‘Goursat’s Lemma’), $K_i := D \cap \text{Ker}(\pi_{3-i}) \trianglelefteq E_i$ for $i = 1, 2$, and there is a well-defined homomorphism $\varphi_D : E_1 \rightarrow E_2/K_2$ with kernel K_1 such that $D = \{(e_1, e_2) : e_1 \in E_1, e_2 \in \varphi_D(e_1)\}$. In particular D contains $K_1 \times K_2$ as a normal subgroup and $D/(K_1 \times K_2) \cong E_1/K_1 \cong E_2/K_2$; and moreover D is a full diagonal subgroup if and only if $K_1 \times K_2 = 1$.

Finally, we will write $\Omega^{(2)}$ for the set of ordered pairs of distinct points of a set Ω , and, for a finite group Y , we will write $\mathcal{S}(Y)$ for the intersection of all nontrivial subnormal subgroups of Y . Note that $\mathcal{S}(Y)$ is nontrivial only if Y has a unique minimal normal subgroup, which is simple.

Theorem 3.8. *Let $G, \Omega, \Sigma, M, R, L, \Delta, s, \rho_{i,j}$, etc. be as in Notation 3.3, and let $X := G^{(2), \Omega}$ and $Y = X_\Delta^\Delta$. Suppose that $G \neq X$ but $L = L^{(2), \Sigma}$.*

- (a) *Then $X = N \rtimes G$ with $N = X \cap B_Y \neq 1$ as in Lemma 3.5, and N is a subdirect subgroup of a base group $B_A = A^s$ for some $A \trianglelefteq Y$ (see Notation 3.3.7).*

- (b) If M is maximal in G , then either
- (i) N is a full diagonal subgroup of B_A ; or
 - (ii) for all distinct $i, j \leq s$, $\rho_{i,j}(N)$ is not a full diagonal subgroup of $A \times A$; moreover, either N contains the base group $B_{S(Y)}$, or Y has a unique minimal normal subgroup which is isomorphic to C_p , for some prime p .
- (c) Furthermore, if M is maximal but not normal in G , and if in addition $p = |R|$ is prime and (b)(i) holds, then $Y = R$ and N is the diagonal subgroup $\{(x, \dots, x) : x \in R\}$ of $B_R = R^s \cong C_p^s$.

Proof. (a) Since N is a subgroup of the base group B_Y of $Y \wr L$ and N is normalized by X , it follows that N is a subdirect subgroup of a base subgroup $B_A = A^s$ for some subgroup $A \trianglelefteq Y$, as in part (a). Note that each element $y \in Y$ acting by conjugation induces an automorphism of A , and we denote this automorphism by $\iota(y)$. For the rest of the proof assume that M is maximal in G , or equivalently, $L = G^\Sigma$ is primitive. This means, by Higman's theorem [17, (1.12)] discussed above, that every nondiagonal L -orbital digraph is connected.

(b) Since N is a subdirect subgroup of $B_A = A^s$, for any distinct i, i' , the image $\rho_{i,i'}(N)$ is a subdirect subgroup of $A \times A$. Suppose first that there exist distinct i, i' such that $\rho_{i,i'}(N)$ is a full diagonal subgroup of $A \times A$, that is to say, $\rho_{i,i'}(N) = \{(x, x^\theta) : x \in A\}$ for some $\theta \in \text{Aut}(A)$. If $g = y\sigma \in G < B_Y \rtimes L$ with $y \in B_Y, \sigma \in L$ and $(j, j') = (i, i')^\sigma$, and if $z \in N$ such that $\rho_{i,i'}(z) = (x, x^\theta)$, then $z^g \in N$ (since G normalizes N) and

$$\rho_{j,j'}(z^g) = (x^{y_i}, (x^\theta)^{y_{i'}}) = (u, u^{\theta'}) \quad \text{where } u = x^{y_i} \quad \text{and} \quad \theta' = \iota(y_i^{-1})\theta\iota(y_{i'}).$$

Since $A \trianglelefteq Y$, we have $u \in A$ and $\theta' \in \text{Aut}(A)$, and since this holds for each $z \in N$, and hence for each $(x, x^\theta) \in \rho_{i,i'}(N)$, it follows that $\rho_{j,j'}(N)$ is a full diagonal subgroup of $A \times A$.

Let Γ be the (nondiagonal) L -orbital digraph containing (i, i') as an edge. As mentioned above Γ is a connected digraph, and so, since $L = G^\Sigma$ is transitive on Σ , there exists an edge of the form $(1, k)$ for some $k > 1$. We claim that $\rho_{1,j}(N)$ is a full diagonal subgroup of $A \times A$, for all $j > 1$. We prove this claim by induction on the distance $d_\Gamma(1, j)$, the length of the shortest path from 1 to j in Γ . By the argument in the previous paragraph the claim holds for all points j with $d_\Gamma(1, j) = 1$. Suppose that $d > 1$, that the claim holds for all j with $d_\Gamma(1, j) \leq d - 1$, and that $d_\Gamma(1, j) = d$. There is a path $i_0, i_1, \dots, i_d$ in Γ with $i_0 = 1, i_d = j$, and set $k = i_{d-1}$. Then by the previous paragraph $\rho_{k,j}(N) = \{(x, x^\theta) : x \in A\}$ for some $\theta \in \text{Aut}(A)$, and by induction, $\rho_{1,k}(N) = \{(x, x^{\theta'}) : x \in A\}$ for some $\theta' \in \text{Aut}(A)$, and it follows that $\rho_{1,j}(N) = \{(x, x^{\theta\theta'}) : x \in A\}$. Thus the claim is proved by induction. It now follows that there exist automorphisms $\theta_2, \dots, \theta_s \in \text{Aut}(A)$ such that $N = \{(x, x^{\theta_2}, \dots, x^{\theta_s}) : x \in A\}$ and (b)(i) holds.

Thus, we may assume that, for all distinct i, j , the subdirect subgroup $\rho_{i,j}(N)$ of $A \times A$ is not a full diagonal subgroup. This is the first assertion of part (b)(ii).

As mentioned before the statement of the proposition, $\mathcal{S}(Y)$ is trivial unless Y has a unique minimal normal subgroup, which is simple. Thus, to complete the proof of (b)(ii), we may assume that Y is almost simple with simple socle $\mathcal{S}(Y)$. Hence also A is almost simple with socle $\mathcal{S}(Y)$. If $\mathcal{S}(Y) = C_p$, for some prime p , then part (b)(ii) holds, so we may assume further that $\mathcal{S}(Y)$ is a nonabelian simple group. Let $N_1 := X \cap B_{\mathcal{S}(Y)} = N \cap B_{\mathcal{S}(Y)}$. Then N_1 is normalized by X , and as in the proof of part (a), N_1 is a subdirect subgroup of $B_{A_1} = A_1^s$ for some $A_1 \trianglelefteq Y$. By the definition of N_1 , $A_1 \subseteq \mathcal{S}(Y)$, so either $A_1 = 1$ or $A_1 = \mathcal{S}(Y)$. If $A_1 = 1$ then $N_1 = 1$, and so the insoluble group $N \cong N/N_1$ is isomorphic to a subgroup of $B_A/B_{\mathcal{S}(Y)} \leq \text{Out}(\mathcal{S}(Y))^s$, which by the ‘Schreier Conjecture’ is soluble, a contradiction. Thus $A_1 = \mathcal{S}(Y)$. Then for distinct i, j , the image $\rho_{i,j}(N_1) = A_1 \times A_1$ (since $\rho_{i,j}(N)$ is not a diagonal subgroup of $A \times A$). Further, the subdirect subgroup N_1 of B_{A_1} is, by ‘Scott’s Lemma’ [32, Theorem 4.16(iii)], a direct product of pairwise disjoint full strips of B_{A_1} ; and by our observation about the $\rho_{i,j}(N_1)$ it follows that each of the strips has support of size 1, that is, it is one of the simple direct factors of B_{A_1} . Thus, in this case $N_1 = B_{A_1}$, and part (b)(ii) holds.

(c) Finally suppose that M is maximal but not normal in G , that $|R| = p$ is prime, and (a)(i) holds. Then $Y = R$ since $R \leq Y \leq R^{(2),\Delta} = R$. Also M is not contained in a proper normal subgroup of G , and so the stated assertion follows from Lemma 3.4. $\square$

The next result is needed to prove some important consequences of Theorem 3.8.

Lemma 3.9. *For $i = 1, 2$, let Δ_i be a finite set and E_i a transitive subgroup of $\text{Sym}(\Delta_i)$. Let $\Omega = \Delta_1 \times \Delta_2$, and consider the product action of $E_1 \times E_2$ on Ω . Suppose that K , a finite group, acts on Δ_i with $K^{\Delta_i} \leq E_i$, for each i , such that the induced K -action on Ω is faithful. Let Γ_i be the set of K -orbits in Δ_i , and let F_i be the (largest) subgroup of E_i fixing setwise each K -orbit in Γ_i . Then*

$$(E_1 \times E_2) \cap K^{(1),\Omega} \leq F_1 \times F_2,$$

and moreover, if $K^{\Delta_i} = F_i$, for $i = 1, 2$, then $K^{(1),\Omega}$ is a subdirect subgroup of $F_1 \times F_2$.

Proof. Write $\Gamma_1 := \{\Lambda_1, \dots, \Lambda_r\}$ and $\Gamma_2 := \{\Pi_1, \dots, \Pi_s\}$. Then the orbits of $K^{\Delta_1} \times K^{\Delta_2}$ in its action on Ω are the sets $\Lambda_i \times \Pi_j$, for $1 \leq i \leq r$, $1 \leq j \leq s$. Since the 1-closure of a permutation group is the stabilizer of each of its orbits in the associated symmetric group, it follows that $(E_1 \times E_2) \cap (K^{\Delta_1} \times K^{\Delta_2})^{(1),\Omega} = F_1 \times F_2$. Then since $K^\Omega \leq (K^{\Delta_1} \times K^{\Delta_2})^\Omega$, we deduce that $(E_1 \times E_2) \cap K^{(1),\Omega} \leq F_1 \times F_2$. Finally, if $K^{\Delta_i} = F_i$, for $i = 1, 2$, then clearly $(E_1 \times E_2) \cap K^{(1),\Omega}$ is a subdirect subgroup of $F_1 \times F_2$. $\square$

We now derive a series of consequences of Theorem 3.8. They show that the question of whether or not G^Ω is 2-closed often reduces to the same question for

the induced action of G on a nontrivial block system. We use the notation and assumptions in Notation 3.3, and recall in particular that this includes the assumption that $G \cong G^\Sigma = L$. The *base size* of a finite transitive permutation group L is the minimum size of a subset S of points such that $\bigcap_{i \in S} L_i = 1$.

The next result we state is the second main theorem in this section. As mentioned in Sec. 1.1, it is not only useful for our proof of Theorem 1.2, but it also serves as a general tool for studying 2-closures of imprimitive groups.

Theorem 3.10. *Let $G, \Omega, \Sigma, M, R, L, \Delta, s, \rho_{i,j}$, etc. be as in Notation 3.3, let $X := G^{(2),\Omega}$ and $Y = X^\Delta_\Delta$, and suppose that $L = L^{(2),\Sigma}$.*

- (1) *If $|\Delta| = 2$, so $R = M^\Delta = \mathbb{Z}/2\mathbb{Z}$ (written additively), then the following are equivalent:*
 - (i) $X \neq G$ (that is, G^Ω is not 2-closed);
 - (ii) for all $i, j \leq s$, $(M_i \cap M_j)G_{\alpha_i} = M_i$, where $\alpha_i \in \Delta_i$;
 - (iii) the element $z = (1, 1, \dots, 1)$ in the base group $B_R = R^s$ of $R \wr L = B_R \rtimes L$ lies in X .
- (2) *If $X \neq G$, then $X = N \rtimes G$ with N a subdirect subgroup of a base group B_A for some nontrivial $A \trianglelefteq Y$, and all A -orbits in Δ have the same length, say a , where a divides $|L_{ij}|$ for all distinct $i, j \leq s$ (where L_{ij} is the stabilizer in L of Δ_i and Δ_j).*
- (3) *If L has base size 2, then $X = G$.*
- (4) *Suppose that $|\Delta| = p$, where p is prime, so that $R = M^\Delta = \mathbb{Z}/p\mathbb{Z}$. Suppose also that $X \neq G$ (that is, G^Ω is not 2-closed). Then for all $i, j \leq s$, and all $\alpha_i \in \Delta_i$, we have $(M_i \cap M_j)G_{\alpha_i} = M_i$.*

Proof. (1) (i) $\Rightarrow$ (ii): Suppose that $X \neq G$ so that, by Lemma 3.5 (all parts of it), $Y = R$ and $X = N \rtimes G \leq R \wr L$, where N consists of all elements $y = (y_1, \dots, y_s) \in R^s = (\mathbb{Z}/2\mathbb{Z})^s$ satisfying (3.1). Suppose also that the condition in part (a)(ii) above fails. (We will derive a contradiction.) Then there exist $i, j \leq s$ such that $(M_i \cap M_j)G_{\alpha_i} \neq M_i$, where $\alpha_i \in \Delta_i$. Since G_{α_i} is a subgroup of M_i of index 2, this means that $M_i \cap M_j \leq G_{\alpha_i}$. In particular $i \neq j$, and $M_i \cap M_j$ acts trivially on Δ_i . Moreover, since G is transitive on Σ , we may assume that $i = 1$. By Lemma 3.5(c), there exists $y = (y_1, \dots, y_s) \in N$ such that both y_1 and y_j are nontrivial. On the other hand, by Lemma 3.5(b), $(y_1, y_j) \in (R \times R) \cap (\rho_{1,j}(M_1 \cap M_j))^{(1), \Delta_1 \times \Delta_j}$, and hence (y_1, y_j) fixes setwise each orbit of $M_1 \cap M_j$ in its induced action on $\Delta_1 \times \Delta_j$. Since $y_1 \neq 0$, this contradicts the fact that $M_1 \cap M_j$ acts trivially on Δ_1 . Thus, the condition in part (a)(ii) holds.

(ii) $\Rightarrow$ (iii): Suppose that the condition in part (a)(ii) holds, and note that z is central in $R \wr L$. Let $U := \langle z \rangle G$, so that $G \leq U \leq R \wr L$ and $|U : G| \leq 2$. We will prove that G and U have the same orbits in $\Omega \times \Omega$, that is $(\alpha, \beta)^G = (\alpha, \beta)^X$ for all $\alpha, \beta \in \Omega$. If $\alpha = \beta$ then $(\alpha, \beta)^G = (\alpha, \beta)^X$ since both X and G are transitive on Ω , so suppose that $\alpha \neq \beta$. If $\{\alpha, \beta\} = \Delta_i$ for some i , then, since M_i

interchanges α and β , the orbit $(\alpha, \beta)^G$ consists of all ordered pairs of points in the same block Δ_j of Σ . Hence $(\alpha, \beta)^G = (\alpha, \beta)^X$ in this case since X preserves Σ , by Lemma 3.1. Thus we may assume that $\alpha \in \Delta_i$ and $\beta \in \Delta_j$ for distinct i, j . The condition $(M_i \cap M_j)G_{\alpha_i} = M_i$ implies that $(M_i \cap M_j)^{\Delta_i} = M_i^{\Delta_i} = R_i$, and similarly (interchanging i and j), $(M_i \cap M_j)^{\Delta_j} = M_j^{\Delta_j} = R_j$. Hence $\rho_{i,j}(M_i \cap M_j)$ is a subdirect subgroup of $R_i \times R_j \cong C_2 \times C_2$, and in particular contains $(1, 1)$. Thus, there exists $x = (z_1, \dots, z_s)\sigma \in M_i \cap M_j \leq G$ such that $z_i = z_j = 1$. Now

$$(R \wr L)_{\alpha\beta} = \{(z_1, \dots, z_s)\sigma : z_i = z_j = 0, \sigma \in L_{ij}\},$$

and

$$U_{\alpha\beta} = G_{\alpha\beta} \cup z\{(z_1, \dots, z_s)\sigma \in G : z_i = z_j = 1, \sigma \in L_{ij}\}. \quad (3.3)$$

Since $x = (z_1, \dots, z_s)\sigma \in M_i \cap M_j \leq G$ with $z_i = z_j = 1$, it follows from (3.3) that $zx \in U_{\alpha\beta} \setminus G_{\alpha\beta}$, and hence $|U_{\alpha\beta}| \geq 2|G_{\alpha\beta}|$. Since $[U : G] \leq 2$, we have

$$|(\alpha, \beta)^U| = |U : U_{\alpha\beta}| = \frac{|U : G| \cdot |G : G_{\alpha\beta}|}{|U_{\alpha\beta} : G_{\alpha\beta}|} \leq |G : G_{\alpha\beta}| = |(\alpha, \beta)^G| \leq |(\alpha, \beta)^U|,$$

and hence equality holds. Thus, we have shown that $U \subseteq G^{(2), \Omega} = X$, and hence that $z \in X$.

(iii) $\Rightarrow$ (i): Finally suppose that $z \in X$. By Notation 3.3.10, G is faithful on Σ , so $G \cap R^s = 1$. Hence $z \in X \setminus G$, whence $X \neq G$.

(2) Since $X \neq G$, it follows from Theorem 3.8(a) that $X = N \rtimes G$ with $N \neq 1$, and N a subdirect subgroup of a base group B_A , where $1 \neq A \trianglelefteq Y$. The A -orbits in Δ all have the same length, say a , since $Y = X^\Delta_\Delta$ is transitive. Choose distinct $i, j \leq s$. We apply Lemma 3.9 with $K, \Delta_1, \Delta_2, E_1, E_2$ being $\rho_{i,j}(M_i \cap M_j), \Delta_i, \Delta_j, Y, Y$, respectively. Note that the hypotheses of Lemma 3.9 hold since $\rho_{i,j}(M_i \cap M_j)$ acts faithfully on $\Delta_i \times \Delta_j$, and the groups it induces on Δ_i and Δ_j are both contained in Y . Thus, Lemma 3.9 implies that $Z := (Y \times Y) \cap (\rho_{i,j}(M_i \cap M_j))^{(1), \Delta_i \times \Delta_j}$ fixes setwise each orbit of $\rho_{i,j}(M_i \cap M_j)$ in $\Delta_i \cup \Delta_j$. Further, it follows from Lemma 3.5(b) that $\rho_{i,j}(N) \leq Z$, and since $\rho_{i,j}(N)$ is a subdirect subgroup of $A \times A$, this implies that each orbit of $\rho_{i,j}(M_i \cap M_j)$ in $\Delta_i \cup \Delta_j$ is a union of A -orbits, and hence has length a multiple of a . In particular a divides $|M_i \cap M_j|$. Finally, since $G \cong G^\Sigma = L$ we have $M_i \cap M_j \cong L_{ij}$. This completes the proof.

(3) Suppose that $L = G^\Sigma$ has base size 2 and let $\{i, j\}$ be a base, that is, $L_{ij} = 1$. This implies, since $G \cong L$, that $M_i \cap M_j \cong L_{ij} = 1$. It now follows from part (2) that $X = G$.

(4) To prove part (4), the argument is identical to the (i) $\Rightarrow$ (ii) argument in (1) above, with 2 replaced by p . $\square$

4. Total 2-Closure of Sporadic Simple Groups

A major objective for the remainder of the paper is to determine precisely which sporadic simple groups are totally 2-closed. In the next few sections we will complete

the proof that Theorem 1.1 holds if T is a sporadic simple group. In Proposition 2.12(b), we reduced the list of possible candidates to just eleven sporadic groups. In this section, we will eliminate the groups ON , Co_2 , Fi_{23} , HN , $\mathbb{B}$, and we will show that five of the remaining six groups, namely, J_1 , J_3 , Ly , Th , $\mathbb{M}$, are totally 2-closed. The remaining group, namely, the fourth Janko group J_4 , requires additional effort and will be treated in Sec. 5.

Our broad strategy is to apply Theorem 3.10 to the 11 sporadic groups listed in Proposition 2.12(b). To do this, we require detailed information, firstly, on the greatest common divisors of the orders of the two-point stabilizers in transitive actions of these groups; and secondly, on the base sizes of these transitive actions. Our main technical result is the following proposition.

Proposition 4.1. *Let T be one of the following simple groups: J_1, J_3, J_4, Ly, Th , or $\mathbb{M}$. Further, let M be a proper subgroup of T , so that T acts faithfully and transitively by right multiplication on the right coset space $[T : M]$. Then*

- (a) *Either $T^{[T:M]}$ has base size at most 2, or (T, M) is as in Table 1, (where the last two subgroups in the J_4 -line may possibly have base size 2).*
- (b) *Further, for each (T, M) in Table 1, let $d(T, M)$ be the greatest common divisor of the orders of the stabilizers in $M^{[T:M]}$ of points of $[T : M] \setminus \{M\}$. Then $d(T, M)$ divides $g(T, M)$, where $g(T, M)$ is as in the third column of Table 1. Moreover, if $T = J_4$ and $M = 2^{10} : L_5(2)$, then $d(T, M) \leq 30$.*

The proof of Proposition 4.1 is spread over six propositions: one for each of the groups $T = J_1, J_3, J_4, Ly, Th$, and $\mathbb{M}$. Most of these propositions will prove that the relevant group T is totally 2-closed, and also that Proposition 4.1 holds for T , namely, Propositions 4.8, 4.9, 4.13, and 4.17 for J_1, J_3, Ly and Th , respectively. For the monster however, we give the proof of Proposition 4.1 in Proposition 4.10, and prove total 2-closure in Proposition 4.11. For the fourth Janko group J_4 , we give the proof of Proposition 4.1 in Proposition 5.6, and prove total 2-closure in Proposition 5.8. Proposition 4.1 will be applied in Sec. 7 where we study the total 2-closure of direct products of sporadic simple groups.

The group $T^{[T:M]}$ has base size 1 if and only if $M = 1$, so we will always assume

Table 1. Information for non-base 2 actions: the greatest common divisor of the stabilizer orders in $M^{[T:M]}$ of points in $[T : M] \setminus \{M\}$ divides $g(T, M)$.

T	M	$g(T, M)$
J_1	$L_2(11)$	1
J_3	$L_2(16), L_2(16).2$	1, 2
J_4	$2^{11} : M_{24}, 2^{1+12}.3.M_{22}.2, 2^{10} : L_5(2),$	24, 2, $2^4.3^2.5.7$,
	$2^{1+12}.3.M_{22}, 2^{1+12}.3.L_3(4), 2^{1+12}.3.L_3(4).2$	2, 2, 2
Ly	$G_2(5), 3.McL, 3.McL.2$	48, 30, 30
Th	$2^5.L_5(2), {}^3D_4(2), {}^3D_4(2).3$	1, 6, 6
$\mathbb{M}$	$2.\mathbb{B}$	2090188800

that $1 < M < T$. For the base sizes of the sporadic groups in their primitive actions, our main reference is [9, Tables 1 and 2]. However, the subgroup M occurring in Notation 3.3.4 is not always maximal in the finite group under investigation, and so we need also to study base sizes of imprimitive actions of sporadic simple groups. The following preliminary result from [9] allows us to do this.

Proposition 4.2 ([9, Corollary 2.4]). *Let G be a transitive permutation group with point stabilizer H , and for a positive integer c let*

$$\widehat{Q}(G, H, c) := \sum_{i=1}^m \frac{|x_i^G \cap H|^c}{|x_i^G|^{c-1}},$$

where $x_1, \dots, x_m$ are representatives for the conjugacy classes of elements of G of prime order. If $\widehat{Q}(G, H, c) < 1$, then G has base size at most c .

Lemma 4.1 and Proposition 4.2 allow us to apply Theorem 3.10 parts (2) and (3) to the relevant sporadic simple groups, to determine whether or not certain of their imprimitive actions are 2-closed. Studying their primitive actions needs a different approach, and for this we use the following lemma. Recall that, for a group G acting on a set Ω , we denote by G^Ω the subgroup of $\text{Sym}(\Omega)$ induced by G . Also, we say that permutation groups G^{Ω_1} and H^{Ω_2} on finite sets Ω_1 and Ω_2 are *permutationally isomorphic* if there exists a bijection $f : \Omega_1 \rightarrow \Omega_2$ and a group isomorphism $\varphi : G^{\Omega_1} \rightarrow H^{\Omega_2}$ such that $f(\alpha^g) = f(\alpha)^{\varphi(g)}$ for all $\alpha \in \Omega_1$, $g \in G^{\Omega_1}$.

Lemma 4.3. *Let T be a finite simple group with proper subgroup M , and let $\Omega = [T : M]$ so that $T \leq \text{Sym}(\Omega)$, acting by right multiplication, and $M = T_\alpha$ for the point $\alpha = M \in \Omega$.*

- (a) *If also $T < G \leq \text{Aut}(T)$, and $M = T \cap K$ for some $K \leq G$ such that $TK = G$ and $G^{[G:K]}$ has base size 2, then the actions of T on Ω and on $[G : K]$ are permutationally isomorphic, and $G^{[G:K]} \not\leq T^{(2), [G:K]}$.*
- (b) *Suppose now that M is maximal in T , that $T^{[T:M]}$ is not 2-transitive, and that $(T, |T : M|)$ is not one of the following:*

- $T = M_{11}, M_{12}, M_{23}, M_{24}$, or $\text{Alt}(9)$, with $|T : M| = 55, 66, 253, 276$, or 120, respectively; or
- $T = G_2(q)$ with $q \geq 3$ a prime power and $|T : M| = \frac{1}{2}q^3(q^3 - 1)$; or
- $T = \Omega_7(q)$ with q a prime power and $|T : M| = \frac{1}{(2, q-1)}q^3(q^4 - 1)$.

Then the following hold:

- (i) *if $T = \text{Aut}(T)$, then $T = T^{(2), \Omega}$;*
- (ii) *if $M \neq T \cap K$ for any $K \leq \text{Aut}(T)$ such that $K \neq M$, then $T = T^{(2), \Omega}$;*
- (iii) *if, for all $K \leq \text{Aut}(T)$ such that $M = T \cap K \neq K$, the group $(TK)^{[TK:K]}$ has base size 2, then $T = T^{(2), \Omega}$.*

Proof. (a) Let M, T, K, G be as in part (a). Since $G^{[G:K]}$ is transitive with base size 2, there exists a base $\{K, Kx\}$ containing the ‘point’ K , for some $x \in G$, and hence

the stabilizer in G of these two points, namely, $K \cap K^x$, is trivial. Further, since $G = TK = KT$, we may assume that $x \in T$. The equality $G = TK$ also implies that T is transitive on $[G : K]$ and the stabilizer in T of the ‘point’ $K \in [G : K]$ is $T \cap K = M$. Thus the actions of T on Ω and on $[G : K]$ are permutationally isomorphic. Since $M = T \cap K$, we have also $M \cap M^x = T \cap K \cap (T \cap K)^x = T \cap K \cap K^x = 1$, that is, the stabilizer in M of the point $Kx \in [G : K]$ is trivial, so the M -orbit containing Kx has size $|M : M \cap M^x| = |M|$, while the K -orbit containing Kx has size $|K : K \cap K^x| = |K|$. Since $|K : M| = |TK : T| = |G : T| > 1$, it follows that K does not leave invariant the M -orbit containing Kx , and hence G does not leave invariant the 2-relation $(K, Kx)^T$ of T . Hence $G^{[G:K]} \not\leq T^{(2),[G:K]}$.

(b) From now on assume that M is maximal in T , and that T is as in part (b), so T^Ω is primitive but not 2-transitive. Then by [26, Theorem 1], the group $T^{(2),\Omega}$ is contained in $N := N_{\text{Sym}(\Omega)}(T^\Omega)$. Since T^Ω is primitive, N is almost simple with socle T^Ω . In particular $T^\Omega \leq N \leq \text{Aut}(T^\Omega)$, and part (b)(i) follows. If $T^{(2),\Omega}$ properly contains T^Ω then, taking $K = (T^{(2),\Omega})_\alpha$, we have $T^{(2),\Omega} = T^\Omega K$ since T^Ω is transitive, $K \leq \text{Aut}(T^\Omega)$ since N has socle T^Ω , and $M \cong M^\Omega = T^\Omega \cap K = T_\alpha^\Omega \neq K$. Thus in part (b)(ii) we conclude that $T^{(2),\Omega} = T$.

Finally, suppose that the assumptions of part (b)(iii) hold, and that $T^{(2),\Omega} \neq T^\Omega$. The previous paragraph shows that $K = (T^{(2),\Omega})_\alpha$ satisfies $K \leq \text{Aut}(T^\Omega)$ and $M \cong M^\Omega = T^\Omega \cap K \neq K$. Further, since T^Ω is transitive we have $T^{(2),\Omega} = T^\Omega K$, and so, by the assumptions in (b)(iii), $T^{(2),\Omega}$ has base size 2. However part (a), applied with $G = T^{(2),\Omega}$, then gives a contradiction. $\square$

Before proceeding further with the general analysis, we illustrate how to use Lemma 4.3 by proving the following useful corollary.

Corollary 4.4. *Each primitive permutation group isomorphic to J_3 or ON is 2-closed.*

Proof. Let T be J_3 or ON, let M be a maximal subgroup of T , and consider T as a primitive permutation group on $\Omega = [T : M]$ acting by right multiplication. Up to permutational isomorphism, each primitive representation of T arises in this way. We need to prove that T is equal to $X := T^{(2),\Omega}$. By [26, Theorem 1], X is almost simple with socle T . In each case $G := \text{Aut}(T) \cong T.2$, so X is equal to either T or G . Note that T has no 2-transitive representations. If M cannot be written as $M = K \cap T$ for any $K \leq G$ with $K \neq M$, then $X = T$ by Lemma 4.3(b)(ii). So assume that such a K exists, and note that $TK = G$. By Lemma 4.3(b)(iii), we may assume further that G has base size larger than 2 in its action on $\Sigma := [G : K]$. Using [9, Tables 1 and 2], the possibilities are as follows.

If $G = J_3 : 2$, then $K \in \{L_2(16) : 4, (3 \times M_{10}) : 2\}$. In these cases, we use GAP [35]. If $K = L_2(16)$, then G^Σ has rank 7, while T^Σ has rank 8. If $K = (3 \times M_{10}) : 2$, then G^Σ has rank 18, while T^Σ has rank 28. Thus, $T^\Sigma = T^{(2),\Sigma}$ in each case.

If $G = ON : 2$, then $K = 4.L_3(4).2.2$. Here, we check using GAP [35] that G^Σ has rank 34, while T^Σ has rank 48. So $T^\Sigma = T^{(2),\Sigma}$. $\square$

We are now ready to outline our strategy to prove Theorem 1.1. Our analysis uses the notion of the depth $\ell_G(H)$ of a proper subgroup H of a finite group G , namely, the least positive integer ℓ such that there exists a subgroup chain $H = G_\ell < \cdots < G_1 < G_0 = G$ with G_i maximal in G_{i-1} for $i = 1, \dots, \ell$. The maximum value of $\ell_G(H)$ over all proper subgroups H of G is the depth of the trivial subgroup, and this is at most the length of the longest subgroup chain in G but may be smaller. For example, $1 < C_2 < C_2^2 < \text{Alt}(4) < \text{Sym}(4)$ is a subgroup chain for $\text{Sym}(4)$ of maximum length four, while $\ell_{\text{Sym}(4)}(1) = 3$ in view of the chain $1 < C_3 < \text{Sym}(3) < \text{Sym}(4)$.

Strategy 4.5. Assume that T is one of the sporadic simple groups in Proposition 2.12(b) so, in particular, T admits no nontrivial factorizations. By Corollary 2.11, T is totally 2-closed if and only if $T^{[T:H]} = T^{(2),[T:H]}$ for all proper subgroups H of G . It follows from Theorem 2.4 that this holds for $H = 1$, so we need to consider proper nontrivial subgroups H , and in these cases we note that $T^{[T:H]}$ has base size at least 2.

We proceed in a series of steps. At Step ℓ we will have remaining a non-empty set $\mathcal{S}_\ell$ of proper nontrivial subgroups of T to consider, where $\mathcal{S}_1$ is the set of all proper nontrivial subgroups of T . Either (a) we complete the analysis at Step ℓ by concluding with certainty that T is, or is not, totally 2-closed, or (b) we remove some subgroups from $\mathcal{S}_\ell$ leaving a set $\mathcal{S}_{\ell+1}$ which, if non-empty, will be considered in Step $\ell + 1$. In particular, for each ℓ , $\mathcal{S}_\ell$ will have the following property:

- (*) If $H \in \mathcal{S}_\ell$, then $\ell_T(H) \geq \ell$, and if $\ell_T(H) = \ell$ and $H < M < T$, then $T^{[T:M]} = T^{(2),[T:M]}$ and $T^{[T:M]}$ has base size at least 3.

Note that $\mathcal{S}_1$ has property (*) since if $\ell_T(H) = 1$ then H is maximal and the last condition holds vacuously. Also, the process will complete in a finite number of steps since T is finite. Finally, we describe what we do in Step ℓ (for an arbitrary value of ℓ).

Step ℓ : For each $H \in \mathcal{S}_\ell$ with $\ell_T(H) = \ell$, we determine $T^{(2),[T:H]}$. To do this when $\ell = 1$ (that is, when H is maximal in T), we use Lemma 4.3 together with some known results about the subdegrees of the primitive almost simple groups. If $\ell \geq 2$ then $T^{[T:H]}$ is imprimitive, and by property (*), $T^{[T:M]} = T^{(2),[T:M]}$ for all M such that $H < M < T$. We can therefore use Theorem 3.10 in our analysis. (This of course is the most difficult part of the procedure.)

- (1) Thus for each $H \in \mathcal{S}_\ell$ with $\ell_T(H) = \ell$, we determine $T^{(2),[T:H]}$.
 - (a) If $T^{[T:H]} < T^{(2),[T:H]}$, for some such H , then we conclude that T is not totally 2-closed, and the analysis is completed.
 - (b) Else we have $T^{[T:H]} = T^{(2),[T:H]}$ for every $H \in \mathcal{S}_\ell$ with $\ell_T(H) = \ell$.

- (2) Next, we check the base size of $T^{[T:H]}$ for each $H \in \mathcal{S}_\ell$ with $\ell_T(H) = \ell$, and construct the new set $\mathcal{S}_{\ell+1}$.
- (a) If, $T^{[T:H]}$ has base size 2, then it follows from Theorem 3.10(3) that, for each $K < H$, we have $T^{[T:K]} = T^{(2),[T:K]}$. Thus proper subgroups of H need no further testing.
 - (b) We remove from $\mathcal{S}_\ell$ each $H \in \mathcal{S}_\ell$ with $\ell_T(H) = \ell$ and, in addition, if $T^{[T:H]}$ has base size 2, we also remove all proper subgroups of H . Let $\mathcal{S}_{\ell+1}$ denote the resulting set of subgroups.
 - (c) If $\mathcal{S}_{\ell+1} = \emptyset$ then we have successfully shown that $T^{[T:H]} = T^{(2),[T:H]}$ for every proper subgroup H of T , we conclude that T is totally 2-closed, and the analysis is completed.
 - (d) On the other hand if $\mathcal{S}_{\ell+1} \neq \emptyset$, then by its construction it is not difficult to check that $\mathcal{S}_{\ell+1}$ has property $(*)$ with $\ell + 1$ in place of ℓ .

We begin by applying this strategy to the group HN, and here we do not need to proceed past stage (1) of Step 1 in Strategy 4.5.

Proposition 4.6. *The Harada Norton group HN is not totally 2-closed.*

Proof. Let $T = \text{HN}$, the Harada Norton group. One of the groups $H \in \mathcal{S}_1$ in Strategy 4.5 is $H = 4.\text{HS}$, and $\text{Aut}(T) = T.2$ has a primitive permutation representation on a set of size 1,539,000, with point stabilizer $H.2$. For the normal subgroup T , this representation is permutationally isomorphic to its transitive action on $[T : H]$, which has rank 9, with all subdegrees distinct, see [19, Lemma 2.18.1]. Thus, in particular, $H.2$ must leave invariant each H -orbit in $[T : H]$. Hence $\text{Aut}(T)$ also has rank 9 on $[T : H]$. We conclude that $T^{(2),[T:H]}$ contains $T.2$, so T is not totally 2-closed. $\square$

We now move on to the group J_1 , where we must use both stages of Steps 1 and 2 of Strategy 4.5.

Remark 4.7. At various stages in the remainder of the paper, we will use the following fact: if K is a proper nontrivial subgroup of a group T and K is contained in a core-free subgroup H of T such that $T^{[T:H]}$ has base size 2, then also $T^{[T:K]}$ has base size 2. (This follows from the fact that $T^{[T:H]}$ having a base of size 2 means that, for some $g \in T$, $H \cap H^g$ acts trivially on $[T : H]$ and hence $H \cap H^g = 1$ since H is core-free in T , whence also its subgroup $K \cap K^g$ is trivial.)

Proposition 4.8. *The Janko group J_1 is totally 2-closed. Also Proposition 4.1, and in particular Table 1, holds for J_1 .*

Proof. Let $T = J_1$. According to Strategy 4.5 our task is to prove that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T . In Step 1 of

Strategy 4.5, for each maximal subgroup H , we conclude that $T^{[T:H]} = T^{(2),[T:H]}$ by Lemma 4.3(b)(i), since $\text{Aut}(T) = T$ and T has no 2-transitive representations. Proceeding to stage (2) of Step 1, we find by [9, Table 1] that, for a maximal subgroup H , $T^{[T:H]}$ has base size 2 unless H is conjugate to $L_2(11)$. Thus, the new subset $\mathcal{S}_2$ consists, up to conjugacy, of all proper nontrivial subgroups of a maximal subgroup $M := L_2(11)$.

We proceed to stage (1) of Step 2 in Strategy 4.5. Let H be a maximal subgroup of M . Then the group $T^{[T:H]} \cong T$ is imprimitive, and we use the notation in Notation 3.3 with $G = T$ and $\Omega = [T : H]$. So $H = G_\omega$, for a point ω contained in a part Δ in a G -invariant partition Σ of Ω , $X = G^{(2),\Omega}$, $Y = X_\Delta^\Delta$, $L = G^\Sigma$, and we may assume that G_Δ (setwise stabilizer) is the subgroup $M = L_2(11)$. Thus $L = L^{(2),\Sigma}$ by the previous paragraph. Using the Web Atlas [41], we see that the M -orbits in Σ have lengths 1, 11, 12, 110, 132. Thus since $|M|/110 = 6$ and $|M|/132 = 5$, the greatest common divisor of the point-stabilizer orders in M^Σ is 1. Thus the first line of Table 1 holds, and it follows from Theorem 3.10(2) that $X = G$, that is, $T^{[T:H]} = T^{(2),[T:H]}$. Next, for each maximal subgroup H of M , we use Magma [5] to compute the expression $\widehat{Q}(T, H, 2)$ of Proposition 4.2 and find that the value is less than 1, and hence that $T^{[T:H]}$ has base size 2. Thus, the subset $\mathcal{S}_3$ constructed in stage (2) of Step 2 is the emptyset, and we conclude that T is totally 2-closed.

To complete the proof that Proposition 4.1 holds for $T = J_1$, note that our argument has so far shown that each proper nontrivial subgroup K of T is either conjugate to $M = L_2(11)$, or is contained in a subgroup H such that $T^{[T:H]}$ has base size 2. In the latter case, as we observed above, $T^{[T:K]}$ also has base size 2. This completes the proof. $\square$

Next, we deal with the group J_3 , where three steps of Strategy 4.5 are required.

Proposition 4.9. *The Janko group J_3 is totally 2-closed. Moreover, Proposition 4.1, and in particular Table 1 holds for J_3 .*

Proof. Let $T := J_3$. According to Strategy 4.5 our task is to prove that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T .

In Step 1 of Strategy 4.5, $\mathcal{S}_1$ is the set of all proper nontrivial subgroups. If $H \in \mathcal{S}_1$ is maximal in T then, by Corollary 4.4, $T^\Omega = T^{(2),\Omega}$, and by [9, Table 1], $H \cong L_2(16) : 2$ is the only maximal subgroup such that T^Ω has base size greater than 2 (and in fact has base size 3). Thus, the set $\mathcal{S}_2$ constructed in Strategy 4.5 consists of all nontrivial, non-maximal subgroups H such that, if $H < M < T$ with M maximal, then M is conjugate to $L_2(16) : 2$.

We see from the Web Atlas [41] that the M -orbits in $[T : M]$ have lengths 1, 85, 120, 510, 680, 1360, 1360, 2040. Thus, since $|M|/1360 = 6$ and $|M|/120 = 68$, it is easily checked that the greatest common divisor of the point stabilizer orders in $M^{[T:M]}$ is 2. The second line of Table 1 then follows for $M = L_2(16) : 2$. From

now on the subgroups H we consider are not maximal, and we use Notation 3.3 with $G = T$, $\Omega = [T : H]$, $X = T^{(2),\Omega}$, $H = G_\omega$, with ω in a part Δ of a G -invariant partition Σ of Ω such that $M = G_\Delta$ (setwise stabilizer), so $\Sigma = [T : M]$ and for $L = G^\Sigma$, we will always have $L = L^{(2),\Sigma}$ by Strategy 4.5.

In Step 2, we consider all maximal subgroups H of $M = L_2(16).2$. Thus M^Δ is primitive. First, we claim that, for each such H , $T^\Omega = T^{(2),\Omega}$. Suppose that this is false, so that, for some H , we have $X \neq G$. Then, by Theorem 3.10(2), $X = N \rtimes G$ with $N = X_{(\Sigma)} \neq 1$ (kernel of the X -action on Σ), and the N^Δ -orbits have constant length a , where a divides the greatest common divisor of the 2-point stabilizers of L . By the previous paragraph a divides 2. Since $1 \neq N^\Delta \trianglelefteq X_\Delta^\Delta$ and $M^\Delta \leq X^\Delta$ is primitive, it follows that N^Δ is transitive, so $a = |\Delta| = 2$. This implies that $|M : H| = 2$ so H is equal to the derived subgroup $M' = L_2(16)$. We check, using Magma [5], that the stabilizer (a subgroup of order 6) in M of a part Δ_j in one of the two M -orbits of length 1360 in Σ lies in H , while the stabilizer of a part Δ_k in the other M -orbit of length 1360 is not contained in H . First, we note that, for $\omega_k \in \Delta_k$, this implies that the two-point stabilizer $G_{\omega,\omega_k} = H_{\omega_k} \cong C_3$, and it follows that the greatest common divisor of the point stabilizer orders in $H^{[T:H]}$ is 1, and hence Line 2 of Table 1 holds also for the subgroup $L_2(16)$. Second, this means that, in the notation of Theorem 3.10, $(M \cap M_j)G_\omega = H \neq M$ so that property (ii) in Theorem 3.10(1) fails, and hence $X = G$, which is a contradiction. Thus $T^{[T:H]} = T^{(2),[T:H]}$ for all maximal subgroups H of M . For those subgroups $H \neq L_2(16)$, we use Magma [5] and compute that the value of the expression $\hat{Q}(T, H, c)$ of Proposition 4.2, with $c = 2$, is strictly less than 1, and hence $T^{[T:H]}$ has base size 2.

Suppose now that $H = L_2(16)$. We compute as follows using GAP [35] to prove that $T^{[T:H]}$ has base size 3. The generators for $T = J_3$ viewed as the primitive group $T^{[T:M]}$ acting on $\Sigma = [T : M]$ are given in the ATLAS [12, 41] and using these we compute with [35] that the point stabilizers of H^Σ have orders 8, 6, 6, 6, 6, 2, 48, 6, 34. This implies in particular that H is transitive on the M^Σ -orbit Σ_0 of length $2040 = |H|/2$. Let $K = H \cap M^g \cong C_2$, the stabilizer in H of a block $\Delta' \in \Sigma_0$. If $T^{[T:H]}$ were to have base size 2, then $H^{[T:H]}$ would have an orbit Ω_0 of length $|H| = 4080$ in $\Omega = [T : H]$, which would have to be the union of the 2040 blocks of Σ in Σ_0 . If this were the case then the stabilizer $J := H \cap H^g$ in H of a point of Ω_0 would be trivial. Thus, we (only) need to check that $J \neq 1$, and we do this using GAP. Now $K = H_{\Delta'}$, the stabilizer in H of $\Delta' \in \Sigma_0$. In GAP we construct $T_{\Delta'}$ (which is M^g) and then the derived subgroup of $T_{\Delta'}$ (which is H^g). Finally, we check (using the 'IsSubgroup' command in GAP) that K is a subgroup of H^g . Since K by definition lies in H we conclude that $K \leq H \cap H^g = J$, and hence $J \neq 1$, as claimed. Thus $T^{[T:H]}$ has base size 3. To complete stage (2) of Step 2 for Strategy 4.5, we remove various subgroups from $\mathcal{S}_2$ to form $\mathcal{S}_3$: which consists, up to conjugacy, of those proper nontrivial subgroups H of $M' = L_2(16)$ such that the only maximal subgroup of M containing H is M' .

In Step 3, for each $H \in \mathcal{S}_3$ such that H is maximal in $M' = L_2(16)$, we use Magma [5] and compute that the value of the expression $\hat{Q}(T, H, c)$ of Proposition 4.2,

with $c = 2$, is strictly less than 1, and hence $T^{[T:H]}$ has base size 2. Finally, we prove that $T^{[T:H]} = T^{(2),[T:H]}$ by applying Theorem 3.10(2) with $M' = L_2(16)$ instead of M , so $L = T^{[T:M']} = T^{(2),[T:M']}$. If $X \neq G$ then by Theorem 3.10(2), $X = N \rtimes G$ with N leaving Δ invariant, and all N -orbits in Δ of constant length $a > 1$ such that a divides the greatest common divisor of the 2-point stabilizer orders in $T^{[T:M']}$. In the previous paragraph we proved that this greatest common divisor is equal to 1, which is a contradiction. Thus, $T^{[T:H]} = T^{(2),[T:H]}$ with base size 2, for all $H \in \mathcal{S}_3$ maximal in $M' = L_2(16)$. When we form the set $\mathcal{S}_4$ we obtain an empty set, and so we conclude that $T = J_3$ is totally 2-closed.

We note that this completes the proof also of Proposition 4.1 for $T = J_3$, for if $H < T$ and $T^{[T:H]}$ has base size at least 3, then also, for all $H < M < T$, $T^{[T:M]}$ has base size at least 3. Thus $H \leq M = L_2(16) : 2$, and if $H \neq M$ then it follows from the proofs of Steps 2 and 3 that $H = M'$. $\square$

Next, we will deal with the case where T is the monster. First, we prove Proposition 4.1 for this group.

Proposition 4.10. *Let $T := \mathbb{M}$ (the monster), and let $K < T$ be such that $T^{[T:K]}$ has base size at least 3. Then $K = 2.\mathbb{B}$, a maximal subgroup of T , and the greatest common divisor of the point-stabilizer orders in $K^{[T:K]}$ is 209,018,880. In particular, Proposition 4.1 (especially the final row of Table 1) holds for $\mathbb{M}$.*

Proof. Let $K \leq M < T$ with M maximal in T . Then also $T^{[T:M]}$ has base size at least 3. Hence by [9, Table 2], $M = 2.\mathbb{B}$. Suppose that $K < M$, and let H be a maximal subgroup of M containing K . Then also $T^{[T:H]}$ has base size at least 3. For all maximal subgroups H of M , we use the GAP Character Table Library [7] (together with the available class fusions) and compute, using GAP [35], that the value of the expression $\widehat{Q}(T, H, 2)$ of Proposition 4.2 is strictly less than 1, and hence that $T^{[T:H]}$ has base size 2. This is a contradiction, and hence $K = M = 2.\mathbb{B}$. Finally, the point stabilizers in $K = 2.\mathbb{B}$ of the nine K -orbits in $[T : K]$ are given in [18, top of p. 101] and we compute that the greatest common divisor of the point-stabilizer orders in $K^{[T:K]}$ is 209,018,880, completing the proof. $\square$

We are now ready to prove that the monster is totally 2-closed.

Proposition 4.11. *The monster $\mathbb{M}$ is totally 2-closed.*

Proof. Let $T = \mathbb{M}$. As usual our task, using Strategy 4.5, is to prove that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T . Since $T = \text{Aut}(T)$ and T has no 2-transitive representations, it follows from Lemma 4.3(b)(i) that this holds for all maximal subgroups H . Thus we proceed to stage (2) of Step 1 in Strategy 4.5, and here, by Proposition 4.10, the only maximal subgroup M such that $T^{[T:M]}$ has base size at least 3 is $M = 2.\mathbb{B}$. Hence the set $\mathcal{S}_2$ formed in Step 1 consists, up to conjugacy, of all proper nontrivial subgroups of M .

In Step 2 of Strategy 4.5, we examine the maximal subgroups H of M . We again use Notation 3.3 with $G = T$, $\Omega = [T : H]$, $X = T^{(2),\Omega}$, $H = G_\omega$, with ω in a part Δ of a T -invariant partition Σ of Ω such that $M = G_\Delta$, so $\Sigma = [T : M]$ and $L = G^\Sigma$ satisfies $L = L^{(2),\Sigma}$, and we set $Y = X^\Delta$. Since H is maximal in M and $Z(M) = \Phi(M) \cong C_2$, we have $Z(M) = \text{Core}_M(H) = \Phi(M)$ and $M^\Delta \cong \mathbb{B}$. This implies that $M^\Delta = M^{(2),\Delta}$, by Lemma 4.3(b)(i), since $\text{Aut}(\mathbb{B}) = \mathbb{B}$ and $\mathbb{B}$ has no 2-transitive representations. It now follows from Lemma 3.1(b) that $Y = \mathbb{B} = M^\Delta$. Suppose now that $X \neq G$. Then by Theorem 3.8(a), $X = N \rtimes G$ with N^Δ a nontrivial normal subgroup of Y , and hence $N^\Delta = Y$. By the definition of the subgroup N from Lemma 3.5(b), this would imply that, for $1 < j \leq s = |\Sigma|$, the group $(M \cap M_j)^\Delta$ is transitive. Since $M \cap M_j$ is a proper subgroup of M and $Z(M) = \text{Core}_M(H) \leq M \cap M_j$, the induced group $(M \cap M_j)^\Delta$ is a proper subgroup of $M^\Delta = Y$, so we have a proper factorization $Y = Y_\omega(M \cap M_j)^\Delta$. This is a contradiction since Y has no such factorizations, by [27]. Thus $X = G$, that is, $T^{[T:H]} = T^{(2),[T:H]}$. We now move to stage (2) of Step 2 for Strategy 4.5. By Proposition 4.10, each of the groups $T^{[T:H]}$ just examined has base size 2. Hence the set $\mathcal{S}_3$ formed in Step 2 is empty, and we conclude that T is totally 2-closed. $\square$

The next two groups we deal with are the O’Nan group and the Lyons group.

Proposition 4.12. *The O’Nan group ON is not totally 2-closed.*

Proof. Let $T = \text{ON}$ and let M be a maximal subgroup of T with $M = L_3(7) : 2$. Also let $H = M'$ (the derived subgroup), so $|M : H| = 2$. By Corollary 4.4, $T^{[T:M]} = T^{(2),[T:M]}$. We check, using GAP [35], that no two-point stabilizer in $M^{[T:M]}$ is contained in H . Hence, $(M \cap M^t)H = M$ for all $t \in T$, that is to say, condition (ii) of Theorem 3.10(1) holds. Since $T^{[T:M]} = T^{(2),[T:M]}$, Theorem 3.10(1) applies, and we conclude that $T^{[T:H]}$ is not 2-closed. Thus, T is not totally 2-closed. $\square$

Proposition 4.13. *The Lyons group Ly is totally 2-closed. Moreover, Proposition 4.1, and in particular Table 1 holds for $T = \text{Ly}$.*

Proof. Let $T = \text{Ly}$. As in the other cases, our task is to prove, using Strategy 4.5, that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T . Since $T = \text{Aut}(T)$ and T has no 2-transitive representations, it follows from Lemma 4.3(b)(i) that this holds for all maximal subgroups H . Thus, we proceed to stage (2) of Step 1 in Strategy 4.5: constructing the set $\mathcal{S}_2$. By [9, Table 2], there are only two conjugacy classes of maximal subgroups M of T such that $T^{[T:M]}$ has base size at least 3, with representatives $M_1 = G_2(5)$ and $M_2 = 3.\text{McL} : 2$. Therefore according to Strategy 4.5, the set $\mathcal{S}_2$ consists of all non-maximal subgroups H such that the only maximal subgroups of T containing H are conjugate to M_1 or M_2 .

We now move on to Strategy 4.5, Step 2, stage (1). Here, we need to prove that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_2$ with $\ell_T(H) = 2$. Let $H \in \mathcal{S}_2$ with $\ell_T(H) = 2$, and assume that $T^{[T:H]} \neq T^{(2),[T:H]}$. Let M be a maximal subgroup of T containing H , so H is maximal in M . Without loss of generality, we may assume that $M = M_i$ for $i = 1$ or 2 . We adopt the rest of the notation in Notation 3.3, with $T = G$. So $\Omega = [T : H]$, $\Sigma = [T : M]$, $L = T^\Sigma$, $M = T_\Delta$ (with $\Delta \in \Sigma$), $H = T_\omega$ (with $\omega \in \Delta$), etc. Note that $L^{(2),\Sigma} = L$, by the previous paragraph, and M^Δ is primitive since H is maximal in M . We also set $X = T^{(2),\Omega}$, so $X > T$, and $Y = X_\Delta^\Delta \leq M^{(2),\Delta}$.

We begin with the case $M = M_1 = G_2(5)$. We note that, by [19, Lemma 2.19.1], it follows that the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$ is 48, and hence the entry in Table 1 for $M = G_2(5)$ is correct. Suppose first that $|M : H| \neq 5^3(5^3 - 1)/2$. Then $M \cong M^\Delta$, so [26, Theorem 1] implies that $Y \leq M^{(2),\Delta}$ is almost simple with socle $G_2(5)$. Since $\text{Out}(G_2(5)) = 1$ and $G_2(5)$ has no 2-transitive representations, it follows from Lemma 4.3(b)(i) that $M^\Delta = M^{(2),\Delta}$. Therefore, $Y \cong G_2(5)$ and so the nontrivial normal subgroup A of Y in Theorem 3.10(2) is equal to Y . In particular, A^Δ is transitive. However this means, by Theorem 3.10(2), that $|M : H| = |\Delta|$ divides the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$, that is, $|M : H|$ divides 48. This is a contradiction since the smallest degree of a transitive permutation representation of $G_2(5)$ is 3906, [12, p. 114]. Thus, $|M : H| = 5^3(5^3 - 1)/2$, and [26, Theorem 1] implies that $Y \leq M^{(2),\Delta} \leq \text{Aut}(\Omega_7(5)) = \Omega_7(5).2$. Since $M \leq Y$ and the only maximal overgroups of $G_2(5)$ in $\Omega_7(5).2$ are $G_2(5)$, $\Omega_7(5)$, and $\Omega_7(5).2$, we have $Y \in \{G_2(5), \Omega_7(5), \Omega_7(5).2\}$. Hence, the nontrivial normal subgroup A of Y in Theorem 3.10(2) satisfies $A \in \{G_2(5), \Omega_7(5), \Omega_7(5).2\}$. In all cases $A \geq G_2(5)$, so A^Δ is transitive since M^Δ is primitive. Then the same argument as in the previous shows that $|\Delta|$ divides 48 and we get the same contradiction. Thus, we have shown that $T = T^{(2),[T:H]}$ for all maximal subgroups H of $M = M_1$, completing Step 2 stage (1) for $M = M_1$.

Now, we consider the case $M = M_2 = 3.\text{McL} : 2$. We note that, by [19, Lemma 2.19.1], it follows that the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$ is 30, and hence the entry in Table 1 for $M = 3.\text{McL} : 2$ is correct. Let $Z = Z(M')$, where $M' = 3.\text{McL}$ is the derived subgroup of M , so $|Z| = 3$. Since $Z \leq \Phi(M)$ and H is maximal in M , we have $Z \leq \text{Core}_M(H)$. Suppose first that $H = M'$, so $|\Delta| = 2$. We will use Theorem 3.10(1): by [19, Lemma 2.19.1], M^Σ has an orbit such that the stabilizer $M \cap M_j$ in M of a point in this orbit has shape $2.\text{Sym}(7)$. A computation using MAGMA shows that $(M \cap M_j)H \neq M$. Thus, the condition in Theorem 3.10(1)(ii) fails, and hence $X = T$, which is a contradiction. Therefore $H \neq M'$, and hence $\text{Core}_M(H) = Z \cong C_3$ and $M^\Delta \cong \text{McL} : 2$. Then $M^\Delta = M^{(2),\Delta}$, by [26, Theorem 1], and since $M^\Delta \leq Y \leq M^{(2),\Delta}$, this means that $Y = M^{(2),\Delta} \cong \text{McL} : 2$. Thus, the nontrivial normal subgroup A of Y in Theorem 3.10(2) is either McL or $\text{McL} : 2$. In particular, A^Δ is transitive, and hence, by Theorem 3.10(2), $|M : H| = |\Delta|$ divides the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$, that is, $|M : H|$ divides 30. This is a

contradiction since the smallest degree of a transitive permutation representation of $\text{McL} : 2$ is 275, [12, p. 100].

This completes stage (1) of Step 2. We now proceed to stage (2) of Step 2: constructing the set $\mathcal{S}_3$. To do this, we prove the following claim.

Claim: *If $H \leq T$ and either (i) $\ell_T(H) = 2$ but $H \not\cong M'_2 = 3.\text{McL}$, or (ii) $H < M'_2$ with H conjugate to $3.M_{22}$ or $3.2^4.A_7$, then $T^{[T:H]}$ has base size 2.*

Proof of Claim: Let M be a maximal subgroup of T containing H . If $T^{[T:M]}$ has base size 2 then so does $T^{[T:H]}$, so we may assume that $T^{[T:M]}$ has base size greater than 2. As we saw above, this means (replacing H, M by some T -conjugate if necessary) that $M = M_i$ for $i = 1$ or 2 . Suppose first that $\ell_T(H) = 2$, that is, H is maximal in M , but $H \neq M'_2$. Then H is either a maximal subgroup of $M_1 = G_2(5)$, or $M = M_2$, H contains $Z := Z(M') \cong C_3$, and H/Z is a maximal subgroup of $M/Z \cong \text{McL} : 2$ of index greater than 2. Such subgroups H can be read off using the Web Atlas [41]. As in Proposition 4.11, an upper bound for the base size of $T^{[T:H]}$ can be computed using Proposition 4.2. Using the character table library (together with the available class fusions), we check using GAP [35] that $\hat{Q}(T, H, 2) < 1$ for each choice of H . Hence by Proposition 4.2, $T^{[T:H]}$ has base size 2. This proves part (i). Finally suppose that $H < M'_2$ with H conjugate to $3.M_{22}$ or $3.2^4.A_7$. The same method, using GAP, shows that $\hat{Q}(T, H, 2) < 1$, and hence that $T^{[T:H]}$ has base size 2, proving part (ii).

It follows from part (i) of the claim, and the definition of $\mathcal{S}_3$, that $\mathcal{S}_3$ consists of those subgroups conjugate to a proper nontrivial subgroup H of $M'_2 = 3.\text{McL}$ such that the only subgroup K containing H with $\ell_T(K) = 2$ is $K = M'_2$. Examining the Atlas [12, p. 100], we see that H must be conjugate to a subgroup of $N_1 := 3.M_{22}$ or $N_2 := 3.2^4.A_7$, since all other proper subgroups of M'_2 are contained in a subgroup K of M_2 with $\ell_T(K) = 2$ and $K \neq M'_2$.

Now, we move on to Strategy 4.5, Step 3, stage (1). We need to prove that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_3$ with $\ell_T(H) = 3$. By the previous paragraph each such subgroup H is conjugate to N_1 or N_2 (both maximal subgroups of $3.\text{McL}$). The argument showing that in these two cases $T^{[T:H]}$ is 2-closed is almost identical to that given in Step 2, stage (1) for maximal subgroups of M_2 other than M'_2 . We assume that $H = N_i$ and $T^{[T:H]}$ is not 2-closed. We have $\text{Core}_{M'_2}(H) = Z \cong C_3$ and (using Notation 3.3) $(M'_2)^\Delta \cong \text{McL}$ where $\Delta = [M'_2 : H]$. We again apply [26, Theorem 1] to deduce that $Y = (M'_2)^{(2),\Delta} = \text{McL}$. This implies that the subgroup A in Theorem 3.10(2) is transitive on Δ and hence that $|\Delta| = |M'_2 : H|$ divides the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M'_2]}$. Hence $|M'_2 : H|$ divides 30, which is a contradiction. This completes Step 3, stage (1).

Finally to complete Step 3, stage (2), we construct $\mathcal{S}_4$. A subgroup $H \in \mathcal{S}_4$ must satisfy $H \in \mathcal{S}_3$ and $\ell_T(H) \geq 4$, and further, if $\ell_T(H) = 4$ and $H < K < T$, then in particular $T^{[T:K]}$ has base size at least 3. Now, $H \in \mathcal{S}_3$ and $\ell_T(H) \geq 4$ imply that H is conjugate to a proper subgroup of N_i for some i , and the second condition implies that $T^{[T:N_i]}$ has base size at least 3. This however contradicts part (ii) of

the claim, and we conclude that $\mathcal{S}_4$ is empty. We have therefore shown that T is totally 2-closed.

Finally, we prove Proposition 4.1 for $T = \text{Ly}$. The information deduced above implies that, if H is a proper nontrivial subgroup of T with $H \notin \{G_2(5), 3.\text{McL}, 3.\text{McL} : 2\}$, then $T^{[T:H]}$ has base size 2. We know from [9, Table 2] that $T^{[T:H]}$ has base size 3 when $H \in \{G_2(5), 3.\text{McL} : 2\}$, and in these cases the entries in Table 1 is correct. Suppose then that $H = 3.\text{McL} < M_2 = 3.\text{McL} : 2$. Then, for each $g \in T$, $|M_2 \cap M_2^g : H \cap H^g|$ divides 4. However, by [19, Lemma 2.19.1], $|M_2 \cap M_2^g| > 4$ for all $g \in T$. Thus $H \cap H^g \neq 1$ for each $g \in T$, and hence $T^{[T:H]}$ has base size greater than 2. Finally, the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:H]}$ divides the analogous number for the action $T^{[T:M_2]}$ which, as we showed above, is 30. Thus the entry in Table 1 is correct also for 3.McL. $\square$

Now, we deal with the Conway and Fischer groups Co_2 and Fi_{23} , and then the baby monster $\mathbb{B}$.

Proposition 4.14. *Neither the second Conway group Co_2 nor the Fischer group Fi_{23} is totally 2-closed.*

Proof. Let $T \in \{\text{Co}_2, \text{Fi}_{23}\}$. Consider the following maximal subgroup M of T , namely, $M = U_6(2) : 2$ if $T = \text{Co}_2$, and $M = O_8^+(3) : \text{Sym}(3)$ if $T = \text{Fi}_{23}$. In each case, $T^{[T:M]}$ has rank 3. Set $H := M'$, the derived subgroup, so that $|M : H| = 2$, and adopt the notation in Notation 3.3, with $T = G$. We argue as in the case of $T = \text{ON}$ in Proposition 4.12: we check, using GAP [35], that each of the 2-point stabilizers $M \cap M_j$ in the T -action on $[T : M]$ satisfies $(M \cap M_j)H = M$, that is to say, condition (ii) of Theorem 3.10(1) holds. On the other hand, $T^{[T:M]} = T^{(2).[T:M]}$ by Lemma 4.3(b)(i) since $T = \text{Aut}(T)$ and $T^{[T:M]}$ is not 2-transitive. Hence Theorem 3.10(1) applies, and we conclude that $T^{[T:H]}$ is not 2-closed. Hence, T is not totally 2-closed. $\square$

Proposition 4.15. *Fischer's baby monster $\mathbb{B}$ is not totally 2-closed.*

Proof. Let $T = \mathbb{B}$, and consider the maximal subgroup $M = 2.^2E_6(2).2$. Then $M/Z(M)$ is an extension of ${}^2E_6(2)$ by a graph automorphism. Let $H = M' \cong 2.^2E_6(2)$, of index 2 in M . The stabilizers in $T^{[T:M]}$ of two distinct points are given in [19, Lemma 2.25.1], namely

$$(2 \times 2^{1+20}.U_6(2)).2, \quad 2^2 \times F_4(2), \quad \text{Fi}_{22} : 2, \quad 2^{1+20}.U_4(3).2^2.$$

We call these groups $E_i = M \cap M_i$, for $i = 2, 3, 4, 5$, respectively. We claim that $HE_i = M$ for $i = 2, \dots, 5$. All assertions in the next two paragraphs are from the short proof of the main theorem in [25]. The groups E_i are the same as in [25], but there the group M is denoted by E .

First, $|M : E_2|$ is odd, so certainly $E_2H = M$. The group E_3 contains $Z(M)$, and $E_3/Z(M)$ is the centralizer in ${}^2E_6(2).2$ of a graph automorphism. In particular, E_3 contains a graph automorphism, so we also have $E_3H = M$. Next, M has precisely two conjugacy classes of subgroups isomorphic to $\text{Fi}_{22} : 2$, and they are interchanged by an automorphism of M which acts trivially on H (see [25, top of p. 1040]). It follows, since E_4 is one of these groups, that $E_4 \not\leq H$, and hence $E_4H = M$.

Finally, the group E_5 can be written as $E_5 = KQ$ as follows (see [25, p. 1040]). We may take $E_2 = C_M(d_2)$ with $d_2 \in E_4$ being a $2A$ -involution of $E_4 = \text{Fi}_{22} : 2$; and $K = C_{E_4}(d_2) \cap C_{E_4}(d'_2)$ with d'_2 a $2B$ -involution of E_4 which does not commute with d_2 (using ATLAS notation). Then Q is a unique index 2 subgroup of $O_2(E_2) = 2 \times 2^{1+20}$ which is normalized by K . A computation in GAP [35] shows that $KE'_4 = E_4$. Hence, since $Q \leq M' = H$, $E'_4 \leq M' = H$ and $E_4H = M$, we deduce that

$$M = E_4H = (KE'_4)H = K(E'_4H) = KH = K(QH) = (KQ)H = E_5H.$$

Thus $E_iH = M$ for all i , proving the claim. Hence the condition of Theorem 3.10(1)(ii) holds for the action $T^{[T:H]}$, and it follows from Theorem 3.10(1) that $T^{[T:H]}$ is not 2-closed. The result follows. $\square$

We now move on to our second to last case: the Thompson group Th . We begin by showing that a particular transitive action of Th is 2-closed.

Proposition 4.16. *Suppose that $H = {}^3D_4(2)$ is a subgroup of $T = \text{Th}$ with $\ell_T(H) = 2$, with H contained in a maximal subgroup $M := {}^3D_4(2).3$. Then $T^{[T:H]}$ is 2-closed.*

Proof. Let T , H , and M be as in the statement, and use the notation in Notation 3.3, with $T = G$. So $\Omega = [T : H]$, $\Sigma = [T : M]$ with $s = |\Sigma|$, $L = T^\Sigma$, $M = T_\Delta$ (with $\Delta \in \Sigma$), $H = T_\omega$ (with $\omega \in \Delta$), etc. By Lemma 4.3(b)(i), $L^{(2),\Sigma} = L$, since $T = \text{Aut}(T)$ and T has no 2-transitive representations. We also set $X = T^{(2),\Omega}$ and $Y = X_\Delta^\Delta \leq M^{(2),\Delta}$. Since $|\Delta| = 3$, we have $R = M^\Delta = Y = M^{(2),\Delta} \cong C_3$.

Assume that $T^{[T:H]}$ is not 2-closed. Then, by Lemma 3.5, $X = N \rtimes T$ with N a nontrivial subdirect subgroup of $B_Y = C_3^s$, and moreover, for each $j = 2, \dots, s$, $(\rho_{1,j}(M \cap M_j))^{\Delta \times \Delta_j}$ is either equal to $Y \times Y$ or is a diagonal subgroup of $Y \times Y$. By Proposition 3.7, this group is diagonal for strictly more than $s/2$ values of j , and hence by Lemma 3.6, N is a diagonal subgroup of B_Y . Thus $N \cong C_3$ and, since $X = N \rtimes T$ with T a nonabelian simple group, the group T centralizes N .

We now examine a part $\Delta_2 \in \Sigma$ such that $M \cap M_2 = 2^2.[2^9].(\text{Sym}(3) \times 3)$. As in the proof of Proposition 3.7, $(\rho_{1,2}(M \cap M_2))^{\Delta \times \Delta_2}$ is a diagonal subgroup of $Y \times Y$. Further, the subdegrees of $T^{[T:M]}$ are pairwise distinct, so each T -orbit in $\Sigma^{(2)} = \{(\alpha, \beta) : \alpha, \beta \in \Sigma, \alpha \neq \beta\}$ is self-paired. It follows that the setwise stabilizer $T_{\{\Delta \cup \Delta_2\}}$ in T of $\Delta \cup \Delta_2$ contains an element, say g , which interchanges Δ and Δ_2 . Thus, the permutation group induced by $T_{\{\Delta \cup \Delta_2\}}$ on $\Delta \cup \Delta_2$ is regular of order 6,

so is either C_6 or $\text{Sym}(3)$. Moreover, since $g \in T_{\{\Delta \cup \Delta_2\}}$ and T centralizes N , it follows from Lemma 3.5(b) that g centralizes $\rho_{1,2}(M \cap M_2)^{\Delta \times \Delta_2}$, and hence the permutation group induced by $T_{\{\Delta \cup \Delta_2\}}$ on $\Delta \cup \Delta_2$ is C_6 , and

$$T_{\{\Delta \cup \Delta_2\}}/(H \cap H^g) \cong C_6. \quad (4.1)$$

Let $P := O_2(M \cap M_2) = 2^2.[2^9]$, and $Q := N_T(P)$. Since $|M \cap M_2 : H \cap H^g| = 3$ it follows that $P < H \cap H^g < H < Q$. Also, since P is characteristic in $M \cap M_2$ and $M \cap M_2 \trianglelefteq T_{\{\Delta \cup \Delta_2\}}$, we have $P < T_{\{\Delta \cup \Delta_2\}} < Q$. By [28, Sec. 2.2], Q has shape $2^5.(2^6 : \text{Sym}(3) \times L_3(2))$ and Q is contained in a maximal subgroup $K := 2^5.L_5(2)$ of T . More precisely, Q contains $O_2(K) \cong 2^5$, and $Q/O_2(K)$ is a maximal parabolic subgroup of $L_5(2)$ of shape $2^6 : \text{Sym}(3) \times L_3(2)$. Further, $O_2(Q) = O_2(M \cap M_2) = P$ and $(M \cap M_2)/P$ has shape $\text{Sym}(3) \times 3$. Hence, since $M \cap M_2$ has index 2 in $T_{\{\Delta \cup \Delta_2\}}$, it follows that $T_{\{\Delta \cup \Delta_2\}}/P$ is isomorphic to a subgroup of $\text{Sym}(3) \times L_3(2)$ of shape $(\text{Sym}(3) \times 3).2$. The only such subgroups of $\text{Sym}(3) \times L_3(2)$ are the Sylow 3-normalizers, and they are isomorphic to $\text{Sym}(3) \times \text{Sym}(3)$. However, since $H \cap H^g$ contains P , $T_{\{\Delta \cup \Delta_2\}}/(H \cap H^g)$ is isomorphic to a quotient of $T_{\{\Delta \cup \Delta_2\}}/P \cong \text{Sym}(3) \times \text{Sym}(3)$. This contradicts (4.1), since $\text{Sym}(3) \times \text{Sym}(3)$ has no cyclic quotients of order 6. Thus $T^{[T:H]}$ is 2-closed. $\square$

Proposition 4.17. *The Thompson group Th is totally 2-closed. Furthermore, Proposition 4.1, and in particular Table 1, holds for $T = \text{Th}$.*

Proof. Let $T = \text{Th}$. Our task is to prove, using Strategy 4.5, that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T . Since $T = \text{Aut}(T)$ and T has no 2-transitive representations, it follows from Lemma 4.3(b)(i) that this holds for all maximal subgroups H . Thus, we proceed to stage (2) of Step 1 in Strategy 4.5: constructing the set $\mathcal{S}_2$. By [9, Table 2], there are only two conjugacy classes of maximal subgroups M of T such that $T^{[T:M]}$ has base size at least 3, with representatives $M_1 = {}^3D_4(2) : 3$ and $M_2 = 2^5.L_5(2)$. Therefore according to Strategy 4.5, the set $\mathcal{S}_2$ consists of all non-maximal subgroups H such that the only maximal subgroups of T containing H are conjugate to M_1 or M_2 .

We now move on to Strategy 4.5, Step 2, stage (1). We need to prove that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_2$ with $l_T(H) = 2$. Let $H \in \mathcal{S}_2$, and assume that $T^{[T:H]} \neq T^{(2),[T:H]}$. Let M be a maximal subgroup of T containing H , so without loss of generality we may assume that $M \in \{M_1, M_2\}$. We use the rest of the notation in Notation 3.3, with $T = G$. So $\Omega = [T : H]$, $\Sigma = [T : M]$, $L = T^\Sigma$, $M = T_\Delta$ (with $\Delta \in \Sigma$), $H = T_\omega$ (with $\omega \in \Delta$), etc. Note that $L^{(2),\Sigma} = L$, by Step 1. We also set $X = T^{(2),\Omega}$, so $X > T$, and $Y = X_\Delta^\Delta \leq M^{(2),\Delta}$. Suppose first that $M = M_2$. We note that, by [19, Lemma 2.20.1(ii)], it follows that the greatest common divisor of the orders of the two-point stabilizers in $T^{[T:M]}$ is 1, and hence the entry in Table 1 for $M = 2^5.L_5(2)$ is correct. This implies, by Theorem 3.10(2), that $X = T$, which is a contradiction. We note here that this argument is valid for all proper subgroups H of M_2 (without the condition $l_T(H) = 2$).

Thus $M = M_1$. At this point, we note that it follows from [19, Lemma 2.20.1(i)] that the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$ is 6, and hence the entry in Table 1 for $M = {}^3D_4(2) : 3$ is correct. Further this also implies correctness of the entry in that table for the derived subgroup $M' = {}^3D_4(2)$ (we show below that $T^{[T:M']}$ has base size at least 3). If $H = M'$ then $T^{[T:H]}$ is 2-closed by Proposition 4.16. Thus, we may assume that $H \neq M'$, and we assume also that $\ell_T(H) = 2$. Then, by [26, Theorem 1], $M^{(2),\Delta}$, and hence also its subgroup Y , is almost simple with socle ${}^3D_4(2)$. Since $X > T$, the nontrivial normal subgroup A of Y in Theorem 3.10(2) must contain ${}^3D_4(2)$. In particular, A^Δ is transitive, and so by Theorem 3.10(2), $|M : H| = |\Delta|$ divides the greatest common divisor of the orders of the 2-point stabilizers in $T^{[T:M]}$, which as noted above is 6. This is a contradiction, and hence $T^{[T:H]}$ is 2-closed, completing Step 2, stage (1).

We now proceed to stage (2) of Step 2: constructing the set $\mathcal{S}_3$. Given the results of Step 1 above, the set $\mathcal{S}_3$ will consist of all conjugates of proper nontrivial subgroups H of $M = M_1$ or $M = M_2$ such that $\ell_T(H) \geq 3$, and if $\ell_T(H) = 3$ then, for any maximal subgroup J of M containing H , $T^{[T:J]}$ has base size at least 3. If $M = M_1$, then one such subgroup J is $M' = {}^3D_4(2)$. However if $H < M'$ is maximal then, by the Atlas [12, p. 89], $H < N_M(H) < M$, and $N_M(H)$ is another maximal subgroup of M containing H . Thus each subgroup $H < M$ with $\ell_T(H) = 3$ is contained in a maximal subgroup J of M with J lying in the union $\mathcal{C}$ of the nine conjugacy classes of maximal subgroups of M of index greater than 3 (that is, maximal subgroups not equal to M'). Thus, if we can prove that $T^{[T:J]}$ has base size 2 for all $J \in \mathcal{C}$, and also for all maximal subgroups J of $M_2 = 2^5.L_5(2)$, then the set $\mathcal{S}_3$ would be empty, and we will have proven that T is totally 2-closed.

So let J be either a maximal subgroup of $M_2 = 2^5.L_5(2)$, or a member of $\mathcal{C}$. We use Proposition 4.2 to show that $T^{[T:J]}$ has base size 2. So let $x_1, \dots, x_m$ be representatives for the T -conjugacy classes of elements of prime order. First, we note the following crude upper bound:

$$\widehat{Q}(T, J, 2) \leq \sum_{p \text{ prime}} \frac{|\{g \in J : |g| = p\}|^2}{\min_i \{|x_i^T| : |x_i| = p\}}. \quad (4.2)$$

We use **Magma** [5] to compute the right-hand side of (4.2), and find that it is less than 1 for each subgroup J , and hence, by Proposition 4.2, $T^{[T:J]}$ has base size 2 for each J . Thus, T is totally 2-closed. (For J a maximal subgroup of $M_2 = {}^3D_4(2) : 3$, we work with the degree 819 permutation representation of ${}^3D_4(2)$ from the Web Atlas to construct ${}^3D_4(2)$ in **Magma** [5], and then we take its normalizer in $\text{Sym}(819)$ to get ${}^3D_4(2) : 3$. For J contained in the *Dempwolff group* $M_2 = 2^5.L_5(2)$, we work with the degree 7440 permutation representation of M_2 given in the Web Atlas.)

Finally, we complete the proof of Proposition 4.1 for the group Th . Suppose that H is a proper subgroup of T for which $T^{[T:H]}$ has base size at least 3. Let M be a maximal subgroup of T containing H , so also $T^{[T:M]}$ has base size at least 3. Then, as above, M is conjugate to M_1 or M_2 , so we may assume that $M \in \{M_1, M_2\}$ and $H \neq M$. Let J be a maximal subgroup of M containing H . If either $M = M_2$

or $J \in \mathcal{C}$, then we showed above that $T^{[T:J]}$ has base size 2, and hence also $T^{[T:H]}$ has base size 2, which is a contradiction. Thus $M = M_1$ and $J = M'$ is the only maximal subgroup of M containing H . As discussed above, by [12, p. 89], the only possibility is $H = J = M'$. Now $|M : H| = 3$ and hence, for each $g \in T$, the index $|M \cap M^g : H \cap H^g|$ divides 9. Moreover, it follows from [19, Lemma 2.20.1(i)] that, for each $g \in T$, $\gcd(|M \cap M^g|, 9)$ is strictly less than $|M \cap M^g|$, and hence $H \cap H^g \neq 1$. Thus, $T^{[T:H]}$ has base size at least 3. This proves Proposition 4.1(a), and the assertions of Proposition 4.1(b) have been verified already in the course of the proof. $\square$

5. The Case $T = J_4$

In this section, we prove that the fourth Janko group J_4 is totally 2-closed, and in so-doing we prove Proposition 4.1 for J_4 . This will complete the proofs of Theorem 1.1 and Proposition 4.1. Our main reference is Janko's original paper [20], and wherever possible we try to be consistent with his notation, or draw attention to it. See Remark 5.1 for more information.

Throughout the section, we set $T := J_4$, and we adopt Strategy 4.5. To complete Step 1 of Strategy 4.5, we need to know all maximal subgroups M of T such that $T^{[T:M]}$ has base size at least 3. By [9, Table 2], each such subgroup M is conjugate to one of the following three maximal subgroups D_1, D_2, D_3 , where z is a representative of the $2A$ conjugacy class of involutions in T , and in fact $T^{[T:M]}$ has base size 3 in each case.

- $D_1 := N_T(V) = V \rtimes F_1 \cong 2^{11} : M_{24}$, where $V = O_2(D_1)$ lies in the unique conjugacy class of elementary abelian subgroups of T of order 2^{11} and $F_1 \cong M_{24}$, [20, Propositions 7 and 18].
- $D_2 := C_T(z) \cong 2^{1+12}.3.M_{22} : 2$. The subgroup $O_2(D_2)$ is not complemented in D_2 , but has a supplement $F_2 \cong 6.M_{22} : 2$ such that $F_2 \cap O_2(D_2) = \langle z \rangle \cong C_2$, [20, Proposition 1].
- $D_3 := N_T(A) = A \rtimes F_3 \cong 2^{10} : L_5(2)$, where $A = O_2(D_3)$ is an elementary abelian self-centralizing subgroup of order 2^{10} , contained in D_2 , and $F_3 \cong L_5(2)$, [20, Propositions 20 and 21].

Maximality of the subgroups D_1, D_2 , and D_3 can also be checked from the Web Atlas [41]. For Strategy 4.5 we also need to examine transitive actions $T^{[T:H]}$ with base size at least 3 for non-maximal subgroups H . For such a subgroup, if $H < M < T$ with M maximal, then $T^{[T:M]}$ also has base size at least 3, and hence M is conjugate to D_i for some i . In similar situations in Sec. 4, we determined whether or not such an action $T^{[T:H]}$ has base size 2 by finding crude upper bounds on $|x^T \cap H|$ for each element $x \in H$ of prime order, and then applying Proposition 4.2. For $T = J_4$, we need to be much more careful when finding upper bounds for $|x^T \cap H|$ in the case where x is an involution. Thus, the first lemma in this section counts involutions in subgroups of the maximal subgroups D_1, D_2 , and D_3 . Since

we make frequent reference to Janko's paper [20], we summarize Janko's notation in Remark 5.1.

Remark 5.1. The Janko group $T = J_4$ has precisely two classes of involutions, denoted class $2A$ and $2B$, and Janko uses z and f , respectively, to denote representatives of these classes, [20, Theorem A(5)]. He also uses the following notation for the subgroups D_1 , D_2 , and D_3 :

- (1) Janko writes $M = D_1$, and $K = F_1$, so $M = V \rtimes K \cong 2^{11} \rtimes M_{24}$ with $V \cap K = 1$, [20, Proposition 18].
- (2) Janko writes $H = D_2 = C_T(z)$, $E = O_2(D_2) \cong 2^{1+12}$, and $H_0 = [D_2, D_2] \cong 2^{1+12}.3.M_{22} : 2$ for the (index 2) derived subgroup of H , [20, Sec. 3]. In [20, last 13 lines of Proposition 1], Janko describes the D_2 -conjugacy classes of involutions in H_0 : there are two classes in $E \setminus \langle z \rangle$, with representatives e and f , and four classes in $H_0 \setminus E$, with representatives t' , $t'z$, t_1 , and t_2 . Moreover all of t' , $t'z$, t_1 , t_2 lie in the same coset of E , [20, p. 570, line-11]. As noted above, $f \in 2B$; by [20, Proposition 9], the elements e, t', t_1 all lie in class $2A$, and $t'z \in 2B$; while by [20, Proposition 15], $t_2 \in 2B$. Finally, by [20, Propositions 6 and 10], there are three D_2 -conjugacy classes of involutions in $H \setminus H_0$, with representatives $\tilde{t}_1$, $\tilde{t}_1z$, and $\tilde{t}_2$, and by [20, Proposition 15], $\tilde{t}_1 \in 2A$, and $\tilde{t}_1z$, $\tilde{t}_2 \in 2B$.
- (3) Janko writes $D_3 = N_T(A) = A \rtimes B$, where $A = 2^{10}$ is self-centralizing with $A < D_2$ and $B = F_3 \cong L_5(2)$, [20, Propositions 20 and 21]. He notes that $O_2(D_2) \cong 2^{1+12}$ normalizes $A = O_2(D_3)$, and $N_{D_2}(A)/A$ is isomorphic to the parabolic subgroup $2^6 : (\text{Sym}(3) \times L_3(2))$ of $D_3/A \cong B \cong L_5(2)$, [20, p. 587, lines 4–7].

To help us find upper bounds for $|x^T \cap H|$ in the case where x is an involution in $H < T$, we introduce the following (non-standard) notation for counting involutions in finite groups.

Notation 5.2. For a subset S of a finite group G , define

$$\text{Inv}(S : \mathcal{P}_1, \dots, \mathcal{P}_r)$$

to be the number of involutions in S with properties $\mathcal{P}_1, \dots, \mathcal{P}_r$. For example, if H is a subgroup of D_j , for some j , then

$$\text{Inv}(H \setminus [D_j, D_j] : 2\text{-central in } D_j)$$

is the number of involutions in $H \setminus [D_j, D_j]$ which are centralized by a Sylow 2-subgroup of D_j . We also define

$$\text{Inv}(S) \text{ to be the total number of involutions in } S,$$

and for $S \subseteq J_4$, we denote by $\text{Inv}(S : 2A)$ and $\text{Inv}(S : 2B)$ the number of involutions in S in the conjugacy class $2A$ and $2B$ of J_4 , respectively, so that

$$\text{Inv}(S) = \text{Inv}(S : 2A) + \text{Inv}(S : 2B).$$

Remark 5.3. In some of our proofs we use Magma [5] to compute certain quantities in $T = J_4$. More precisely, we use the 112-dimensional representation for T over $\mathbb{F}_2$. The Magma code for this representation is available from the Web Atlas [41], and it conveniently includes constructions of the subgroups D_1 , D_2 , D_3 , and F_3 defined above.

Lemma 5.4. For $j = 1, 2, 3$, let D_j be as described above, let $E_j = O_2(D_j)$ and consider a subgroup H satisfying $E_j \leq H \leq D_j$. For each such subgroup let $\overline{H} = H/E_j$. Then the following assertions hold:

(1) If $j = 1$, then

$$\begin{aligned} \text{Inv}(H : 2A) &= 7.11.23 + 16 \text{Inv}(\overline{H} : 2\text{-central in } M_{24}) \\ &\quad + 32 \text{Inv}(\overline{H} : \text{non-2-central in } M_{24}), \quad \text{and} \\ \text{Inv}(H : 2B) &= 4.3.23 + 7.16 \text{Inv}(\overline{H} : 2\text{-central in } M_{24}) \\ &\quad + 32 \text{Inv}(\overline{H} : \text{non-2-central in } M_{24}). \end{aligned}$$

(2) For $j = 2$, and for $k \geq 1$, let $\mathcal{P}_k$ be the condition on an involution $\bar{h} \in \overline{H}$ that $|C_{E_2}(h)| = 2^k$ for some preimage $h \in D_2$ of $\bar{h}$, and set $H_0 := [D_2, D_2]$. Then

$$\begin{aligned} \text{Inv}(H : 2A) &= 1387 + 112 \text{Inv}(\overline{H}_0) + 64 \text{Inv}(\overline{H} \setminus \overline{H}_0 : \mathcal{P}_7), \quad \text{and} \\ \text{Inv}(H : 2B) &= 2772 + 208 \text{Inv}(\overline{H}_0) + 128 \text{Inv}(\overline{H} \setminus \overline{H}_0 : \mathcal{P}_6) \\ &\quad + 64 \text{Inv}(\overline{H} \setminus \overline{H}_0 : \mathcal{P}_7). \end{aligned}$$

(3) If $j = 3$, then

$$\begin{aligned} \text{Inv}(H : 2A) &= 5.31 + 64 \text{Inv}(\overline{H} : 2\text{-central in } L_5(2)) \\ &\quad + 16 \text{Inv}(\overline{H} : \text{non-2-central in } L_5(2)), \quad \text{and} \\ \text{Inv}(H : 2B) &= 4.7.31 + 64 \text{Inv}(\overline{H} : 2\text{-central in } L_5(2)) \\ &\quad + 48 \text{Inv}(\overline{H} : \text{non-2-central in } L_5(2)). \end{aligned}$$

Proof. For $i \in \{1, 2, 3\}$, and $X \in \{A, B\}$, let $[H : E_i]$ denote a set of representatives for the E_i -cosets in H . Then we have

$$\text{Inv}(H : 2X) = \sum_{h \in [H : E_i]} \text{Inv}(E_i h : 2X). \quad (5.1)$$

If h_1 and h_2 are conjugate in D_i , then $\text{Inv}(E_i h_1 : 2X)$ and $\text{Inv}(E_i h_2 : 2X)$ are equal. If $h \in D_i \setminus E_i$ and $E_i h$ contains an involution, then we may assume that h is an involution, and also in this case $\bar{h}$ is an involution in $\overline{D}_i$. On the other hand, if $i = 1$ or 3 , then $D_i = E_i \rtimes F_i$, and in these cases if $\bar{h}$ is an involution then we may assume that $h \in F_i$ and hence h is also an involution. If $i = 2$, then $D_i = E_i F_i$ with $E_i \cap F_i = Z(E_i) = \langle z \rangle$ of order 2. In this case, if $E_2 h$ is such that $\bar{h}$ is an involution,

then we have checked using Magma (see Remark 5.3) that there is an involution $x \in D_2$ such that $E_2h = E_2x$. Thus, for all i , a nontrivial coset E_ih contains an involution if and only if $\bar{h}$ is an involution. We use these facts frequently throughout the proof. We now prove parts (1)–(3) separately.

(1) Here, we are counting involutions in $D_1 = 2^{11} : M_{24}$ with $E_1 \cong 2^{11}$. By (5.1), we just need to compute $\text{Inv}(E_1h)$ for each $h \in [H : E_2]$. We first deal with the case $h = 1$. Here, $\text{Inv}(E_1 : 2A) = 7.11.23$ and $\text{Inv}(E_1 : 2B) = 4.3.23$, by [20, Proposition 9] (see also Remark 5.1). Suppose now that $1 \neq h \in [H : E_1]$. As mentioned above, we may assume that h is an involution and also that $\bar{h}$ is an involution. Since E_1 is an elementary abelian 2-group, the involutions in E_1h are precisely the elements of $C_{E_1}(\bar{h})h$. Thus, $\text{Inv}(E_1h) = |C_{E_1}(\bar{h})|$. We now use [20, Proposition 18]: if $\bar{h}$ is 2-central in $\overline{D_1} \cong M_{24}$, then $|C_{E_1}(\bar{h})| = 2^7$, and 2^4 of the 2^7 involutions in E_1h are of type 2A, with the other 7.2^4 of type 2B. If $\bar{h}$ is not 2-central in M_{24} , then $|C_{E_1}(\bar{h})| = 2^6$, and E_1h contains 2^5 involutions of type 2A and 2^5 of type 2B. The stated formulae in (1) then follow from (5.1).

(2) Here, we are counting involutions in $D_2 = 2^{1+12}.3.M_{22} : 2$ with $E_2 \cong 2^{1+12}$. As mentioned above, for each coset E_2h in H we may assume that $h \in F_2 = 6.M_{22} : 2$, where $F_2 \cap E_2 = \langle z \rangle$. By Remark 5.1(2), there are two D_2 -conjugacy classes of involutions in $E_2 \setminus \langle z \rangle$, with representatives e and f , in 2A and 2B, respectively, and from the information in [20, Proposition 1 (from line-13)] about $|C_{D_2}(e)|$, $|C_{D_2}(f)|$, we compute their class sizes as $|e^{D_2}| = 1386$ and $|f^{D_2}| = 2772$. Thus, we have the following, which will also be important for part (3):

$$\begin{aligned} \text{The group } E_2 \text{ contains } 4159 \text{ involutions, with } 1 + 1386 = 1387 \\ \text{of type 2A and } 2772 \text{ of type 2B.} \end{aligned} \quad (5.2)$$

Next, we count the involutions in $H_0 \setminus E_2$, where $H_0 = [D_2, D_2]$. By Remark 5.1(2), there are four D_2 -conjugacy classes of involutions in $H_0 \setminus E_2$ with representatives t', t_1 (in class 2A) and $t'z, t_2$ (in class 2B), and all four of these elements lie in the same coset E_2t' . Thus, by our remarks above, the only nontrivial E_2 -cosets containing involutions are E_2h with h conjugate in D_2 to t' . By [20, p. 569, line-3], E_2t' contains 16 involutions conjugate to t' and 16 conjugate to $t'z$, and by [20, p. 570, line-11, line-10], E_2t' contains $2^5.3$ involutions conjugate to t_1 and $2^6.3$ conjugate to t_2 . Hence, for each involution $\bar{h}$ in $\overline{H_0}$,

$$\begin{aligned} E_2h \text{ contains } 2^6.5 \text{ involutions, with } 2^4 + 2^5.3 = 112 \text{ of type 2A, and} \\ 2^4 + 2^6.3 = 208 \text{ of type 2B.} \end{aligned} \quad (5.3)$$

Finally, we count the involutions in $H \setminus H_0$. Again, as above, the only E_2 -cosets in $H \setminus H_0$ containing involutions are those of the form E_2h for an involution $h \in H \setminus H_0$. By Remark 5.1(2), there are three D_2 -conjugacy classes of involutions in $H \setminus H_0$ with representatives $\tilde{t}_1$ (in class 2A) and $\tilde{t}_1z, \tilde{t}_2$ (in class 2B). By [20, Proposition 6], $\tilde{t}_1$ and $\tilde{t}_1z$ have property $\mathcal{P}_7$ while $\tilde{t}_2$ has property $\mathcal{P}_6$. Let $s \in \{\tilde{t}_1, \tilde{t}_1z, \tilde{t}_2\}$, so s has property $\mathcal{P}_k$ with $k = 6$ or 7 . An element $xs \in E_2s$ is an involution if and only if

$x^s = x^{-1}$ and this, in turn, is equivalent to either (i) $x \in C_{E_2}(s)$ with $x^2 = 1$, or (ii) $|x| = 4$ and s inverts x . Now, for each $x \in E_2$ of order 4, we have $x^{-1} = zx$. We claim that, for $s = \tilde{t}_1$, there are no possibilities in case (ii). Assume to the contrary that x is such an element. Then

$$\tilde{t}_1^x = x^{-1}\tilde{t}_1x = zx\tilde{t}_1x = z\tilde{t}_1x^{\tilde{t}_1}x = z\tilde{t}_1,$$

contradicting the fact that $\tilde{t}_1$ and $\tilde{t}_1z = z\tilde{t}_1$ are not conjugate. Thus, for $s = \tilde{t}_1$, the involutions in $E_2\tilde{t}_1$ are precisely the elements $x\tilde{t}_1$, where $x \in C_{E_2}(\tilde{t}_1)$ and $x^2 = 1$. Since $C_{E_2}(\tilde{t}_1)$ is elementary abelian of order 2^7 by [20, Proposition 6], we deduce that $E_2\tilde{t}_1$ contains 2^7 involutions. Since $D_2 = C_T(z)$ and $C_{E_2}(\tilde{t}_1)$ contains both $\tilde{t}_1$ and $\tilde{t}_1z$, exactly half (that is, 2^6) of these involutions are D_2 -conjugate to $\tilde{t}_1$, and the other 2^6 involutions are conjugate to $\tilde{t}_1z$.

Finally consider $s = \tilde{t}_2$. The element $\tilde{t}_2$ inverts 2^6 elements of E_2 of order 4 (this can be checked using Magma [5], see Remark 5.3). In particular, there exists $x \in E_2$ of order 4 such that $x^{\tilde{t}_2} = xz$. Also, $C_{E_2}(\tilde{t}_2)$ is elementary abelian of order 2^6 by [20, Propositions 6 and 10]. Thus, there are $2^6 + 2^6 = 2^7$ involutions in $E_2\tilde{t}_2$, consisting of the elements of the form $x\tilde{t}_2$, where either $x \in C_{E_2}(\tilde{t}_2)$, or $|x| = 4$ and $x^{\tilde{t}_2} = xz$. If $x \in C_{E_2}(\tilde{t}_2)$, then $C_{E_2}(x\tilde{t}_2) = C_{E_2}(\tilde{t}_2) \cong 2^6$, so $x\tilde{t}_2$ cannot be D_2 -conjugate to $\tilde{t}_1$ or $\tilde{t}_1z$ (since these elements have centralizers of order 2^7 in E). If $|x| = 4$ and $x^{\tilde{t}_2} = x^{-1}$, then $x\tilde{t}_2$ inverts the element x of order 4, so again $x\tilde{t}_2$ cannot be D_2 -conjugate to $\tilde{t}_1$ or $\tilde{t}_1z$. Hence all of the 2^7 involutions in $E_2\tilde{t}_2$ are D_2 -conjugate to $\tilde{t}_2$.

It follows from the last two paragraphs above that if $h \in D_2 \setminus H_0$ and h is an involution, then

- $\text{Inv}(Eh : 2A) = \text{Inv}(Eh : 2B) = 2^6$ if $|C_{E_2}(h)| = 2^7$; and
- $\text{Inv}(Eh : 2A) = 0$ and $\text{Inv}(Eh : 2B) = 2^7$ if $|C_{E_2}(h)| = 2^6$.

Putting all these counts together and applying (5.1) yields part (2).

(3) Here, we are counting involutions in $D_3 = 2^{10} : L_5(2)$ (so $E_3 \cong 2^{10}$). Recall that $D_3 = E_3 \rtimes F_3$ with $F_3 \cong L_5(2)$. Thus by our discussion above, the E_3 -cosets in D_3 containing involutions are precisely the cosets E_3h with $h \in F_3$ such that $h^2 = 1$. We first consider the case $h = 1$. By [20, Proposition 20], we have

$$\text{Inv}(E_3 : 2A) = 5.31 \quad \text{and} \quad \text{Inv}(E_3 : 2B) = 4.7.31. \quad (5.4)$$

Assume now that $h \in F_3 \setminus \{1\}$ and h is an involution. As mentioned in Remark 5.1, the group E_3 is normalized by E_2 . Furthermore, $|E_2E_3| = 2^{16}$, and $N_{D_2}(E_3) = E_3Q$, where Q is a subgroup of $F_3 \cong L_5(2)$ of shape $2^6 : (L_3(2) \times \text{Sym}(3))$ (see [20, Sec. 5, fourth paragraph]).

Now, $F_3 \cong L_5(2)$ has two conjugacy classes of involutions: the 2-central involutions and the non-2-central involutions. Moreover, each such involution is F_3 -conjugate to an element of the normal subgroup $O_2(Q) \cong 2^6$ of $Q \leq N_{D_2}(E_3)$. Indeed, 21 of the $2^6 - 1$ involutions in $O_2(Q)$ are 2-central in F_3 , and 42 are

non-2-central. Note that $|E_2E_3 : E_2| = 2^3$, and $E_2E_3 \leq [D_2, D_2]$. Let $\{h_1 = 1, h_2, \dots, h_8\} \subseteq [D_2, D_2]$ be a set of representatives for the E_2 -cosets in E_2E_3 . Then, for $X \in \{A, B\}$,

$$\begin{aligned} \text{Inv}(E_2E_3 \setminus E_3 : 2X) &= \text{Inv}(E_2E_3 : 2X) - \text{Inv}(E_3 : 2X) \\ &= \left(\sum_{i=1}^8 \text{Inv}(E_2h_i : 2X) \right) - \text{Inv}(E_3 : 2X). \end{aligned}$$

It then follows from (5.2)–(5.4) that

$$\text{Inv}(E_2E_3 \setminus E_3 : 2A) = 1387 + 7.112 - 5.31 = 2016.$$

In exactly the same way, we calculate that $\text{Inv}(E_2E_3 \setminus E_3 : 2B) = 3360$. On the other hand, $E_2E_3 = E_3O_2(Q)$, so

$$\text{Inv}(E_2E_3 \setminus E_3) = \sum_{g \in O_2(Q) \setminus \{1\}} \text{Inv}(E_2g).$$

As mentioned above, 21 of the $2^6 - 1$ involutions in $O_2(Q) \cong 2^6$ are 2-central in F_3 , with 42 non-2-central. It follows that, for $x, y \in O_2(Q) \leq F_3 \cong L_5(2)$ with x a 2-central involution, and y a non-2-central involution,

$$\begin{aligned} 21 \text{Inv}(E_3x : 2A) + 42 \text{Inv}(E_3y : 2A) &= 2016, \quad \text{and} \\ 21 \text{Inv}(E_3x : 2B) + 42 \text{Inv}(E_3y : 2B) &= 3360. \end{aligned} \tag{5.5}$$

In particular, $\text{Inv}(E_2E_3 \setminus E_3) = 5376 = 21 \text{Inv}(E_3x) + 42 \text{Inv}(E_3y)$. Since $\text{Inv}(E_3x) = |C_{E_3}(x)|$ and $\text{Inv}(E_3y) = |C_{E_3}(y)|$ are powers of 2, it is an easy exercise to show that the only possibility is $\text{Inv}(E_3x) = 2^7$ and $\text{Inv}(E_3y) = 2^6$.

Finally, we use **Magma** [5] to compute the J_4 -conjugacy classes of involutions in which the involutions in D_3 lie (see Remark 5.3). We find that the group D_3 contains 134705 involutions in the class $2A$, and 343108 involutions in the class $2B$. Since $L_5(2)$ has 465 2-central involutions and 6510 non-2-central involutions, we deduce from (5.1) and (5.4) that

$$\begin{aligned} 465 \text{Inv}(E_3x : 2A) + 6510 \text{Inv}(E_3y : 2A) + 5.31 &= 134705, \quad \text{and} \\ 465 \text{Inv}(E_3x : 2B) + 6510 \text{Inv}(E_3y : 2B) + 4.7.31 &= 343108. \end{aligned} \tag{5.6}$$

The system of equations given by (5.5) and (5.6) can be solved to give $\text{Inv}(E_3x : 2A) = 64$, $\text{Inv}(E_3y : 2A) = 16$, $\text{Inv}(E_3x : 2B) = 64$, and $\text{Inv}(E_3y : 2B) = 48$. Part (3) then follows from (5.1). $\square$

Corollary 5.5. *Let D_1, D_2, D_3 be as described above, and let H be a subgroup of $T = J_4$. Then $T^{[T:H]}$ has base size 2 for each of the following subgroups H :*

- (1) $O_2(D_1) < H < D_1$, with $H/O_2(D_1)$ maximal in $D_1/O_2(D_1) \cong M_{24}$;
- (2) $O_2(D_2) < H < D_2$ such that one of the following holds:
 - (a) $H/O_2(D_2)$ is maximal in $D_2/O_2(D_2) \cong 3.M_{22} : 2$, and $H/O_2(D_2)$ is not of shape $3.L_3(4).2$ or $3.M_{22}$;

- (b) $H/O_2(D_2)$ is a maximal subgroup of $3.L_3(4).2 < 3.M_{22} : 2$, and $H/O_2(D_2) \neq 3.L_3(4)$;
- (c) $H/O_2(D_2)$ is a maximal subgroup of $3.M_{22} < 3.M_{22} : 2$ isomorphic to $3.Alt(7)$;
- (3) $O_2(D_3) < H < D_3$, with $H/O_2(D_3)$ maximal in $D_3/O_2(D_3) \cong L_5(2)$;
- (4) H conjugate to a maximal subgroup M_{24} of D_1 (in one of two D_1 -conjugacy classes); or the maximal $F_2 \cong 6.M_{22} : 2$ in D_2 ; or the maximal $F_3 \cong L_5(2)$ in D_3 ;
- (5) H maximal of shape $[2^7].3.L_3(4).2$ in the maximal subgroup of the form $2^{1+12}.3.L_3(4).2$ of D_2 .

Proof. Let H be one of the subgroups in the statement. We will show that

$$\widehat{Q}(T, H, 2) := \sum_{i=1}^m \frac{|x_i^T \cap H|^2}{|x_i^T|} \text{ is less than } 1, \quad (5.7)$$

where $x_1, \dots, x_m$ are representatives for the conjugacy classes of elements of T of prime order. By Proposition 4.2, this suffices to show that $T^{[T:H]}$ has base size 2. Let $\mathcal{X} = \{x_1, \dots, x_m\}$.

To prove (5.7), we compute upper bounds for the quantities $\frac{|x^T \cap H|^2}{|x^T|}$ for each $x \in \mathcal{X}$. For any fixed prime p , the following crude bound is available:

$$\sum_{x \in \mathcal{X}, |x|=p} \frac{|x^T \cap H|^2}{|x^T|} \leq \frac{|\{y \in H : |y| = p\}|^2}{\min\{|x^T| : x \in \mathcal{X}, |x| = p\}}. \quad (5.8)$$

The right-hand side above can be found, for each H and each prime p , using Magma (see Remark 5.3). The bounds at (5.8) in fact suffice to prove (5.7) for all subgroups H in parts (4) and (5) of the corollary.

In parts (1)–(3), however, we need to be more careful. We can use Magma [5] (see Remark 5.3) and Lemma 5.4, to compute the quantities

$$\frac{|z^T \cap H|^2}{|z^T|} \quad \text{and} \quad \frac{|f^T \cap H|^2}{|f^T|}$$

precisely for each subgroup H , (recalling that $z \in 2A$ and $f \in 2B$). Putting these together with the bounds for odd prime orders coming from (5.8) above then yields $\widehat{Q}(T, H, 2) < 1$ in each case, as required. $\square$

We now have the tools needed to prove Lemma 4.1 for J_4 . Recall the notion of the depth $\ell_G(H)$ of a subgroup H of a group G introduced just before Strategy 4.5.

Proposition 5.6. *Let H be a proper subgroup of $T = J_4$.*

- (1) *If $T^{[T:H]}$ has base size at least 3, then one of the following holds:*
 - (a) $\ell_T(H) = 1$ and H is T -conjugate to one of D_1 , D_2 , or D_3 ;

- (b) $\ell_T(H) = 2$ and either H is T -conjugate to $[D_2, D_2] \cong 2^{1+12}.3.M_{22}$; or H is T -conjugate to a maximal subgroup of D_2 containing $O_2(D_2)$ of shape $2^{1+12}.3.L_3(4) : 2$.
- (c) $\ell_T(H) = 3$ and H is T -conjugate to a second maximal subgroup of D_2 containing $O_2(D_2)$ of shape $2^{1+12}.3.L_3(4)$.
- (2) If $\ell_T(H) \geq 4$, then H is properly contained in a subgroup K of T such that $T^{[T:K]}$ has base size 2.
- (3) Proposition 4.1, and in particular Table 1 holds for $T = J_4$.

Since it is not necessary for our purposes, we have not attempted to resolve whether or not the permutation group $T^{[T:H]}$ has base size 2 in the cases where H has shape $2^{1+12}.3.L_3(4) : 2$ (in Part (1)(b)), or $2^{1+12}.3.L_3(4)$ (in Part (1)(c)).

Proof. We first prove part (1) for subgroups H with $\ell(H) \leq 3$. Let H be a proper subgroup of T such that $T^{[T:H]}$ has base size larger than 2. Then by [9], and as noted at the beginning of the section, H is conjugate to a subgroup of D_1 , D_2 , or D_3 . Thus if $\ell_T(H) = 1$ then Part (1)(a) holds. Therefore we may assume both that $\ell_T(H) \geq 2$, and that $H < D_i$ for some $i = 1, 2, 3$. Recall that, if $H < K < T$ then also $T^{[T:K]}$ has base size larger than 2. Also $D_i = E_i F_i$, where $E_i = O_2(D_i)$, and $E_i \cap F_i$ is trivial with E_i elementary abelian and F_i acting irreducibly on it, if $i = 1$ or 3, and $E_2 \cap F_2 = Z(E_2) = \langle z \rangle$ of order 2 with F_2 irreducible on $E_2/Z(E_2)$, (see Remark 5.1). Also using Magma [5], we find that there are two D_1 -conjugacy classes of complements to E_1 , with representatives F_1 and F_{11} , and we obtain, for each i , a complete set $\mathcal{M}(D_i)$ of representatives for the D_i -conjugacy classes of maximal subgroups of D_i , as

$$\mathcal{M}(D_1) = \{F_1, F_{11}\} \cup \{E_1 I : I < F_1 \text{ maximal}\} \cup \{E_1 I : I < F_{11} \text{ maximal}\},$$

$$\mathcal{M}(D_2) = \{F_2\} \cup \{E_2 I : I < F_2 \text{ maximal}\},$$

$$\mathcal{M}(D_3) = \{F_3\} \cup \{E_3 I : I < F_3 \text{ maximal}\}.$$

Assume now that $\ell_T(H) = 2$ so that $H \in \mathcal{M}(D_i)$ for some i . It then follows from Corollary 5.5 that $i = 2$ and, up to T -conjugacy, $H = E_2 I$ with $I/\langle z \rangle$ maximal in $F_2/\langle z \rangle \cong 3.M_{22}.2$ of shape $3.L_3(4).2$ or $3.M_{22}$. This proves (1)(b).

Suppose next that $\ell_T(H) = 3$. Then by part (1)(b), H is T -conjugate to a maximal subgroup of either $[D_2, D_2]$ or a subgroup $K = E_2 I$ with $I \cong 3.L_3(4).2 < 3.M_{22}.2$. Consider first the latter case. Using Magma, we see that $E_2/Z(E_2)$ has composition length 2 as an $\mathbb{F}_2[I]$ -module, with both composition factors of dimension 6. Let $E_0/Z(E_2)$ be an I -submodule of $E_2/Z(E_2)$ of dimension 6. Then using Magma again, we see that the set

$$\{E_0 I\} \cup \{E_2 I_1 : I_1 < I \text{ maximal}\} \quad (5.9)$$

is a complete set of representatives for the K -conjugacy classes of maximal subgroups of K . The subgroup $E_0 I$ has shape $[2^7].3.L_3(4).2$, and by Corollary 5.5(5),

$T^{[T:E_0I]}$ has base size 2. Thus, H is K -conjugate to E_2I_1 for some maximal subgroup I_1 of I . By Corollary 5.5(2)(b), it follows that $I_1 \cong 3.L_3(4)$, as in Part (1)(c). Suppose now that H is maximal in $[D_2, D_2]$. By using Magma, we find that the set

$$\{[F_2, F_2] \cong 3.M_{22}\} \cup \{E_2J : J < [F_2, F_2] \text{ maximal}\} \quad (5.10)$$

is a complete set of representatives for the $[D_2, D_2]$ -conjugacy classes of maximal subgroups of $[D_2, D_2]$. Since, by Corollary 5.5(4), $T^{[T:F_2]}$ has base size 2, it follows that also $T^{[T:L]}$ has base size 2, where $L = [F_2, F_2] < F_2$. Thus H is not conjugate to $[F_2, F_2]L$, and we may assume that $H = E_2J$ for some maximal subgroup J of $[F_2, F_2] \cong 3.M_{22}$. If $J < M < 3.M_{22} : 2$ with M maximal and $M \not\cong 3.L_3(4).2$ or $3.M_{22}$, then we would have $T^{[T:H]}$ of base size 2 on applying Corollary 5.5(2)(a). Thus this is not the case, and, by using the Web Atlas, we see that the only possibilities for J are the subgroups of shape $3.Alt(7)$ or $3.L_3(4)$. By Corollary 5.5(2)(c), it follows that $H \cong E_2J$, where $J = 3.L_3(4)$. This proves completes the proof of Part (1) when $\ell(H) \leq 3$.

Next, we complete the proof of (1) in the case of subgroups H with $\ell(H) \geq 4$, and prove (2). Note that, if (2) holds then, for any subgroup H with $\ell_T(H) \geq 4$, the base size of $T^{[T:H]}$ is at most 2, and hence Part (1) lists all possibilities for H where the base size of $T^{[T:H]}$ could possibly be greater than 2. So assume that H is a subgroup of T , $\ell_T(H) \geq 4$, and H is not properly contained in a subgroup K of T with $T^{[T:K]}$ having base size 2. Then, by our arguments up to this point, H is T -conjugate to a subgroup of $K_1 := E_2I_1$, where $I_1 \cong 3.L_3(4)$. Since $\ell_T(H) \geq 4$, we may assume that $H < K_1 = E_2I_1$. Using Magma, we find that

$$\{E_0I_1\} \cup \{E_2I_2 : I_2 < I_1 \text{ maximal}\}$$

is a complete set of representatives for the K_1 -conjugacy classes of maximal subgroups of K_1 , where E_0 is as defined in the previous paragraph. Consider $M = E_2I_2$, for some maximal subgroup I_2 of I_1 . We can check using Magma that, for every subgroup $I_2 < I_1 < [F_2, F_2] \cong 3.M_{22}$, I_2 is contained in either a maximal subgroup of F_2 which is not isomorphic to $3.M_{22}$ or $3.L_3(4).2$; or a maximal $3.Alt(7)$ subgroup of $3.M_{22}$. Thus, $T^{[T:M]}$ has base size 2 by Corollary 5.5, and hence H is not contained in any such subgroup M . Thus, we may assume that $H \leq E_0I_1$ and H projects onto I_1 . Now E_0I_1 is an index 2 subgroup of E_0I , with $I = 3.L_3(4).2$ as above, and by Corollary 5.5(5), the permutation group $T^{[T:E_0I]}$ has base size 2. Hence also $T^{[T:J]}$ has base size 2, for all subgroups J of E_0I_1 . This is a contradiction to our assumption that $T^{[T:H]}$ has base size larger than 2, and hence Part (2) is proved.

Finally, we complete the proof of Part (3). If $M = D_1 \cong 2^{11} : M_{24}$, then the subdegrees of $T^{[T:M]}$ are given in [19, Lemma 2.23.1]. Thus, we can compute $d(T, M)$ precisely (the greatest common divisor of the orders of the stabilizers in $M^{[T:M]}$ of points in $[T : M] \setminus \{M\}$): we get $d(T, M) = 24$. The first entry in the final column of the third row of Table 1 follows.

Next, suppose that $M = D_3 = E_3 F_3 = 2^{10} : L_5(2)$. As noted above, the maximal subgroups of M are T -conjugate to either F_3 , or $E_3 I$, where I is a maximal subgroup of F_3 . Now, since $(|M|, |T : M| - 1) = 1$, there exists, for each prime p dividing $|M|$, a point stabilizer $M_{(p)} := M \cap M^{t_p}$ in $M^{[T:M]}$ containing a Sylow p -subgroup of M . If L is a maximal subgroup of M containing a Sylow 31-subgroup of M , then either L is T -conjugate to F_3 , or L contains E_3 and $|L/E_3| = 5.31$ (one can see this by inspection of the maximal subgroups of $L_5(2) \cong M/O_2(M)$). Thus, $|M_{(31)}|$ divides $2^{10}.3^2.5.7.31$. Similarly, $M_{(2)}$ has order dividing $2^{20}.3^2.5.7$. Thus, the greatest common divisor of the orders of the point stabilizers in $M^{[M:T]}$ divides $2^{10}.3^2.5.7$. On the other hand, we can compute the rank of $T^{[T:M]}$ using the permutation character and GAP [35] – it is 27. Thus, at least one of the M -orbits in $[T : M]$ must be of size greater than or equal to $(|T : M| - 1)/26$. It follows that at least one point stabilizer in M has order less than $26|M|/(|T : M| - 1) < 32$. Thus, $d(T, M)$ divides $2^{10}.3^2.5.7$ and is less than 32. It follows that $d(T, M)$ divides $2^4.3^2.5.7$ and $d(T, M) \leq 30$, as required.

To prove Part (3), suppose that

$$M \in \{D_2, [D_2, D_2], 2^{1+12}.3.L_3(4) : 2, 2^{1+12}.3.L_3(4)\}.$$

Then M is T -conjugate to a subgroup of D_2 . Thus, $d(T, M)$ divides $d(T, D_2)$. The subdegrees of $T^{[T:D_2]}$ are given in [6, Table 2], and the number $d(T, D_2)$ can then be computed immediately: we get $d(T, D_2) = 2$. The remaining entries in the third column of the J_4 -row of Table 1 follow. In the case where $M = D_2$, the base size of $T^{[T:M]}$ is 3, by [9, Table 2]. This action is equivalent to the conjugation action of T on the class of 2A-involutions of T . The orders of the 2-point stabilizers in this action have been computed by Bray *et al.* in [6] and are listed in the column headed s_2 in [6, Table 2]. The smallest order is 16. If now $H = [D_2, D_2]$, of index 2 in M , then each 2-point stabilizer $H \cap H^x$ in the permutation group $T^{[T:H]}$ has index at most 4 in $M \cap M^x$, and since $|M \cap M^x| \geq 16$, it follows that $H \cap H^x \neq 1$. Thus $T^{[T:H]}$ also has base size greater than 2. $\square$

We remark that by proving Proposition 5.6(3), we have completed the proof of Proposition 4.1. As final preparation for proving that $T = J_4$ is totally 2-closed, we deal separately with a specific coset action.

Proposition 5.7. *Let $T = J_4$ and $D_1 = E_1 \rtimes F_1 = 2^{11} : M_{24}$ as above, and consider $H = E_1 \rtimes M_{23} < D_1$ (so $\ell_T(H) = 2$). Then $T^{[T:H]}$ is 2-closed.*

Proof. We use the notation in Notation 3.3, with $G = T$ and $M = D_1$. So $\Omega = [T : H]$, $\Sigma = [T : M]$ with $s = |\Sigma|$, $L = T^\Sigma$, $M = T_\Delta$ (with $\Delta \in \Sigma$), $H = T_\omega$ (with $\omega \in \Delta$), etc. By Lemma 4.3(b)(i), $L^{(2),\Sigma} = L$, since $T = \text{Aut}(T)$ and T has no 2-transitive representations. We also set $X = T^{(2),\Omega}$ and $Y = X_\Delta^\Delta \leq M^{(2),\Delta}$. The group $R = M^\Delta \cong M_{24}$ is 2-transitive of degree 24. Hence, we have $M_{24} \cong R = M^\Delta \leq Y \leq M^{(2),\Delta} \cong \text{Sym}(24)$. The only overgroups of M_{24} in $\text{Sym}(24)$ are M_{24} , $\text{Alt}(24)$, and $\text{Sym}(24)$. Thus, $Y \in \{M_{24}, \text{Alt}(24), \text{Sym}(24)\}$.

Assume that $T^{[T:H]}$ is not 2-closed. Then, by Lemma 3.5, $X = N \rtimes T$ with N a nontrivial subdirect subgroup of B_A , for some nontrivial normal subgroup A of Y . Hence, $A \in \{M_{24}, \text{Alt}(24), \text{Sym}(24)\}$.

We now examine a part $\Delta_2 \in \Sigma$ such that $M \cap M_2 = L_2(23)$ (the groups $M \cap M_j$ are listed in [18, Lemma 2.23.1]). Suppose that the subdirect subgroup $\rho_{1,2}(N) \leq A \times A$ is not a full diagonal subgroup of $A \times A$. Then since A is almost simple, $\rho_{1,2}(N)$ must contain $\text{soc}(A) \times \text{soc}(A)$. Since $\text{soc}(A)$ acts transitively on Δ , it follows that $\rho_{1,2}(N)^{\Delta \times \Delta_2}$ is transitive. On the other hand, by Lemma 3.5(b), the group $\rho_{1,2}(N)^{\Delta \times \Delta_2}$ leaves invariant each $(M \cap M_2)^{\Delta \times \Delta_2}$ -orbit. Since $M \cap M_2 \cong L_2(23)$ cannot act transitively on a set of size 24^2 , we have a contradiction. Thus $\rho_{1,2}(N) \leq A \times A$ is a full diagonal subgroup of $A \times A$. It then follows from Theorem 3.8(b) that N is a full diagonal subgroup of B_A . In particular, since $X \cong N \rtimes T$ with $N \cong A \in \{M_{24}, \text{Alt}(24), \text{Sym}(24)\}$ and $T \cong J_4$, we have $X = N \times T$.

We examine further the part $\Delta_2 \in \Sigma$ with $M \cap M_2 = L_2(23)$. We claim that

$$C_T(M \cap M_2) \text{ has even order.} \quad (5.11)$$

To prove this, consider the setwise stabilizer $T_{\{\Delta \cup \Delta_2\}}$ of $\Delta \cup \Delta_2$ in T . Note first that the subdegrees of $T^{[T:M]}$ are pairwise distinct (see [18, Lemma 2.23.1]), so each T -orbit in $\Sigma^{(2)} := \{(\Delta', \Delta'') : \Delta', \Delta'' \in \Sigma, \Delta' \neq \Delta''\}$ is self-paired. It follows that $T_{\{\Delta \cup \Delta_2\}}$ contains an element, say g , which interchanges Δ and Δ_2 . Moreover, $T_{\{\Delta \cup \Delta_2\}}$ contains $M \cap M_2 \cong L_2(23)$. Since $M \cap M_2 \cap O_2(M) = 1$, and the unique conjugacy class of subgroups of M_{24} isomorphic to $L_2(23)$ acts transitively on the cosets of M_{23} in M_{24} , the group $(M \cap M_2)^\Delta$ is transitive. It follows that $T_{\{\Delta \cup \Delta_2\}}^{\Delta \cup \Delta_2}$ is transitive, and that the kernel of the action of $T_{\{\Delta \cup \Delta_2\}}$ on the block system $\{\Delta, \Delta_2\}$ is the group $M \cap M_2 \cong L_2(23)$.

Now, $(M \cap M_2)^{\Delta \cup \Delta_2} = \rho_{1,2}(M \cap M_2)^{\Delta \cup \Delta_2}$, by the definition of $\rho_{1,2}$. Since $M \cap M_2 < X$ and X normalizes N , we have that $\rho_{1,2}(M \cap M_2)^{\Delta \cup \Delta_2}$ normalizes $\rho_{1,2}(N)^{\Delta \cup \Delta_2}$. As shown above, $\rho_{1,2}(N)^{\Delta \cup \Delta_2}$ is a full diagonal subgroup of $A \times A \leq \text{Sym}(24) \times \text{Sym}(24)$, that is to say, for some $\theta \in \text{Aut}(A)$ we have $\rho_{1,2}(N)^{\Delta \cup \Delta_2} = \{(a, a^\theta) : a \in A\}$. Now $A \leq \text{Sym}(24)$ is almost simple and primitive, so it has trivial centralizer in $\text{Sym}(24)$, and $N_{\text{Sym}(24)}(A) = \text{Aut}(A) \cong \text{PGL}_2(23), \text{Sym}(24), \text{Sym}(24)$ for $A = L_2(23), \text{Alt}(24), \text{Sym}(24)$, respectively. Hence the normalizer of $\rho_{1,2}(N)^{\Delta \cup \Delta_2}$ in $(\text{Sym}(\Delta \cup \Delta_2))_{\Delta, \Delta_2}$ is $\{(b, b^\theta) : b \in \text{Aut}(A)\}$. It follows that $\rho_{1,2}(M \cap M_2)^{\Delta \cup \Delta_2} = \{(d, d^\theta) : d \in D\}$, for some subgroup D of $\text{Sym}(24)$ with $D \cong L_2(23)$.

Next, consider the natural epimorphism $\mu : X_{\{\Delta \cup \Delta_2\}}^{\Delta \cup \Delta_2} \rightarrow X_{\{\Delta \cup \Delta_2\}}^{\Delta \cup \Delta_2} \leq \text{Sym}(24) \wr 2$. Since the element g (which interchanges Δ and Δ_2) centralizes N , its image $\mu(g)$ centralizes $\mu(N) = \rho_{1,2}(N)^{\Delta \cup \Delta_2} = \{(a, a^\theta) : a \in A\}$. Thus, writing $\mu(g) = (y_1, y_2)\sigma$ (where $\sigma \in \text{Sym}(2) \setminus \{1\}$ and $y_i \in \text{Sym}(24)$), we have

$$(a, a^\theta) = (a, a^\theta)^{\mu(g)} = (a^{\theta y_2}, a^{y_1}) \quad \text{for all } a \in A.$$

It follows that $a^{\theta y_2} = a = a^{\theta y_1^{-1}}$, for all $a \in A$. Thus, since $C_{\text{Sym}(24)}(A) = 1$, we have $y_1 = y_2^{-1} = \theta$. Hence, for each $(d, d^\theta) \in \rho_{1,2}(M \cap M_2)^{\Delta \cup \Delta_2} = \mu(M \cap M_2)$,

$$(d, d^\theta)^{\mu(g)} = (d^{\theta y_2}, d^{y_1}) = (d, d^\theta),$$

and therefore $\mu(g)$ centralizes $\mu(M \cap M_2)$. Since $M \cap M_2$ is simple and $\mu(M \cap M_2) \neq 1$, the kernel of the restriction of μ to $M \cap M_2$ is trivial. Hence, g centralizes $M \cap M_2$, and since g has even order, claim (5.11) follows.

This leads to a contradiction, since $M \cap M_2$ contains elements of order 23, and the unique T -conjugacy class of such elements has centralizer of order 23 (see the Web Atlas [41]). Thus, $T^{[T:H]}$ is 2-closed. $\square$

We now prove that $T = J_4$ is totally 2-closed.

Proposition 5.8. *The Janko group J_4 is totally 2-closed.*

Proof. We use the notation fixed at the beginning of the section, so that $T = J_4$, $D_1 \cong 2^{11} : M_{24}$, etc. Our task is to prove, using Strategy 4.5, that $T^\Omega = T^{(2),\Omega}$, where $\Omega = [T : H]$, for each proper nontrivial subgroup H of T . Since $T = \text{Aut}(T)$ and T has no 2-transitive representations, it follows from Lemma 4.3(b)(i) that this holds for all maximal subgroups H . Thus, we proceed to stage (2) of Step 1 in Strategy 4.5: constructing the set $\mathcal{S}_2$. By [9, Table 2], there are precisely three conjugacy classes of maximal subgroups M of T such that $T^{[T:M]}$ has base size at least 3, with representatives $D_1 \cong 2^{11} : M_{24}$, $D_2 \cong 2^{1+12}.3 : M_{22}.2$, and $D_3 = 2^{10} : L_5(2)$ as defined at the beginning of the section. Therefore according to Strategy 4.5, the set $\mathcal{S}_2$ consists of all non-maximal subgroups H such that the only maximal subgroups of T containing H are conjugate to D_1 , D_2 , or D_3 .

We now move on to Strategy 4.5, Step 2, stage (1). We need to prove that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_2$ with $\ell_T(H) = 2$. Let $H \in \mathcal{S}_2$, and assume that $T^{[T:H]} \neq T^{(2),[T:H]}$. Let M be a maximal subgroup of T containing H so that, without loss of generality, $M = D_i$, for some $i = 1, 2, 3$ and H is maximal in M . We use the rest of the notation in Notation 3.3, with $T = G$. So $\Omega = [T : H]$, $\Sigma = [T : M]$, $L = T^\Sigma$, $M = T_\Delta$ (with $\Delta \in \Sigma$), $H = T_\omega$ (with $\omega \in \Delta$), etc. Note that $L^{(2),\Sigma} = L$, by Step 1. We also set $X = T^{(2),\Omega}$, so $X > T$, and $Y = X_\Delta^\Delta \leq M^{(2),\Delta}$.

Suppose first that $i = 1$ or 3 . Then $M = E_i \rtimes F_i$, where E_i is elementary abelian, and F_i is a simple group acting faithfully and irreducibly on E_i . Then, as mentioned in the proof of Proposition 5.6(1)(b), since H is maximal in M , either $H \cong F_i$ or $H = E_i I$, where I is a maximal subgroup of F_i . It follows that either $M^{[M:H]} \cong E_i F_i$ is a primitive group of affine type, of degree $|E_i|$; or $M^{[M:H]} \cong F_i$ is a primitive almost simple group, of degree $|F_i : I|$. Since $T^{[T:H]}$ is not 2-closed, it follows that the normal subgroup A of $Y \leq M^{(2),\Delta}$ from Theorem 3.10(2) is transitive. On the other hand, Theorem 3.10(2) and Proposition 4.1 imply that the size of an A -orbit in Δ divides $d(T, M)$, with $d(T, M)$ as in Proposition 4.1(b),

that is to say, $|M : H| = |\Delta|$ divides $d(T, M)$. Now, by Proposition 4.1(b), $d(T, M)$ divides $g(T, M)$, with $g(T, M)$ as in Table 1. In both cases $g(T, M) < |E_i|$, so $M^{[M:H]} \cong F_i$ is nonabelian simple and primitive. If $i = 1$ then $g(T, M) = 24$ and it follows that $|M : H| = 24$ and $I \cong M_{23}$ maximal in $F_1 \cong M_{24}$; however, by Proposition 5.7, $T^{[T:H]}$ is 2-closed in this case, which is a contradiction. Thus, $i = 3$, and by Proposition 4.1(b), $d(T, M) \leq 30$. This means that $|M : H| = |F_3 : I| \leq 30$, which is not possible for any maximal subgroup I of $F_3 \cong L_5(2)$. Thus $T^{[T:H]}$ is 2-closed for all maximal subgroups H of $M \in \{D_1, D_3\}$.

Next, assume that H is maximal in $M = D_2$. Then, as mentioned in the proof of Proposition 5.5, either H is M -conjugate to a supplement $F_2 \cong 6.M_{22} : 2$ for $E_2 = O_2(M)$ in M , or $H = E_2 I$, for some $I < F_2$ maximal. Since $T^{[T:H]}$ is not 2-closed, it follows that the normal subgroup A of the primitive permutation group $Y \leq M^{(2), \Delta}$, from Theorem 3.10(2), is transitive. On the other hand, the size of an A -orbit divides 2, by Proposition 4.1(b). Hence, we must have $|M : H| = |\Delta| = 2$, and it follows that H is the maximal subgroup $[M, M] \cong 2^{1+12}.3.M_{22}$. By Theorem 3.10(1), for each $\Delta_2 \in \Sigma \setminus \{\Delta\}$, the stabilizer $M_2 = T_{\Delta_2}$ satisfies $M = (M \cap M_2)H$. We shall obtain a contradiction by proving that

$$\text{for some } \Delta_2 \in \Sigma \setminus \{\Delta\}, \text{ the stabilizer } M \cap M_2 = T_{\Delta, \Delta_2} \text{ is contained in } H. \quad (5.12)$$

We prove the claim using Magma and the 112-dimensional representation of T over $\mathbb{F}_2$. In particular, we use the generating 112-dimensional matrices a and b of T given in the Web Atlas [41], of orders 2 and 4, respectively. Using this notation, we have that M is T -conjugate to $C_T(a)$, so we may assume that $M = C_T(a)$. Define

$$t := (ab^2)^4; \quad c := ab; \quad y := t^{c^2}.$$

Using the excellent computations done in [6, Sec. 2], we have that

$$M = \langle [a, b]^5 (ab^2)^6, ba(b^2)ab[a, ba(b^2)ab]^5 ababab[a, ababab]^5 \rangle.$$

Then the action of T on the set of T -cosets of M is permutation isomorphic to the action of T on the set of T -conjugates of a . Thus, the 2-point stabilizers $M \cap M_2$ in $T^{[T:M]}$ are the groups $C_T(a) \cap C_T(a^g)$, for $g \in T$. Consider the 2-point stabilizer $C_M(a^y) = C_T(a) \cap C_T(a^y)$. One can quickly check using Magma [5] that $(aa^y)^6$ is an element of M of order 2, and that $C_M((aa^y)^6)$ is contained in $[M, M] = H \cong 2^{1+12}.3.M_{22}$. Hence, since $C_M(a^y) \leq C_M((aa^y)^6)$, we have that $C_T(a) \cap C_T(a^y) = C_M(a^y) \leq [M, M] = H$. Thus, (5.12) is proved, yielding a contradiction. We conclude that $T^{[T:H]}$ is 2-closed for all maximal subgroups H of $M = D_2$, and hence that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_2$ with $\ell_T(H) = 2$.

This completes stage (1) of Step 2 of Strategy 4.5. For stage (2) of Step 2, we construct the set $\mathcal{S}_3$. By Proposition 5.6, we remove from $\mathcal{S}_2$ all (conjugates of) proper subgroups of D_1 , D_2 , and D_3 except those (conjugates of) subgroups H of D_2 such that $\ell_T(H) \geq 3$, and if $H < K < T$, then K is conjugate to one of the

groups in

$$\{D_2, [D_2, D_2], 2^{1+12}.3.L_3(4).2\}.$$

The set $\mathcal{S}_3$ consists precisely of these remaining subgroups H .

We now commence Strategy 4.5, Step 3, stage (1). We need to prove that $T^{[T:H]}$ is 2-closed for all $H \in \mathcal{S}_3$ with $\ell_T(H) = 3$. We may assume that such a subgroup H is contained in $M := D_2$. Then H must be a maximal subgroup in either $[M, M]$, or a subgroup of shape $2^{1+12}.3.L_3(4).2$. We listed conjugacy class representatives for these subgroups in the proof of Proposition 5.6(1)(c), namely, in (5.9) and (5.10). We also showed that every maximal subgroup of $[M, M]$, apart from those conjugate to $2^{1+12}.3.Alt(7)$ or $2^{1+12}.3.L_3(4)$, is contained in a maximal subgroup K of M with $T^{[T:K]}$ of base size 2. Thus, the subgroups H which remain to be considered are the following:

- (i) the maximal subgroups H of $2^{1+12}.3.L_3(4).2 < M$; and
- (ii) the subgroup $H := 2^{1+12}.3.Alt(7) < [M, M] < M$.

Let H be one of these subgroups, and let $K := 2^{1+12}.3.L_3(4).2$ in case (i), and $K = [M, M]$ in case (ii), so that H is maximal in K . In either case the quantities $g(T, K) = 2$ and $d(T, K) \leq 2$ (see Table 1 and Proposition 4.1(b)). Assume that $T^{(2), [T:H]} \neq T$. Then, by Theorem 3.10(2) applied to $\Sigma = [T : K]$, and Proposition 4.1(b), the usual argument shows that the only possibility is for K, H as in case (i) with $H = [K, K] \cong 2^{1+12}.3.L_3(4)$. So assume that we are in this case. Then by (5.12), there exists an element t of T such that $M \cap M^t \leq [M, M]$. Hence, $K \cap K^t \leq K \cap [M, M] \cong 2^{1+12}.3.L_3(4) = [K, K]$. It then follows from Theorem 3.10(1) that $T^{[T:H]}$ is 2-closed, which is a contradiction. This completes stage (1) of Step 3. For stage (2) of Step 3, we simply note that the set $\mathcal{S}_4$ is empty, by Proposition 5.6(2). Thus, it follows from Strategy 4.5 that $T = J_4$ is totally 2-closed. $\square$

6. The Exceptional Groups

In this section, we prove that none of the finite simple exceptional groups of Lie type listed in Proposition 2.12(a) can be totally 2-closed. To do this, we assume throughout this section that

$$T \in \{{}^3D_4(q), {}^2F_4(q)', F_4(q), {}^2E_6(q), E_6(q), E_7(q), E_8(q)\}, \quad \text{with } q = p^f, \quad p \text{ prime.} \quad (6.1)$$

Our aim is to prove the following.

Proposition 6.1. *Let T be one of the groups at (6.1). Then T is not totally 2-closed.*

Recall that the subdegrees of a transitive permutation group are the orbit lengths of a point stabilizer. Our first lemma collates the subdegrees of the minimal

degree permutation representations of T , which were determined in [36, Theorem 2] for $F_4(q)$, in [37, Theorems 1–3] for $E_6(q), E_7(q), E_8(q)$, and in [38, Theorems 3–6] for ${}^3D_4(q), {}^2E_6(q), {}^2F_4(q)$ ($q > 2$), and ${}^2F_4(2)'$, respectively.

Lemma 6.2. *Let M be a point stabilizer for T in a minimal degree (necessarily primitive) permutation representation. Then M and the point stabilizers in $M^{[T:M]}$ are known, and M , the degree $|T : M|$, the rank, and the subdegrees are given in Table 2.*

Proposition 6.3. *Suppose that either T is twisted; or T is untwisted with $f > 1$, where $q = p^f$ as in (6.1). Let M be a point stabilizer for T in a minimal degree permutation representation, so M is as in Table 2. Then T is not totally 2-closed.*

Proof. By Lemma 6.2, the group M is a maximal parabolic subgroup of T . More precisely, we write $M = P_I$, where I is a subset of the set of node-labels of the Dynkin diagram for T , and P_I is the parabolic subgroup obtained by deleting the nodes with labels in I . In particular, in each case the group $G := TN_{\text{Aut}(T)}(M)$ is strictly larger than T . Indeed, if T is twisted, then T has a nontrivial graph automorphism which normalizes M , while if T is untwisted and $f > 1$, then T has a nontrivial field automorphism which normalizes M . See [16, Secs. 2.5 and 2.6]. In particular, we use the definition of graph, field, and graph-field automorphisms of T given by [16, Definition 2.5.13].

Now, since M is maximal in T , we have $M = N_T(M)$. Hence, $|G : N_{\text{Aut}(T)}(M)| = |T : M|$, and $T^{[T:M]}$ is permutationally isomorphic to the per-

Table 2. Information on the minimal degree permutation representations for exceptional groups in (6.1).

T	M	$ T : M $	Rank	Subdegrees
${}^3D_4(q)$	P_2	$(q^8 + q^4 + 1)(q + 1)$	4	$1, q(q^3 + 1), q^9, q^5(q^3 + 1)$
${}^2F_4(q),$ $q > 2$	$P_{2,3}$	$(q^6 + 1)(q^3 + 1)(q + 1)$	5	$1, q(q^2 + 1), q^{10}, q^4(q^2 + 1),$ $q^7(q^2 + 1)$
${}^2F_4(2)'$	$A_2(3) : 2$	1600	3	$1, 351, 1248$
$F_4(q)$	P_1 or P_4	$\frac{(q^{12}-1)(q^4+1)}{q-1}$	5	$1, \frac{(q^4-1)(q^3+1)}{q-1}, q^{15}, q^5 \frac{q^6-1}{q-1},$ $q^8 \frac{(q^4-1)(q^3+1)}{q-1}$
${}^2E_6(q)$	P_4	$\frac{(q^{12}-1)(q^6-q^3+1)(q^4+1)}{q-1}$	5	$1, q(q^5 + 1)(q^3 + 1)(q + 1), q^{21},$ $q^6(q^5 + 1)(q^4 + q^2 + 1),$ $q^{11}(q^5 + 1)(q^3 + 1)(q + 1)$
$E_6(q)$	P_1 or P_6	$\frac{(q^9-1)(q^8+q^4+1)}{q-1}$	3	$1, q \frac{(q^8-1)(q^3+1)}{q-1}, q^8 \frac{(q^5-1)(q^4+1)}{q-1}$
$E_7(q)$	P_1	$\frac{(q^{14}-1)(q^9+1)(q^5+1)}{q-1}$	4	$1, q \frac{(q^9-1)(q^8+q^4+1)}{q-1},$ $q^{10} \frac{(q^9-1)(q^8+q^4+1)}{q-1}, q^{27}$
$E_8(q)$	P_1	$\frac{(q^{30}-1)(q^{12}+1)(q^{10}+1)(q^8+1)}{q-1}$	5	$1, q \frac{(q^{14}-1)(q^9+1)(q^5+1)}{q-1}, q^{57},$ $q^{29} \frac{(q^{14}-1)(q^9+1)(q^5+1)}{q-1},$ $q^{12} \frac{(q^{14}-1)(q^{12}+q^6+1)(q^9+q^4+1)}{q-1}$

mutation group $T^{[G:N_{\text{Aut}(T)}(M)]}$. By inspection of Table 2, we see that in each case, any two subdegrees of $T^{[G:N_{\text{Aut}(T)}(M)]}$ involve different q -parts, and hence the subdegrees are pairwise distinct. Thus T and G have the same orbits on ordered pairs of points from $[G:N_{\text{Aut}(T)}(M)]$, and it follows from Theorem 2.4 that $G \leq T^{(2),[G:N_{\text{Aut}(T)}(M)]}$. Therefore T is not totally 2-closed. $\square$

In proving Proposition 6.1, it is convenient to deal with the cases $E_6(2)$, $E_7(2)$ and $E_8(2)$ separately. We do this in the next proposition, where we use the following labeling of the Dynkin diagrams for these groups. This is consistent with the labeling in the computer algebra system **Magma** [5].

$$\begin{array}{c}
 \begin{array}{ccccccc}
 & & \bullet & & & & \\
 & & | & & & & \\
 \bullet & - & \bullet & - & \bullet & - & \bullet \\
 1 & & 3 & & 4 & & 5 & & 6 & & E_6
 \end{array}
 \quad
 \begin{array}{ccccccc}
 & & \bullet & & & & \\
 & & | & & & & \\
 \bullet & - & \bullet & - & \bullet & - & \bullet & - & \bullet & - & \bullet \\
 1 & & 3 & & 4 & & 5 & & 6 & & 7 & & E_7
 \end{array}
 \\
 \\
 \begin{array}{ccccccc}
 & & \bullet & & & & \\
 & & | & & & & \\
 \bullet & - & \bullet & - & \bullet & - & \bullet & - & \bullet & - & \bullet & - & \bullet \\
 1 & & 3 & & 4 & & 5 & & 6 & & 7 & & 8 & & E_8.
 \end{array}
 \end{array} \tag{6.2}$$

Proposition 6.4. *The simple exceptional groups $E_6(2)$, $E_7(2)$, and $E_8(2)$, are not totally 2-closed.*

Proof. Here, $T = E_m(2)$ for some $m \in \{6, 7, 8\}$. Let (B, N) be a (B, N) -pair for T . Since T is defined over a field of order 2, the group N is isomorphic to the Weyl group W of T , with respect to this (B, N) -pair. Let Φ be the associated root system for T , with fundamental roots $\Pi := \{a_1, \dots, a_m\}$. As mentioned above, we label the roots and Dynkin diagram for T as in **Magma** [5], see (6.2). Let $M := P_{a_3}$ be the maximal parabolic subgroup of T obtained by deleting the node labeled by 3 in the Dynkin diagram. Then $M = U \rtimes L$, where U is the unipotent radical of M and L is a Levi subgroup.

For each root a_i , we denote by Φ_i the set of roots $r \in \Phi$ with the property that the a_i -coordinate of r is nonzero; and we set $\Phi_{-i} = \Phi \setminus \Phi_i$. For any subset Λ of Φ , we write Λ^+ for the set of positive roots in Λ . Finally, we write U_r for the root subgroup of the unipotent radical U associated with r .

Then U is the 2-group $U = \langle U_r : r \in \Phi_3^+ \rangle$, and we may take L to be the standard Levi subgroup $L := \langle U_r : r \in \Phi_{-3} \rangle$. It is clear from the Dynkin diagram (6.2), that

$$L = \langle U_{\pm a_1} \rangle \times \langle U_{\pm a_2}, U_{\pm a_4}, U_{\pm a_5}, \dots, U_{\pm a_m} \rangle \cong A_1(2) \times A_{m-2}(2). \tag{6.3}$$

Let $\pi_1 : L \rightarrow A_1(2)$ and $\pi_2 : L \rightarrow A_{m-2}(2)$ denote the canonical projections. Set $H = U \rtimes [L, L]$, so that $[M : H] = 2$.

Now, a straightforward computation with **Magma** [5] shows that, for each $w \in W$,

$$(\Phi_3^+ \cup \Phi_{-3})^w \cap \Phi_{-3} \tag{6.4}$$

contains a root $r_w \in \Phi$ with nonzero a_1 -coordinate (and zero a_3 -coordinate). Recall that $N \cong W$, as we noted above. Choose $n \in N$, and suppose that n corresponds to

$w \in W$. Let $r := r_w$ be a root in the set in (6.4) (with nonzero a_1 -coordinate and zero a_3 -coordinate). Then it follows from (6.4) that $M^n \cap L$ contains the root subgroup U_r . Since U_r has nonzero a_1 -coordinate, $\pi_1(U_r)$ contains a Sylow 2-subgroup of $A_1(2) \cong \text{Sym}(3)$. Thus, $\pi_1(M^n \cap L)$ contains a Sylow 2-subgroup of $A_1(2)$, and hence $(M \cap M^n)H = M$ for all $n \in N$.

Now, by Bruhat Normal Form [16, Theorem 2.3.5], we have $T = BNB$, so the element $t = xny$ for some $x, y \in B < M$ and $n \in N$. Then, noting that $y \in N_T(H)$ since $|M : H| = 2$, we have

$$(M \cap M^t)H = (M \cap M^{ny})H = ((M \cap M^n)H)^y = M^y = M.$$

It now follows from Theorem 3.10(1) applied with $G, \Omega, \Sigma, M, \Delta$ and $L = L^{(2), \Sigma}$ being $T, [T : H], [T : M], M, [M : H]$ and $T^{[T:M]}$, that at least one of $T^{[T:M]}$ or $T^{[T:H]}$ is not 2-closed. In either case T is not totally 2-closed. $\square$

We are now ready to prove Proposition 6.1.

Proof of Proposition 6.1. By Proposition 6.3, we may assume that T is untwisted and that $q = p$ is prime. Thus, by Propositions 2.12(a) and 6.4, we may assume that $T \in \{F_4(p), E_6(p), E_7(p), E_8(p)\}$, and that p is an odd prime.

Let S be a maximally split torus in T , with (B, N) a (B, N) -pair for T , with respect to S . Then B has the form $B = U \rtimes S$, where U is a Sylow p -subgroup of T . If $T^{[T:B]}$ is not 2-closed there is nothing further to prove for this group, so we may assume that $T^{[T:B]}$ is 2-closed. Since p is odd, the group S has even order, and hence has a subgroup R of index 2. Let $H := UR$. Then $|B : H| = 2$.

For the permutation group $T^{[T:B]}$, the subgroup B is the stabilizer of the ‘point’ B . Each B -orbit in $[T : B] \setminus \{B\}$ is of the form $\{Bah \mid h \in B\}$ for some $a \in T \setminus B$. Now, by Bruhat Normal Form [16, Theorem 2.3.5], we have $T = BNB$, so $a = xny$ for some $x, y \in B$ and $n \in N$, and it follows that the B -orbit containing Ba also contains $Bay^{-1} = Bn$, and the stabilizer in B of the point Bn is $B \cap B^n$. Since $N = N_T(S)$, we have $S \leq B \cap B^n$, and it follows that $B = (B \cap B^n)H$. This holds for all $n \in N$.

Finally, by again using the fact that $T = BNB$, and arguing in exactly the same way as in the final paragraph of the proof of Proposition 6.4, we can see that $(B \cap B^t)H = B$ for all $t \in T$, and hence, after applying Theorem 3.10(1), that $T^{[T:H]}$ is not 2-closed. Hence, T is not totally 2-closed. $\square$

6.1. Proof of Theorem 1.1

By Proposition 2.12, together with our results in Secs. 4–6, on sporadic simple groups, and Proposition 6.1 in this section for the exceptional groups, we know that each of the groups $J_1, J_3, J_4, \text{Th}, \text{Ly}, \mathbb{M}$ is totally 2-closed, and that these are the only totally 2-closed simple groups. Thus Theorem 1.1 is proved.

7. Direct Products

In this section, our aim is to prove Theorem 1.2. For this reason, we assume throughout the section that G is a nontrivial finite totally 2-closed group with $F(G) = 1$. By Theorem 2.5, the group G must have the form $G = T_1 \times \cdots \times T_r$ where for each $i \leq r$, T_i is a finite totally 2-closed nonabelian simple group, and T_i is not a section of $\prod_{i \neq j} T_j$. Moreover, by Theorem 1.1, each of the T_i lies in the set

$$\mathcal{S} := \{J_1, J_3, J_4, \text{Th}, \text{Ly}, \mathbb{M}\}.$$

By inspection of the sections of these sporadic simple groups, we see that there exist distinct $S, T \in \mathcal{S}$ such that S is isomorphic to a section of T if and only if $S = \text{Th}$ and $T = \mathbb{M}$, see Remark 2.6. Thus not both of the groups Th and $\mathbb{M}$ lie in $\{T_i : 1 \leq i \leq r\}$, and hence $r \leq 5$ and one of the conditions (3)(i) or (3)(ii) of Theorem 1.2 holds. This proves that each of the assertions (1)–(3) of Theorem 1.2 holds.

Thus to complete the proof of Theorem 1.2 we must show that, if $G = \prod_{i=1}^r T_i$ with the T_i pairwise non-isomorphic, and if one of the conditions (3)(i) or (3)(ii) of Theorem 1.2 holds, then G is totally 2-closed. Since none of the groups in $\mathcal{S}$ admits a nontrivial factorization (Corollary 2.3), it follows from Proposition 2.10 that it is sufficient to prove that $\varphi(G)$ is 2-closed for each permutation representation $\varphi : G \rightarrow \text{Sym}(\Omega)$ (faithful and non-faithful) with $|\Omega| > 1$ and $\varphi(G)$ transitive. With this in mind, we introduce the following hypothesis and notation, which we will assume, for the remainder of the section.

Hypothesis 7.1. (a) The group $G = T_1 \times \cdots \times T_r$ with the T_i pairwise non-isomorphic, and either

- (i) $T_i \in \{J_1, J_3, J_4, \text{Th}, \text{Ly}\}$ for each $i \leq r$; or
- (ii) $T_i \in \{J_1, J_3, J_4, \text{Ly}, \mathbb{M}\}$ for each $i \leq r$.

- (b) We extend the notation for direct products introduced in Hypothesis 2.7. Namely, for a direct product $G_1 \times \cdots \times G_s$ of arbitrary finite groups G_i , and each subgroup L_i of G_i , we write

$$\widehat{L_i} = \{(x_1, \dots, x_s) : x_i \in L_i, \text{ and } x_j = 1 \text{ for } j \neq i\} \leq G_1 \times \cdots \times G_s.$$

- (c) Seeking a contradiction, we assume that G is *not* totally 2-closed, and that the number r of simple direct summands is minimal with respect to this property. Note that $r > 1$ by Theorem 1.1, and by Proposition 2.10, G has a proper subgroup H such that $G^{[G:H]}$ is not 2-closed. Without loss of generality, we may assume that the proper subgroup H is maximal by inclusion such that $G^{[G:H]}$ is not 2-closed. By [39, Theorem 5.12 and Example 5.13], it follows that $H \neq 1$. Further, H is core-free in G (since otherwise $G/\text{Core}_G(H) \cong G^{[G:H]}$ is not totally 2-closed and is a direct product of less than r of the simple groups in (i) or (ii), contradicting the minimality of r).

- (d) Next, write $\mu_i : G \rightarrow T_i$ for the natural coordinate projection, for $i \leq r$. Note that $\mu_i(\widehat{T}_i \cap H)$ is normal in $\mu_i(H)$, but to avoid cumbersome notation, we will usually just say that $\widehat{T}_i \cap H$ is normal in $\mu_i(H)$.
- (e) Our final piece of notation is defined with the minimality of r in mind: we set $G_1 := T_1$, and $G_2 := T_2 \times \cdots \times T_r$, so that both G_1 and G_2 are finite totally 2-closed groups (by the minimality of r). We write the coordinate projections here as $\pi_1 : G \rightarrow G_1$ and $\pi_2 : G \rightarrow G_2$ (so in particular $\pi_1 = \mu_1$). We also set $M_i := \pi_i(H)$ and $H_i = \widehat{G}_i \cap H$, for $i \in \{1, 2\}$, where $\widehat{G}_i$ is as defined in (b) relative to the direct product $G_1 \times G_2$. Note that H is a subdirect subgroup of $M := M_1 \times M_2 \leq G$ and $\pi_i(H_i) \leq M_i$ for each i . As in part (d), we often omit the π_i , and just say that $H_i \leq M_i$. Since H is a subdirect subgroup of $M = M_1 \times M_2$, it is important to note that $M_1/H_1 \cong M_2/H_2$ (Goursat's Lemma, see [32, Theorem 4.8(i)]).

Remark 7.2. The final piece of notation in Hypothesis 7.1 is introduced for convenience only: there will be nothing special about the labeling of the simple direct factors of G , and we will sometimes vary the direct factor which is chosen to be T_1 in our analysis. We warn the reader however, that the labeling of the direct factors of G changes in this way, then the group M defined in Hypothesis 7.1(e) may also change.

In order to prove that each group G satisfying Hypothesis 7.1 is totally 2-closed, we use the following strategy.

Strategy 7.3. Let G, T_i, H , etc. be as in Hypothesis 7.1. We follow the four steps below.

Step 1: We prove (Lemma 7.4) that the subgroup $M = M_1 \times M_2$ in Hypothesis 7.1(e) is a proper overgroup of H , that M is core-free in G , and that $G^{[G:M]}$ is 2-closed. This enables us to use the tools developed in Sec. 3, (see Notation 3.3).

Step 2: The tools developed in Sec. 3 also require information concerning the transitive permutation group $M^{[M:H]}$ with point stabilizer $H^{[M:H]}$. Note that H is a subdirect subgroup of $M = M_1 \times M_2$. Step 2 comprises two results (Lemmas 7.5 and 7.6) concerning subdirect subgroups of a direct product of finite groups.

Step 3: We prove (Lemma 7.7) that, for some $i \leq r$, the permutation group $T_i^{[T_i:\mu_i(H)]}$ has base size greater than 2, and so, by Proposition 4.1, this pair $(T_i, \mu_i(H))$ lies in Table 1.

Step 4: For each pair $(T_i, \mu_i(H))$ in Table 1, we examine the structure of $X := G^{(2), [G:H]}$ using Theorem 3.10(2), seeking a contradiction. We use the same notation as that introduced in Notation 3.3 with some small changes given in Notation 7.9.

We begin with Step 1.

Lemma 7.4. *Let G, r, T_i, H, M be as in Hypothesis 7.1. Then,*

- (i) $\mu_i(H) \neq T_i$ for any $i \leq r$, M is core-free in G , $G^{[G:M]}$ is 2-closed, and $H < M < G$; and
- (ii) *For each $i \leq r$, and each composition factor S of $\mu_i(H)/(\hat{T}_i \cap H)$, there exists $j \neq i$ such that $\mu_j(H)$ has a composition factor isomorphic to S .*

Proof. (i) Note that Hypothesis 2.7 also holds for G (with projection maps $G \rightarrow T_i$ now called μ_i). Hence, since H is core-free in G by Hypothesis 7.1(c), it follows from Lemma 2.8 that $\mu_i(H) \neq T_i$ for any $i \leq r$. Then, since $M \leq \mu_1(H) \times \cdots \times \mu_r(H)$, the group M does not contain $\hat{T}_i$, for any $i \leq r$, and so M is core-free in G . Also, (see Hypothesis 7.1(e)), $M = M_1 \times M_2 < G_1 \times G_2$, and by the minimality of r , $G_i^{[G_i:M_i]}$ is 2-closed for $i \in \{1, 2\}$. Hence $G^{[G:M]}$ is 2-closed, by [10, Theorem 5.1]. In particular, since by assumption $G^{[G:H]}$ is not 2-closed, we have $H < M < G$. This proves part (i).

(ii) In order to prove part (ii) we will prove the following claim:

Claim: *Let S be a finite simple group, and let $X_1, \dots, X_t$ be finite groups ($t \geq 1$) such that no X_i has a section isomorphic to S . If X is a subdirect subgroup of $X_1 \times \cdots \times X_t$, then X has no section isomorphic to S .*

First, we see that part (ii) follows from the claim. Indeed, assume that the claim holds, and that S is a composition factor of $\mu_i(H)/(\hat{T}_i \cap H)$, for some $i \leq r$. Without loss of generality assume that $i = 1$, so S is a composition factor of $\mu_1(H)/(\hat{T}_1 \cap H) = M_1/H_1$. Since H is a subdirect subgroup of $M = M_1 \times M_2$ (Hypothesis 7.1(e)), we have $M_1/H_1 \cong M_2/H_2$, and hence S is a composition factor of M_2 . Then since $M_2 = \pi_2(H)$ is a subdirect subgroup of $\mu_2(H) \times \cdots \times \mu_r(H)$, it follows from the claim that $\mu_j(H)$ has a composition factor isomorphic to S for some $j \geq 2$.

Proof of the Claim: The proof is by induction on t . The case $t = 1$ is trivially true, so assume that $t > 1$ and that the claim holds for direct products of less than t groups. Suppose that X is a subdirect subgroup of $X_1 \times \cdots \times X_t$, that no X_i has a section isomorphic to S , but that X has a section isomorphic to S , say $S \cong Y/N$ with $N \trianglelefteq Y \leq X$. We will derive a contradiction. For each $i \leq r$, let $\nu_i : X \rightarrow X_i$ be the natural projection map and $Y_i = \nu_i(Y)$, so Y is a subdirect subgroup of $Z := Y_1 \times \cdots \times Y_t$ and no Y_i has a section isomorphic to S . Further let $\nu'_i : Z \rightarrow \prod_{j \neq i} Y_j$ be the natural projection map, for each i , and note that $\nu'_1(Y)$ is a subdirect subgroup of $Y_2 \times \cdots \times Y_t$. Hence, if $\nu'_1(Y)$ has a section isomorphic to S , then by induction Y_i has a section isomorphic to S , for some $i \geq 2$, which is a contradiction. Hence $\nu'_1(Y)$ has no section isomorphic to S . Since $\nu'_1(Y)/\nu'_1(N)$ is isomorphic to a quotient of the simple group $Y/N \cong S$, this implies that $\nu'_1(N) = \nu'_1(Y)$. Now, the kernel of $\nu'_1|_Y$ is the group $K := Y \cap \hat{Y}_1$, and so $\nu'_1(N) = \nu'_1(Y)$ implies that $Y = NK$, whence $K/(N \cap K) \cong NK/N = Y/N \cong S$. Thus, $\hat{Y}_1 \cong Y_1$ has a section isomorphic to S , which is a contradiction. This completes the inductive proof of the claim, and hence completes the proof of the lemma. $\square$

We now move to Step 2, and prove two lemmas concerning subdirect subgroups of direct products.

Lemma 7.5. *Let $X_1, \dots, X_t$ be finite groups, and let X be a subdirect subgroup of $X_1 \times \dots \times X_t$. Then the following assertions hold:*

- (i) *If $t = 2$, and $\widehat{X}_i \leq X$ for some $i \in \{1, 2\}$, then $X = X_1 \times X_2$.*
- (ii) *Let p be a prime, and suppose that each group X_i is a transitive permutation group of p -power degree. Then the Fitting subgroup $F(X)$ is a (possibly trivial) p -group.*

Proof. (i) This follows immediately from [32, Lemma 4.7].

(ii) For each $i \leq t$, assume that X_i is a transitive permutation group on a set Ω_i with $|\Omega_i|$ a power of p , and let $\nu_i : X \rightarrow X_i$ be the natural projection map. It is sufficient to show that $\nu_i(F(X))$ is a p -group for each i . Note that, for each i , $\nu_i(F(X))$ is a nilpotent normal subgroup of $\nu_i(X) = X_i$ (since X is subdirect), and hence $\nu_i(F(X)) \leq F(X_i)$. If $\nu_i(F(X)) = 1$ there is nothing to prove so assume that $\nu_i(F(X)) \neq 1$ and let q be a prime dividing $|\nu_i(F(X))|$. Then the unique (nontrivial) Sylow q -subgroup Q of $\nu_i(F(X))$ is characteristic in $\nu_i(F(X))$ and hence normal in X_i . Since X_i is transitive on Ω_i , all the orbits of its normal subgroup Q have the same size, say m . Thus m divides $|\Omega_i|$ which is a power of p , and also m divides $|Q|$ which is a power of q . It follows that $p = q$, and hence that $\nu_i(F(X))$ is a p -group. Thus part (ii) is proved. $\square$

Lemma 7.6. *Let E be a finite group, let $\theta \in \text{Aut}(E)$, and let*

$$J := \{(e, e^\theta) : e \in E\},$$

a full diagonal subgroup of $X := E \times E$. Then:

- (i) $\text{Core}_X(J) = \{(z, z^\theta) : z \in Z(E)\} = Z(X) \cap J$.
- (ii) *The group $Y := X^{[X:J]}$ is a central product $E_1 \circ E_2$, where $E_i \cong E$ for each i , and $E_1 \cap E_2 \cong Z(E)$. In particular $Y = X/(Z(X) \cap J)$.*
- (iii) *Writing $\overline{U}$ for the image of a subgroup $U \leq Y$ under the natural projection map $Y \rightarrow Y/(E_1 \cap E_2)$, we have $\overline{Y} \cong \overline{E_1} \times \overline{E_2}$, and $\overline{J}$ is a subdirect subgroup such that*

$$\overline{J} \cap \widehat{\overline{E_1}} = \overline{J} \cap \widehat{\overline{E_2}} = 1.$$

Proof. (i) and (ii): Since part (ii) follows immediately from (i), we only need to prove (i). Let $x \in \text{Core}_X(J) = \bigcap_{y \in X} J^y$. Then $x = (e, e^\theta)$ for some $e \in E$ (since $x \in J$), and also $x^y \in J$ for all $y = (e_1, e_2) \in X$. In particular, taking $y = (1, e_2)$, we have $x^y = (e, e^{\theta e_2}) \in J$, so $e^{\theta e_2} = e^\theta$, and this holds for all $e_2 \in E$. Hence $e^\theta \in Z(E)$, and since θ is an automorphism, it follows that $e \in Z(E)$. Thus, $\text{Core}_X(J) \subseteq \{(z, z^\theta) : z \in Z(E)\}$. The reverse inclusion clearly holds, so equality holds and hence part (i), and also part (ii), are proved.

(iii): The fact that $\overline{Y} \cong \overline{E_1} \times \overline{E_2}$ follows from part (ii), and by definition, $\overline{J}$ projects onto each direct factor $\overline{E_i}$. Noting that $Z(X) = Z(E) \times Z(E)$, it follows that $E_1 \cap E_2 = Z(X)/(Z(X) \cap J)$, and hence $\overline{Y} = Y/(E_1 \cap E_2) \cong X/Z(X)$. Under this isomorphism $\widehat{E_1}$ corresponds to the subgroup of cosets $Z(X)x$ with x of the form $x = (e, z)$ for some $e \in E, z \in Z(E)$. If such a coset $Z(X)x$ corresponds to an element of $\overline{J}$, then $x = x'z'$ with $x' \in J, z' \in Z(X)$, so $x = (e, z) = (e_1, e_1^\theta)(z_1, z_2)$ for some $e_1 \in E, z_1, z_2 \in Z(E)$. This implies firstly that $e_1^\theta = zz_2^{-1} \in Z(E)$ whence $e_1 \in Z(E)$; and secondly that $e = e_1z_1 \in Z(E)$. It follows that the coset $Z(X)x$ is trivial, and hence that $\overline{J} \cap \widehat{E_1} = 1$. The proof that $\overline{J} \cap \widehat{E_2} = 1$ is similar. $\square$

Now, we proceed with Step 3, examining the base sizes of the simple direct factors T_i in their coset actions on $[T_i, \mu_i(H)]$.

Lemma 7.7. *Let G, r, T_i, μ_i, H, M be as in Hypothesis 7.1. Then there exists $i \leq r$ such that the permutation group $T_i^{[T_i: \mu_i(H)]}$ has base size greater than 2, and in particular $(T_i, \mu_i(H))$ is one of the pairs (T, M) in Table 1.*

Proof. By Lemma 7.4(i), H is a proper subgroup of M , and hence H is a proper subgroup of $N := \mu_1(H) \times \cdots \times \mu_r(H)$ since $M \leq N$ (Hypothesis 7.1(e)). Since H is core-free in G (Hypothesis 7.1(c)), it follows from Lemma 2.8 that $\mu_i(H)$ is a proper subgroup of T_i for all $i \leq r$. Thus, N is core-free in G . If $T_i^{[T_i: \mu_i(H)]}$ has base size at most 2 for all i , then $G^{[G:N]}$ has base size at most 2, and it follows from Theorem 3.10(3) that $G^{[G:H]}$ is 2-closed, contradicting Hypothesis 7.1(c). Thus, for some $i \leq r$, $T_i^{[T_i: \mu_i(H)]}$ has base size greater than 2, and hence, by Proposition 4.1(a), $(T_i, \mu_i(H))$ is one of the pairs (T, M) in Table 1. $\square$

We now move on to Step 4 of Strategy 7.3. We examine each possibility for $(T_i, \mu_i(H))$ from Table 1. First, we deal with the unique pair in Table 1 involving the Monster, namely, $(\mathbb{M}, 2.\mathbb{B})$.

Lemma 7.8. *For T_i, μ_i, H as in Hypothesis 7.1, no pair $(T_i, \mu_i(H))$ is equal to $(\mathbb{M}, 2.\mathbb{B})$.*

Proof. Suppose to the contrary that $(T_i, \mu_i(H)) = (\mathbb{M}, 2.\mathbb{B})$, for some i . As discussed in Remark 7.2, we may assume that $i = 1$, so $M_1 = 2.\mathbb{B}$, and by Hypothesis 7.1, $T_j \in \{J_1, J_3, J_4, \text{Ly}\}$ for each $j \geq 2$. Now, H is a subdirect subgroup of $M = M_1 \times M_2$, and by Lemma 7.4(i), H is a proper subgroup of M . Thus, by Lemma 7.5(i), H does not contain $\widehat{M_1}$, so $H_1 = \widehat{T_1} \cap H = \widehat{M_1} \cap H$ is a proper normal subgroup of $M_1 = 2.\mathbb{B}$. It follows that $\mathbb{B}$ is a composition factor of M_1/H_1 , but this contradicts Lemma 7.4(ii) since, for each $j \geq 2$, $T_j \in \{J_1, J_3, J_4, \text{Ly}\}$ and none of these simple groups has a section isomorphic to $\mathbb{B}$. $\square$

To deal with the other pairs $(T_i, \mu_i(H))$ from Table 1, our main tool is Theorem 3.10(2), and we extend the notation used there slightly as follows.

Notation 7.9. We continue to adopt the notation and hypotheses set out in Hypothesis 7.1, and in order to apply Theorem 3.10(2), we recall and slightly extend the notation from Notation 3.3 (notably parts (2), (3), and (5)–(10)):

- (a) $\Omega = [G : H]$, $\Sigma = [G : M]$ of size $s = |\Sigma|$, $\Delta = [M : H]$, $L = G^\Sigma$ (which equals $L^{(2),\Sigma}$ by Lemma 7.4), $R = M^\Delta$, $Y = X_\Delta^\Delta$, and $G \leq W = R \wr L$ acting on Ω . For a subgroup $A \leq R$ we write B_A for the corresponding base group A^s of the base group $B = R^s$ of W . Note that M is core-free in G (by Lemma 7.4(i)), so $L = G^\Sigma \cong G$.
- (b) Since $M = M_1 \times M_2$ and $G = G_1 \times G_2$ (Hypothesis 7.1(e)), the set $\Sigma = \Sigma_1 \times \Sigma_2$ and $s = s_1 s_2$, where $\Sigma_i = [G_i : M_i]$ of size s_i , for $i = 1, 2$. For this reason, and as we will see below, it is more natural to label the G -invariant partition Σ of Ω in Notation 3.3(2) as $\Sigma = \{\Delta_{(i,j)} : 1 \leq i \leq s_1; 1 \leq j \leq s_2\}$, and we choose $\Delta := \Delta_{(1,1)}$, and $\omega \in \Delta$, (see Notation 3.3(2) and (3)).
- (c) In Notation 3.3(5) and (6), we identify Σ with the set $\{(i, j) : 1 \leq i \leq s_1; 1 \leq j \leq s_2\}$, and we assume that $G \leq W = R \wr L$ is acting on $\Omega = \Delta \times \{(i, j) : 1 \leq i \leq s_1; 1 \leq j \leq s_2\}$; where we identify $\Delta_{(i,j)}$ with $\Delta \times \{(i, j)\}$. The elements $(x_{(1,1)}, \dots, x_{(s_1, s_2)}) \in B_R$ and $\sigma = (\sigma_1, \sigma_2) \in L \cong G_1^{[G_1 : M_1]} \times G_2^{[G_2 : M_2]}$ (with $\sigma_i \in G_i^{[G_i : M_i]}$) act as follows on a point $(\delta, (i, j)) \in \Omega$:
- $$(\delta, (i, j))^{(x_{(1,1)}, \dots, x_{(s_1, s_2)})} = (\delta^{x_{(i,j)}}, (i, j)) \quad \text{and} \quad (\delta, (i, j))^\sigma = (\delta, (i^{\sigma_1}, j^{\sigma_2})).$$
- (d) In Notation 3.3(7)–(9), we write

$$R_{(i,j)} = \{(x_{(1,1)}, \dots, x_{(s_1, s_2)}) : x_{(k,\ell)} = 1 \text{ if } (k, \ell) \neq (i, j)\};$$

for $A \leq \text{Sym}(\Delta)$, we replace W_i by $W_{(i,j)} = B_A \rtimes L_{(i,j)}$; the map ρ_i by

$$\rho_{(i,j)} : W_{(i,j)} \rightarrow A \quad \text{given by } \rho_{(i,j)} : x \rightarrow x_{(i,j)}$$

for $x = (x_{(1,1)}, \dots, x_{(s_1, s_2)}) \in B_A$; the map $\rho_{i,j}$ by

$$\rho_{(i,j),(k,\ell)} : W_{(i,j)} \cap W_{(k,\ell)} \rightarrow A \times A \quad \text{given by } \rho_{(i,j),(k,\ell)} : x \rightarrow (x_{(i,j)}, x_{(k,\ell)})$$

for distinct $(i, j), (k, \ell)$; and in applying this to notation to $W = R \wr L$, we write $M_{(i,j)} = W_{(i,j)} \cap G$, so $\rho_{(i,j)}(M_{(i,j)}) = M_{(i,j)}^{\Delta_{(i,j)}} = R_{(i,j)} \cong R$, etc. Moreover, if, for $k \in \{1, 2\}$, $M_k^{(1)}, \dots, M_k^{(s_k)}$ are the point stabilizers in the transitive permutation group $G_k^{[G_k : M_k]}$, then, relabeling if necessary, the stabilizer $M_{(i,j)} = M_1^{(i)} \times M_2^{(j)}$, for $1 \leq i \leq s_1$, and $1 \leq j \leq s_2$.

- (e) By Theorem 3.10(2), $X = N \rtimes G$ with N a subdirect subgroup of a base group $B_A = A^s$ for some nontrivial $A \trianglelefteq Y$, and the A -orbits in Δ have a constant length $a = a(G, M, H)$ such that a divides $|L_{(i,j),(k,\ell)}|$, for all distinct $\Delta_{(i,j)}, \Delta_{(k,\ell)} \in \Sigma$.

Our strategy for proceeding is to prove that the parameter a in Notation 7.9(e) divides a product of r positive integers related to the simple groups $T_1, \dots, T_r$ (see

Lemma 7.10). We then prove (see Lemma 7.11) that a has the form $a = d_1 d_2$, where the d_i are orders of certain normal sections of M_1/H_1 . Finally, in the proof of Theorem 1.2, we prove that in most cases, $d_1 d_2$ cannot divide a , yielding a contradiction in those cases.

Lemma 7.10. *Assume that the notation and conditions from Notation 7.9 hold, noting in particular that $L = L^{(2),\Sigma}$ and $X \neq G$. In addition extend the notation $g(T, M)$ from Proposition 4.1 to all pairs (T, M) with T a finite simple group and $M < T$, by defining $g(T, M) := 1$ if (T, M) is not in Table 1. Then the parameter $a = a(G, M, H)$ divides $\prod_{i=1}^r g(T_i, \mu_i(H))$.*

Proof. By Notation 7.9(e), a divides $|L_{(i,j),(k,\ell)}|$, for all $i, k \leq s_1$ and all $j, \ell \leq s_2$ with $(i, j) \neq (k, \ell)$. Moreover, since G is faithful on Σ (see Notation 7.9(a)) we have $|L_{(i,j),(k,\ell)}| = |M_{(i,j)} \cap M_{(k,\ell)}|$, and by Notation 7.9(d), $|M_{(i,j)} \cap M_{(k,\ell)}| = |M_1^{(i)} \cap M_1^{(k)}| \cdot |M_2^{(j)} \cap M_2^{(\ell)}|$. Thus, for fixed j and ℓ , we see that a divides $|M_1^{(i)} \cap M_1^{(k)}| \cdot |M_2^{(j)} \cap M_2^{(\ell)}|$ for all $i, k \leq s_1$, and hence a divides $d_1 |M_2^{(j)} \cap M_2^{(\ell)}|$, where $d_1 = \gcd\{|M_1^{(i)} \cap M_1^{(k)}| \mid i \neq k\}$.

By Hypothesis 7.1(e), $\mu_1(H) = M_1$, and by Lemma 7.4(i), $\mu_1(H) \neq T_1$, and hence M_1 is a proper subgroup of T_1 . If (T_1, M_1) is not a pair in Table 1, then we have defined $g(T_1, M_1) = 1$; moreover by Proposition 4.1(a), the permutation group $T_1^{[T_1, \mu_1(H)]}$ has base size at most 2 and hence $M_1^{(i)} \cap M_1^{(k)} = 1$ for some i, k . Thus $d_1 = 1 = g(T_1, \mu_1(H))$ in this case. On the other hand, if (T_1, M_1) is a pair in Table 1, then d_1 is the quantity $d(T_1, \mu_1(H))$ in Proposition 4.1(b), and by that result, d_1 divides the quantity $g(T_1, \mu_1(H))$. Thus, in either case a divides $g(T_1, \mu_1(H)) \cdot |M_2^{(j)} \cap M_2^{(\ell)}|$, for all $j, \ell \leq s_2$.

We complete the proof by induction on r . If $r = 2$, then $\mu_2(H) = M_2$ and the argument of the previous paragraph shows that a divides $g(T_1, \mu_1(H)) \cdot g(T_2, M_2)$, proving the lemma in this case. Assume now that $r \geq 3$, and that the result holds for direct products of fewer than r simple groups. Set $G'_1 := T_2$, $G'_2 := T_3 \times \cdots \times T_r$, and $G' := G'_1 \times G'_2$. Also, let $\pi'_i : G' \rightarrow G'_i$ be the coordinate projections and $M'_i := \pi'_i(M_2)$, for $i = 1, 2$. Note that, since $\pi_2(H) = M_2$, it follows from the definition of π'_2 that $M'_1 = \mu_2(H)$ and this is a proper subgroup of $G'_1 = T_2$ by Lemma 7.4(a). Set $M' := M'_1 \times M'_2$. Then $M_2^{(j)} \cap M_2^{(\ell)}$ is a subgroup of $G' = G'_1 \times G'_2$ contained in M' . Now, the greatest common divisor of the orders $|M_2^{(j)} \cap M_2^{(\ell)}|$, for all $j, \ell \leq s_2$ with $j \neq \ell$, divides the greatest common divisor of the orders $|(M')^x \cap (M')^y|$, for all $x, y \in G'$. The argument in the first two paragraphs above shows that $|M_2^{(j)} \cap M_2^{(\ell)}|$ divides $g(T_2, M'_1) \cdot |(M'_2)^x \cap (M'_2)^y|$ for every fixed pair of elements $x, y \in G'_2$. Since $M'_1 = \mu_2(H) < T_2$, we have that a divides $g(T_1, \mu_1(H)) \cdot g(T_2, \mu_2(H)) \cdot |(M'_2)^x \cap (M'_2)^y|$, for all $x, y \in G'_2$, and by induction we conclude that a divides $g(T_1, \mu_1(H)) \cdot g(T_2, \mu_2(H)) \cdots g(T_r, \mu_r(H))$, completing the proof. $\square$

Next, we prove Lemma 7.11 which provides a factorization of the parameter $a = a(G, M, H)$.

Lemma 7.11. *Assume that the notation and conditions from Notation 7.9 hold, and recall that $X = G^{(2), \Omega}$. Then*

- (i) *there exist normal subgroups V of the center $Z(M_1/H_1)$, and U of $(M_1/H_1)/Z(M_1/H_1)$, such that $a = a(G, M, H) = |U| \cdot |V|$; and*
- (ii) *If a is a power of a prime p , then $F(X)$ is a p -group.*

Recalling from Hypothesis 7.1(e) that $M_1/H_1 \cong M_2/H_2$, we see that in the decomposition $a = |U| \cdot |V|$ in part (i), each of U, V is (isomorphic to) a normal section of $M_1/H_1 \cong M_2/H_2$.

Proof. (i) By Notation 7.9(e), $A \trianglelefteq Y = X^\Delta$, so the set of A -orbits in Δ is a Y -invariant partition of Δ with parts of size $a > 1$, and since $M^\Delta = R \leq Y$ (Notation 7.9(a)) this partition is also M^Δ -invariant. By Notation 7.9(a), H is the stabilizer in M of a point $\omega \in \Delta$, and so the setwise stabilizer K in M of the A -orbit ω^A satisfies $H < K \leq M$ and $|K : H| = a$.

Recall the definitions of the M_i, H_i in Hypothesis 7.1(e) and in particular that $M_1/H_1 \cong M_2/H_2 \cong E$, say. Now $S := H_1 \times H_2 \trianglelefteq M = M_1 \times M_2$ and $S < H < M$, so the permutation groups induced on Δ by M and by $\overline{M} := M/S \cong (M_1/H_1) \times (M_2/H_2) \cong E \times E$ are the same. In the latter group, $\overline{H} = H/S$ is the stabilizer of ω and $\overline{K} = K/S$ is the setwise stabilizer of ω^A . In particular $a = |\overline{K} : \overline{H}|$. Since H is a subdirect subgroup of $M = M_1 \times M_2$ it follows that $\overline{H}$ is a subdirect subgroup of $\overline{M} = E \times E$, and moreover from the definition of S it follows that $\overline{H}$ meets each direct factor of $\overline{M} = E \times E$ trivially, so $\overline{H}$ is a full diagonal subgroup of $E \times E$ and we have $\overline{H} = \{(z, z^\theta) \mid z \in E\}$ for some $\theta \in \text{Aut}(E)$ and $\overline{H} \cap (E \times 1) = 1$.

For $i = 1, 2$, let $\nu_i : \overline{M} \rightarrow E$ denote the natural projection map onto the i th direct factor. Since $\overline{H}$ is a subdirect subgroup, $\nu_i(\overline{H}) = E$ and hence also $\nu_i(\overline{K}) = E$, for each i . Let $\widehat{K}_1 := \overline{K} \cap \text{Ker}(\nu_2)$, and note that $\text{Ker}(\nu_2) = E \times 1$. Then $\widehat{K}_1 \trianglelefteq \overline{K}$ and $\overline{K}/\widehat{K}_1 \cong \nu_2(\overline{K}) = E$. Also $\widehat{K}_1 \cap \overline{H} = \overline{H} \cap \text{Ker}(\nu_2) = 1$. It follows that $\overline{K} = \widehat{K}_1 \rtimes \overline{H}$, and hence $|\widehat{K}_1| = |\overline{K} : \overline{H}| = |K : H| = a$. Now, $E \cong M_1/H_1$ has normal subgroups $\widehat{K}_1$ and $Z(E)$, and hence also $\widehat{K}_1 \cap Z(E) \trianglelefteq E$. Thus, $a = d_1 d_2$, where $d_1 = |\widehat{K}_1/(\widehat{K}_1 \cap Z(E))|$ and $d_2 = |\widehat{K}_1 \cap Z(E)|$. Part (i) follows on noting that $V := \widehat{K}_1 \cap Z(E)$ is a normal subgroup of $Z(E)$, and $U := (\widehat{K}_1 Z(E))/Z(E) \cong \widehat{K}_1/(\widehat{K}_1 \cap Z(E))$ is a normal subgroup of $E/Z(E)$.

(ii) Suppose now that a is a power of a prime p . Since $X = N \rtimes G$ with N a subdirect subgroup of the base group $B_A = A^s$ (Notation 7.9(e)), and since G is a direct product of nonabelian simple groups, it follows that $F(X) = F(N) \leq F(A)^s$. Now, A is a permutation group on Δ with all orbits of length a (a power of p), and hence A is a subdirect subgroup of $X_1 \times \cdots \times X_t$ with the X_i satisfying the

conditions of Lemma 7.5(ii). Hence $F(A)$ is a p -group, and therefore also $F(X)$ is a p -group. $\square$

Remark 7.12. We make a preliminary remark about how we may exploit Lemma 7.11(ii) in the proof of Theorem 1.2: suppose that, for a fixed pair (T_1, M_1) , we have used Lemmas 7.10 and 7.11 to show that $a(G, M, H)$ is a power of a prime p , and hence, by Lemma 7.11(ii) that $F(X)$ is a p -group. Suppose that we then consider a pair $(T_j, \mu_j(H))$ where $2 \leq j \leq r$. By changing the labeling, we can replace (T_1, M_1) by $(T_j, \mu_j(H))$, but in doing this we change the overgroup M of H , say to $\widetilde{M}$. It may be possible to use Lemmas 7.10 and 7.11 to show that $a(G, \widetilde{M}, H)$ is a power of a prime s , with $s \neq p$, whence by Lemma 7.11(ii), $F(X)$ is an s -group, implying that $F(X) = 1$. This strategy proves to be very useful. We emphasize that, when we change the labeling, even though the subgroup M may change, the group $X = G^{(2), \Omega}$ and its subgroup H do not change.

We are now ready to prove Theorem 1.2.

Proof of Theorem 1.2. Given our discussion at the beginning of Sec. 7, to complete the proof of Theorem 1.2, it is sufficient to prove that each of the groups $G = \prod_{i=1}^r T_i$ satisfying (i) or (ii) of Theorem 1.2 is totally 2-closed.

In pursuit of a contradiction, we assume that one of these groups G is not totally 2-closed. We then fix $G, H, \Omega, \Sigma, M, R, L, \Delta, s_1, s_2, \rho_{(i,j),(k,\ell)}$, etc., and also $X := G^{(2), \Omega}$ and $Y = X_{\Delta}^{\Delta}$, as in Notation 7.9. Recall that $L = L^{(2), \Sigma}$, and also that $X = N \rtimes G$ with N a subdirect subgroup of a base group $B_A = A^s$ for some nontrivial $A \trianglelefteq Y$, so all A -orbits in Δ have the same length $a = a(G, M, H)$. As in the statement of Lemma 7.10, we extend the notation $g(T, M)$ from Lemma 4.1 to all pairs (T, M) with T a finite simple group and $M < T$, by defining $g(T, M) := 1$ if (T, M) is not in Table 1, and of course $g(T, M)$ is equal to the appropriate entry in the third column of Table 1 if (T, M) does occur. By Lemma 7.10, $a = a(G, M, H)$ divides $\prod_{i=1}^r g(T_i, \mu_i(H))$.

Relabeling the T_i if necessary, we may without loss of generality assume, by Lemma 7.7, that the pair $(T_1, \mu_1(H))$ lies in Table 1, and by Lemma 7.8, that $(T_i, \mu_i(H)) \neq (\mathbb{M}, 2\mathbb{B})$ for any i . We consider each of the other pairs in Table 1, and show that none of them has the conditions required for $(T_1, \mu_1(H))$. Note that H is a subdirect subgroup of $M = M_1 \times M_2$ and that $H_i \trianglelefteq M_i$ for each i (Hypothesis 7.1(e)); moreover, by Lemmas 7.4(i) and 7.5(i), M_1/H_1 is nontrivial. After some general comments in the next paragraph, we subdivide our arguments according to the structure of M_1/H_1 , recalling that $M_1 = \mu_1(H) > H_1$.

For each $i \leq r$, since $(T_i, \mu_i(H))$ is not $(\mathbb{M}, 2\mathbb{B})$, it follows from our comments in the second paragraph of the proof, and from Table 1, that $g(T_i, \mu_i(H))$ has the form $2^k 3^{\ell} 5^m 7^n$. Hence by Lemma 7.10, a is a $\{2, 3, 5, 7\}$ -number, and then by Lemma 7.11(i), $a = d_1 d_2$ where d_1, d_2 is the order of a normal subgroup of $(M_1/H_1)/Z(M_1/H_1)$ or $Z(M_1/H_1)$, respectively. In particular, since $a > 1$, at

least one of $(M_1/H_1)/Z(M_1/H_1)$ or $Z(M_1/H_1)$ has a nontrivial normal $\{2, 3, 5, 7\}$ -subgroup, say S , with $|S|$ dividing a . Furthermore, if $|S|$ has order divisible by 7 then 7 divides a and hence, by Lemma 7.10, 7 divides $g(T_i, \mu_i(H))$ for some i , and this in turn implies, by Table 1 that the pair $(T_i, \mu_i(H)) = (J_4, 2^{10}.L_5(2))$.

Claim 1. M_1/H_1 is not almost simple. Suppose to the contrary that M_1/H_1 is almost simple. Inspecting the almost simple quotients M_1/H_1 of the groups in the second column of Table 1, we see that the only examples with socle a $\{2, 3, 5, 7\}$ -group have socle $L_3(4)$, and that $M_1/H_1 = L_3(4)$ or $L_3(4).2$. In particular, M_1/H_1 has trivial center, and so by Lemma 7.11(i), $a = \delta|L_3(4)|$ with $\delta = 1$ or 2. Since $|L_3(4)|$ has order divisible by 7, it follows from the previous paragraph that $(T_i, \mu_i(H)) = (J_4, 2^{10}.L_5(2))$ for some $i \geq 2$. Interchanging T_1 and T_i as discussed in Remark 7.2, we obtain a new group $\widetilde{M} = \widetilde{M}_1 \times \widetilde{M}_2$ with $\widetilde{M}_1 = 2^{10}.L_5(2)$. Lemma 7.11(i) now implies that $a = \delta|L_3(4)|$ is equal to a product of the orders of some composition factors of $\widetilde{M}_1$, which is a contradiction. This proves Claim 1.

Claim 2. $|M_1/H_1| > 2$. Suppose to the contrary that $M_1/H_1 \cong C_2$, and recall that $M_2/H_2 \cong M_1/H_1$. Thus (see Hypothesis 7.1 and Lemma 7.4(i)) $H_1 \times H_2 \leq H < M = M_1 \times M_2$ with $M/(H_1 \times H_2) \cong C_2^2$, and $\pi_1(H) = M_1$, $\pi_2(H) = M_2$. It follows that H is the unique index 2 subgroup of M containing $H_1 \times H_2$ and projecting onto each M_i , and in particular $|\Delta| = |M : H| = 2$. Moreover, for $i = 1, 2$, G_i acts faithfully and imprimitively on $[G_i : H_i]$ preserving a nontrivial partition corresponding to $[G_i : M_i]$ with blocks of size $2 = |M_i/H_i|$. Thus, the conditions of Notation 3.3 hold for G_i . Moreover, G_i is totally 2-closed by Hypothesis 7.1(e), and in particular $G_i^{[G_i:H_i]}$ is 2-closed, and also $G_i^{[G_i:M_i]}$ is 2-closed (since M_i is core-free in G_i by Lemma 7.4(i)). Thus the conditions of Theorem 3.10(1) hold for G_i , and it follows that there exists k_i with $2 \leq k_i \leq s_i$ such that $(M_i \cap M_i^{(k_i)})H_i \neq M_i$, and hence $M_i \cap M_i^{(k_i)} \leq H_i$ (since $|M_i : H_i| = 2$). It follows that $M_{(k_1, k_2)} = (M_1 \cap M_1^{(k_1)}) \times (M_2 \cap M_2^{(k_2)}) \leq H_1 \times H_2 \leq H$, and this implies that $M_{(k_1, k_2)}H = H \neq M$. Since $|\Delta| = 2$, this contradicts Theorem 3.10(1) applied to G , and Claim 2 is proved.

In the rest of the proof we deal in turn with each of the remaining possibilities (T, M) in Table 1 for (T_1, M_1) . Note that, by Claims 1 and 2, and in the light of Remark 7.12, if $(T_i, \mu_i(H))$ occurs in Table 1, then $|\mu_i(H)/(\widehat{T}_i \cap H)| > 2$ and $\mu_i(H)/(\widehat{T}_i \cap H)$ is not almost simple; and in particular $T_i \neq J_1$ or J_3 , and we have already seen that $T_i \neq \mathbb{M}$.

Case: (T_1, M_1) with $T_1 = \text{Ly}$ and $M_1 \in \{G_2(5), 3.\text{McL}, 3.\text{McL} : 2\}$. By the comments above and Table 1, it follows that M_1/H_1 is 3.McL or 3.McL : 2. Since a is a $\{2, 3, 5, 7\}$ -number and is a product of the orders of some composition factors of M_1/H_1 it follows that $a = 3$ so $A \leq \text{Sym}(3)$. Hence, by Lemma 7.11(ii), $F(X)$ is a 3-group. Moreover, since $N \trianglelefteq X$ and $N \leq B_A = A^s$ it follows that $F(X) \neq 1$.

Next, we note, by Lemma 7.4(ii), that $\mu_k(H)$ has a composition factor isomorphic to McL, for some $k \geq 2$. However the only finite simple group in

$\{J_1, J_3, J_4, \text{Th}, \text{Ly}, \mathbb{M}\}$, apart from Ly , with McL as a section is $\mathbb{M}$ (see [12]), so $T_k = \mathbb{M}$. Moreover, inspecting the maximal subgroups of $\mathbb{M}$ shows that the only one with McL as a section is $2^{1+24}.\text{Co}_1$, and hence $\mu_k(H)$ is contained in a maximal subgroup $2^{1+24}.\text{Co}_1$ of $\mathbb{M}$. Further, inspecting the subgroups of Co_1 we find that there are exactly two conjugacy classes of subgroups with McL as a composition factor, and both are second maximal subgroups of Co_1 : the classes are represented by the subgroups $\text{McL} < \text{Co}_2 < \text{Co}_1$ and $\text{McL} : 2 < \text{Co}_3 < \text{Co}_1$. Thus $\mu_k(H)$ has shape $I.S$, where I is a 2-group, and $S \in \{\text{McL}, \text{McL} : 2\}$. If we now relabel the T_i as discussed in Remark 7.12 so that $T_1 = \mathbb{M}$ and $M_1 = I.S$, then by Lemma 7.11, $a = 3$ is a product of the orders of some composition factors of (the new) M_1/H_1 , which is not the case. This contradiction completes our analysis of all cases with $T_1 = \text{Ly}$.

Thus, in the light of Remark 7.12, if $(T_i, \mu_i(H))$ occurs in Table 1, then $T_i \neq \mathbb{M}, \text{Ly}, J_3, J_1$, and hence T_i is J_4 or Th . Also $\mu_i(H)/(\hat{T}_i \cap H)$ is not almost simple, and has order at least 3.

Case: $(T_1, M_1) = (J_4, 2^{11} : M_{24})$ or $(J_4, 2^{10} : L_5(2))$ or $(\text{Th}, 2^5.L_5(2))$. Suppose first that $T_1 = J_4$, so $M_1 = O_2(M_1).S$ with $S = M_{24}$ or $L_5(2)$. Then $g(T_1, M_1)$ divides $2^4.3^2.5.7$ by Table 1. Now, for each $i \geq 2$, if $(T_i, \mu_i(H))$ does not occur in Table 1, then $g(T_i, \mu_i(H)) = 1$, while if it does occur, then, as we noted above, $T_i = \text{Th}$ (since T_1 is the only occurrence of J_4), and so by Table 1, $g(T_i, \mu_i(H))$ divides 6. Thus by Lemma 7.10 the integer a divides $2^5.3^3.5.7$. However, since $M_1 = O_2(M_1).S$ acts irreducibly on the elementary abelian group $O_2(M_1)$, it follows from Lemma 7.11(i) that a is divisible by $|O_2(M_1)|$ or $|S|$, which is a contradiction. Thus $(T_i, \mu_i(H))$ is not $(J_4, 2^{11} : M_{24})$ or $(J_4, 2^{10} : L_5(2))$ for any i .

In the light of the results above, an analogous argument with $T_1 = \text{Th}$ and $M_1 = O_2(M_1).L_5(2) = 2^5.L_5(2)$ gives $g(T_1, M_1) = 1$ from Table 1, and for each $i \geq 2$, if $g(T_i, \mu_i(H)) > 1$ then, by Table 1, $g(T_i, \mu_i(H)) = 2$ and $T_i = J_4$ (and there is at most one such i). Hence by Lemma 7.10, the integer a divides 2, so $a = 2$. However, since M_1 acts irreducibly on $O_2(M_1)$, it follows from Lemma 7.11(i) that a is divisible by $|L_5(2)|$ or 2^5 , which is a contradiction. Thus, for each $i \geq 1$, $(T_i, \mu_i(H))$ is not one of the three pairs in this case.

Case: $(T_1, M_1) = (\text{Th}, {}^3D_4(2))$ or $(\text{Th}, {}^3D_4(2).3)$. By the previous case and Claim 1, the only possibility is $M_1 = {}^3D_4(2).3$ and $H_1 = {}^3D_4(2)$. Thus (see Hypothesis 7.1 and Lemma 7.4(i)), $M_1/H_1 \cong M_2/H_2 \cong C_3$ and $H_1 \times H_2 \leq H < M = M_1 \times M_2$ with $M/(H_1 \times H_2) \cong C_3^2$, $\pi_1(H) = M_1$, and $\pi_2(H) = M_2$. Hence $|\Delta| = |M : H| = 3$, $H \trianglelefteq M$, and also $a = 3$ (see Theorem 3.10(2)).

Suppose that, for some $i \geq 2$, $(T_i, \mu_i(H))$ occurs in Table 1, so as observed above, $T_i = J_4$ (since T_1 is the only occurrence of Th). Then by the discussion in Remark 7.12 and Lemma 7.11(i), $a = 3$ is a product of the orders of some composition factors of $\mu_i(H)/(\hat{T}_i \cap H)$, and it follows from Table 1 that $\mu_i(H) = 2^{1+12}.3.R$ for some $R \in \{M_{22}, M_{22}.2, L_3(4), L_3(4).2\}$. Moreover, in this case $g(T_i, \mu_i(H)) = 2$ by Table 1. Now, at most one of the T_i is equal to J_4 , and for all other $i \geq 2$,

$T_i^{[T_i:\mu_i(H)]}$ has base size 2.

By [19, Lemma 2.20.1], there exists $t_1 \in T_1$ such that $M_1 \cap M_1^{t_1} \cong \text{Sym}(3)$. For each $i \geq 2$, if $T_i^{[T_i:\mu_i(H)]}$ has base size 2, then choose $t_i \in T_i$ such that $\mu_i(H) \cap \mu_i(H)^{t_i} = 1$; while if $T_i = J_4$ and $\mu_i(H) = 2^{1+12}.3.R$ with $R \in \{M_{22}, M_{22}.2, L_3(4), L_3(4).2\}$, then we may choose $t_i \in T_i$ such that $\mu_i(H) \cap \mu_i(H)^{t_i}$ has order dividing 16 (see [6, last line of Table 2]). Let $g_1 = t_1 \in G_1$, and $g_2 = (t_2, \dots, t_r) \in G_2$, and set $g := (g_1, g_2) \in G$. Then since $M = M_1 \times M_2$, we see that $M \cap M^g = (M_1 \cap M_1^{g_1}) \times (M_2 \cap M_2^{g_2})$. From the definition of the t_i we see that $M_2 \cap M_2^{g_2}$ is a 2-group, and $M_1 \cap M_1^{g_1} \cong \text{Sym}(3)$. It follows that $M \cap M^g$ has no quotient of order 3. Thus, $(M \cap M^g)H \neq M$, since $H \trianglelefteq M$ and $M/H \cong C_3$. This however contradicts Theorem 3.10(4).

Thus, we now know that, if $(T_i, \mu_i(H))$ occurs in Table 1, then $T_i = J_4$ and $\mu_i(H) = 2^{1+12}.3.R$ for some $R \in \{M_{22}, M_{22}.2, L_3(4), L_3(4).2\}$. In particular these are the only possibilities for (T_1, M_1) , and we complete the proof by showing that none of these four pairs is possible.

Case: $(T_1, M_1) = (J_4, 2^{1+12}.3.R)$ with $R \in \{M_{22}, M_{22}.2, L_3(4), L_3(4).2\}$.

In each of these cases, $g(T_1, M_1) = 2$, by Table 1. Moreover, for each $i \geq 2$, $T_i^{[T_i:\mu_i(H)]}$ has base size 2 and $g(T_i, \mu_i(H)) = 1$ (since now the only case where $(T_j, \mu_j(H))$ occurs in Table 1 is for $j = 1$). Thus $a = 2$ by Lemma 7.10.

We claim that $H_1 = 1$. Note that, by Claim 2, $|M_1/H_1| > 2$. Moreover, by Lemma 7.11(i), $a = |U| \cdot |V|$ for some $U \trianglelefteq (M_1/H_1)/Z(M_1/H_1)$ and $V \leq Z(M_1/H_1)$. Since $a = 2$, inspecting the possibilities for M_1/H_1 from the groups M_1 listed, we see that the only possibility for H_1 in each case is $H_1 = 1$ (with $U = 1$ and $V \cong Z(M_1)$), proving the claim.

Since $H \leq M_1 \times M_2 = M$, the subgroup H centralizes $Z(M_1)$. Consider the subgroup $K := HZ(M_1)$ of M . The condition $H_1 = 1$ implies that K properly contains H . Moreover since, by Hypothesis 7.1(c), H is maximal by inclusion such that $G^{[G:H]}$ is not 2-closed, it follows that $G^{[G:K]}$ is 2-closed. Since M_1 is core-free in T_1 there exists $t_1 \in T_1$ such that $M_1 \cap M_1^{t_1}$ does not contain $Z(M_1)$. Also, for each $i \geq 2$, $T_i^{[T_i:\mu_i(H)]}$ has base size 2, and so there exists $t_i \in T_i$ such that $\mu_i(H) \cap \mu_i(H)^{t_i} = 1$. Set $g_1 := t_1$, $g_2 := (t_2, \dots, t_r)$, and $g := (g_1, g_2) \in G$. Then $\mu_i(K \cap K^g) = 1$ for all $i \geq 2$, and hence $K \cap K^g \leq K \cap \widehat{T}_1 = Z(M_1) \cong 2$. On the other hand, $\pi_1(K \cap K^g) \leq \pi_1(M \cap M^g) = \pi_1(M_1 \cap M_1^{t_1}) = M_1 \cap M_1^{t_1}$, and $M_1 \cap M_1^{t_1}$ does not contain $Z(M_1)$. Thus, $K \cap K^g = 1$, so $G^{[G:K]}$ has base size 2. It now follows from Theorem 3.10(3) that $G^{[G:H]}$ is 2-closed. This final contradiction completes the proof of Theorem 1.2. $\square$

8. Concluding Remarks: The General Case

As a result of the work in this paper, the task of classifying all finite totally 2-closed groups G can now be reduced to the case where G is insoluble and $F(G) > 1$ (see Question 1 in Sec. 1.1). We conclude the paper with some remarks about this case,

and a reduction theorem.

We note first that a crucial step in our classification was the essential reduction to classifying those finite groups (with trivial Fitting subgroup) which are 2-closed in all of their transitive permutation representations (see Proposition 2.10). Although we weren't able to reduce our study of arbitrary finite totally 2-closed groups to transitive actions, the following theorem (and subsequent corollary) significantly reduces the types of faithful actions that need to be examined. In order to state it, we need some additional terminology: Permutation representations of a group G on sets Γ and Γ' are said to be *permutationally equivalent* if there exists a bijection $f : \Gamma \rightarrow \Gamma'$ such that, for all $\gamma \in \Gamma$ and all $g \in G$, the image $(\gamma^g)f = (\gamma f)^g$. Note that f induces a bijection $\Gamma \times \Gamma \rightarrow \Gamma' \times \Gamma'$, namely, $f : (\alpha, \beta) \rightarrow (\alpha f, \beta f)$, and that this induced map sends G -orbits in $\Gamma \times \Gamma$ to G -orbits in $\Gamma' \times \Gamma'$.

Our general reduction theorem can now be stated as follows.

Theorem 8.1. *Let $G \leq \text{Sym}(\Omega)$, and suppose that Γ, Γ' are distinct G -orbits on which the G -actions are permutationally equivalent relative to a bijection $f : \Gamma \rightarrow \Gamma'$. Let $\Omega' := \Omega \setminus \Gamma$, so $\Omega = \Omega' \cup \Gamma$. Then*

- (a) *both G and $G^{(2),\Omega}$ act faithfully on Ω' , and*
- (b) *The restriction $G^{(2),\Omega}|_{\Omega'} = G^{(2),\Omega'}$.*
- (c) *If G is finite, then $G = G^{(2),\Omega}$ if and only if $G^{\Omega'} = G^{(2),\Omega'}$.*

Proof. (a) Note first that since $G^{(2),\Omega}$ has the same orbits as G in Ω , the set Ω' is $G^{(2),\Omega}$ -invariant. Also, $G \leq G^{(2),\Omega}$. To prove (a), it is therefore sufficient to show that $G^{(2),\Omega}$ is faithful on Ω' . So assume that $x \in G^{(2),\Omega}$ such that x fixes Ω' pointwise, and let $\gamma \in \Gamma$. Then $\gamma f \in \Gamma' \subseteq \Omega'$, so $(\gamma f)^x = \gamma f$. By Theorem 2.4, there exists $g \in G$ such that $\gamma^x = \gamma^g$ and $(\gamma f)^x = (\gamma f)^g$. Thus $\gamma f = (\gamma f)^g$, and by the defining property of f , $(\gamma f)^g = (\gamma^g)f$. This implies that $\gamma f = (\gamma^g)f = (\gamma^x)f$, and since f is a bijection we deduce that $\gamma = \gamma^x$. It follows that x fixes Γ pointwise, and hence that $x = 1$. This proves part (a).

(b) Let $g \in G^{(2),\Omega}$. Then g leaves invariant each G -orbit in $\Omega \times \Omega$. In particular, $g^{\Omega'}$ leaves invariant each G -orbit in $\Omega' \times \Omega'$ and hence (see [40, Remark 4.2]) $g^{\Omega'} \in G^{(2),\Omega'}$. Thus $G^{(2),\Omega}|_{\Omega'} \leq G^{(2),\Omega'}$. To prove the reverse inclusion let $x \in G^{(2),\Omega'}$. We extend the definition of x to a permutation $y \in \text{Sym}(\Omega)$ by defining $\gamma^y = \gamma^x$ if $\gamma \in \Omega'$, and $\gamma^y = ((\gamma f)^x)f^{-1}$ if $\gamma \in \Gamma$. Since $f : \Gamma \rightarrow \Gamma'$ is a bijection, y is a permutation of Ω .

We claim that $y \in G^{(2),\Omega}$. Note that this claim implies that each element of $G^{(2),\Omega'}$ lies in $G^{(2),\Omega}|_{\Omega'}$, and hence a proof of the claim completes the proof of part (b). By definition, y leaves invariant all the G -orbits in $\Omega' \times \Omega'$, so we need to consider G -orbits Δ contained in $\Gamma_1 \times \Gamma_2$ where the Γ_i are G -orbits in Ω and at least one is equal to Γ .

Suppose first that $\Gamma_1 = \Gamma_2 = \Gamma$ and define $\Delta' = \{(\alpha f, \beta f) \mid (\alpha, \beta) \in \Delta\}$. To see that Δ' is G -invariant let $g \in G$ and $(\alpha f, \beta f) \in \Delta'$. Then, using the defining

property of f ,

$$(\alpha f, \beta f)^g = ((\alpha f)^g, (\beta f)^g) = (\alpha^g f, \beta^g f),$$

which lies in Δ' since Δ is G -invariant. A similar argument shows that Δ' is a G -orbit in $\Gamma' \times \Gamma' \subseteq \Omega' \times \Omega'$. Thus y leaves Δ' invariant (since $y^{\Omega'} = x \in G^{(2),\Omega'}$). Now take an arbitrary element $(\alpha, \beta) \in \Delta$. Then by the definition of y , $(\alpha, \beta)^y = ((\alpha f)^x f^{-1}, (\beta f)^x f^{-1})$ and we have to prove that this lies in Δ . By the definition of Δ' , $(\alpha f, \beta f) \in \Delta'$, and since y leaves Δ' invariant and $y^{\Omega'} = x$, Δ' contains $(\alpha f, \beta f)^y = ((\alpha f)^x, (\beta f)^x)$. Applying Theorem 2.4 to $x \in G^{(2),\Omega'}$, we see that there exists $h \in G^{\Omega'}$ such that $(\alpha f)^x = (\alpha f)^h$ and $(\beta f)^x = (\beta f)^h$, and by part (a), there exists a unique element $g \in G$ such that $g^{\Omega'} = h$, and so $(\alpha f)^x = (\alpha f)^g$ and $(\beta f)^x = (\beta f)^g$. Thus, we have

$$(\alpha f)^x f^{-1} = (\alpha f)^g f^{-1} = (\alpha^g) f f^{-1} = \alpha^g,$$

and similarly $(\beta f)^x f^{-1} = \beta^g$. Thus

$$(\alpha, \beta)^y = ((\alpha f)^x f^{-1}, (\beta f)^x f^{-1}) = (\alpha^g, \beta^g) = (\alpha, \beta)^g,$$

which lies in Δ since Δ is G -invariant. It follows that y leaves invariant each G -orbit $\Delta \subseteq \Gamma \times \Gamma$.

To deal with the remaining G -orbits we assume that $\Gamma_1 \neq \Gamma_2 = \Gamma$ (since the case $\Gamma_1 = \Gamma \neq \Gamma_2$ is similar). This time we define $\Delta' = \{(\alpha, \beta f) \mid (\alpha, \beta) \in \Delta\}$. A similar argument shows that Δ' is a G -orbit in $\Omega' \times \Omega'$ and hence is left invariant by $y^{\Omega'} = x$. For an arbitrary pair $(\alpha, \beta) \in \Delta$, we have $(\alpha, \beta)^y = (\alpha^x, (\beta f)^x f^{-1})$ and we have to prove that this lies in Δ . By Theorem 2.4 applied to $x \in G^{(2),\Omega'}$, there exists $g \in G$ such that $\alpha^x = \alpha^g$ and $(\beta f)^x = (\beta f)^g$, and by the property of f , $(\beta f)^g = (\beta^g) f$. Thus

$$(\alpha, \beta)^y = (\alpha^x, (\beta f)^x f^{-1}) = (\alpha^g, (\beta^g) f f^{-1}) = (\alpha^g, \beta^g) = (\alpha, \beta)^g \in \Delta,$$

since Δ is G -invariant. Thus y leaves Δ invariant in this case also. It follows that y leaves invariant each G -orbit in $\Omega \times \Omega$ and hence $y \in G^{(2),\Omega}$. Part (b) follows.

(c) Now assume that G is finite. Suppose that $G = G^{(2),\Omega}$. By part (a), $G \cong G^{\Omega'}$ and $G^{(2),\Omega} \cong G^{(2),\Omega}|_{\Omega'}$, and by part (b) the latter group is $G^{(2),\Omega'}$. Thus, $G^{\Omega'} \cong G^{(2),\Omega'}$, and since G is finite it follows that $G^{\Omega'} = G^{(2),\Omega'}$. Conversely suppose that $G^{\Omega'} = G^{(2),\Omega'}$. By part (b), $G^{\Omega'} = G^{(2),\Omega}|_{\Omega'}$, and by part (a), we have $G \cong G^{(2),\Omega}$, and since G is finite equality follows: $G = G^{(2),\Omega}$. $\square$

Corollary 8.2. *Let G be a finite group. Then G is totally 2-closed if and only if $G = G^{(2),\Omega}$, for each faithful G -action on a set Ω such that, for distinct G -orbits Γ, Γ' in Ω , the G -actions on Γ, Γ' are not permutationally equivalent.*

Proof. By definition, G is totally 2-closed if and only if $G = G^{(2),\Omega}$, for each faithful G -action on a set Ω . If there exist distinct G -orbits Γ, Γ' in Ω such that the G -actions on Γ, Γ' are permutationally equivalent, then by Theorem 8.1, G acts faithfully on $\Omega' := \Omega \setminus \Gamma$ and the $G^{(2),\Omega}$ is isomorphic to $G^{(2),\Omega'}$. Since G is finite, it follows that G^{Ω} is 2-closed if and only if $G^{\Omega'}$ is 2-closed. Thus to check that G

is totally 2-closed we only need to check 2-closure for faithful actions in which the G -actions on distinct orbits are not permutationally equivalent. $\square$

The significance of Corollary 8.2 is that in order to decide whether or not a given finite group G is totally 2-closed, we no longer have to concern ourselves with studying all faithful permutation representations of G (of which there are infinitely many). We can now determine the total 2-closure or non-total 2-closure of G by deciding whether or not G is 2-closed in all of its faithful permutation representations in which no two distinct G -orbits admit permutationally equivalent actions. Since the number of permutational equivalence classes of transitive G -actions is equal to the number $c(G)$ of G -conjugacy classes of subgroups, there are finitely many (at most $2^{c(G)} - 1$) such representations.

We conclude the paper by making a few remarks about how to check the total 2-closure of a given “small” finite insoluble group G with nontrivial Fitting subgroup. One of the quickest ways to prove that such a group G is **not** totally 2-closed is to use Lemma 2.2: if G has a factorization $G = HK$ with $\text{Core}_G(H) \cap \text{Core}_G(K) = 1$ and $G \neq \text{Core}_G(H) \times \text{Core}_G(K)$, then G is not totally 2-closed. As an example, if $G = 3 \cdot \text{Alt}(6)$ is a triple cover of $\text{Alt}(6)$, then G has subgroups H and K with $H \cong \text{Alt}(5)$ and $K \cong \text{GL}_2(4)$ satisfying the above properties. (We remark that Lemma 2.2 is deduced from (the much stronger) Theorem 2.1, so Theorem 2.1 can also be a highly useful tool in proving that a group cannot be totally 2-closed.)

If no such factorization exists (or more generally, if Theorem 2.1 is not available), then the question of whether or not G is totally 2-closed becomes more difficult. Although one can use direct computation and Corollary 8.2 when $|G|$ is small enough, the number of faithful permutation representations coming from the corollary, and the degrees involved, get large quite quickly. Thus, checking the 2-closure of the relevant permutation representations becomes computationally heavy, even for relatively small $|G|$. For instance, if G is a finite insoluble group with nontrivial Fitting subgroup, and if either $|G| < 1440$, or $|G| = 1440$ and G is not isomorphic to the group `SmallGroup(1440,4594)` (which is a double cover $2 \cdot \text{PGL}_2(9)$ of $\text{PGL}_2(9)$) in the GAP small groups library [35], then one can directly compute that G has a factorization $G = HK$ with $\text{Core}_G(H) \cap \text{Core}_G(K) = 1$, but $G \neq \text{Core}_G(H) \times \text{Core}_G(K)$. It then follows from Lemma 2.2 that no such group is totally 2-closed. The group $G := \text{SmallGroup}(1440,4594)$ has no factorization $G = HK$ with $\text{Core}_G(H) \cap \text{Core}_G(K) = 1$, however, and so, although one can check that G is 2-closed in all of its faithful transitive permutation representations, the question of whether or not G is totally 2-closed remains open. (There are 30 conjugacy classes of subgroups of G in this case, with subgroups in 26 of them containing $Z(G) \cong 2$, and subgroups in the other four classes being core-free. Thus, if G is totally 2-closed, then proving this fact using Corollary 8.2 would require one to check the total 2-closure of $2^{30} - 2^{26} = 1006632960$ permutation representations, of increasingly large degree.)

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